

5 Catalogue of Etruscan and Roman-era dental appliances

PART I: CATALOGUE OF ETRUSCAN AND ROMAN-ERA GOLD-BAND DENTAL APPLIANCES

Three examples that may be in Buffalo, New York: the Barrett and Van Marter appliances

Background: history and identification

Dr. William C. Barrett, an editor of the *Dental Register* and first Dean of the Dental Department at the University of Buffalo, was friend and colleague to Dr. James Gilbert Van Marter, who was practicing in Rome, Italy during the 1880s. Barrett received his DDS from the University of Pennsylvania in 1881 and became a founder of the dental school at Buffalo c.1891. The source of details of Van Marter's life and training in dentistry and how these two came to correspond, remain unknown. Van Marter was to obtain three Etruscan dental appliances, which eventually came to his friend's collection in Buffalo, New York.

In the January 1885 issue of the *Independent Practitioner*, Waite (1885a: 508) refers to an article by a Dr. Van Marter of Rome (probably Van Marter 1885a).¹ This suggests that by 1885 Van Marter had made drawings of the two prostheses then in the Corneto (the old name for Tarquinia) Museum (nos. 11–12). This information may have stimulated Barrett to ask Van Marter to act as his agent in the purchase of other examples from the Etruscan tombs then being systematically stripped of their offerings.

Van Marter's letters to Barrett describe the excavations of tombs in the area of ancient Tarquinia, claiming that gold dental prostheses were often recovered. During this important period in the history of dentistry, with the creation of formal schools and professional societies, there was great interest in dental hygiene as well as dental history. By 1886 Van Marter had located and sent two appliances to Buffalo. After the first (Barrett I, no. 1) arrived, Barrett requested that another be purchased for him, together with any archaeological materials from the same tomb. This prescient concern for archaeological context reflects the scientific orientation of these medical practitioners. The second dental appliance

(Barrett II, no. 2) that Van Marter purchased for Barrett is described in a letter of 1886. A third appliance that appears to have been sent to America, now identified as the Van Marter example (see no. 3), may have been excavated about 1889 during the period of systematic mining of major Etruscan tombs (Becker 1999a).

Both Barrett gold appliances are simple bands, within which only some of the teeth appear to have survived. It is possible that teeth were added to the Barrett I appliance in order to provide some verisimilitude to this artifact. The records suggest that three teeth were with the Barrett II appliance when it was sent to America. The third appliance, the Van Marter, held an original false tooth. The reason for Barrett's purchase of a third example may have been to own a true pontic with a false tooth, as he already had two simple band "retainers." The source(s) for these three appliances was always unclear and remains largely unknown since they had disappeared by the time of Barrett's death in 1903 (but, see Becker 1999a).

Some confusion arose in the late nineteenth century between that first pair of appliances owned by Barrett and the Liverpool pair (nos. 13–14). The confusion apparently derives from W. H. Waite's note (1885a: 508) indicating that Waite had borrowed both (Liverpool) pieces in 1885, and exhibited them before two unnamed New York dental societies. The pair must be the same two brought to Hereford, England in August of 1885 (Waite 1885a: 508): thus, the Liverpool pair, and not the Barrett pieces. Mr. Spence Bate, of the British Dental Association (Western Branch) suggested using funds at hand to illustrate these pieces, inasmuch as *The Independant Practitioner* "did not circulate largely in this country." The resulting drawings (Waite 1885a: facing p. 512) clearly depict the Liverpool pair of dental appliances collected by English goldsmith Joseph Meyer in 1857 (nos. 13–14).

Edlund (1981) found that Barrett's books and dental instruments had been bequeathed to the University at the time of his death in 1903; while his "Etruscan relics" were given to the Buffalo and Erie County Historical Society, where she examined the three vases. They were a small jug, a *plumpe Kanne* (beaked jug) in impasto, imitating bronze vases with palmette handles, and a bucchero ovoid oinochoe with animal heads framing the handle. A date c.500 BCE fits these surviving vases, and presumably the burial with the dental appliance.

A cursory search of *Publications of the Buffalo Historical Society* for the years around 1903 did not locate any information regarding the Barrett estate, but subsequent findings may explain this failure. The estate, or parts of it, did not formally become part of the collections until 1930 (Edlund 1981). The present locations of these three important examples of the history of dentistry remain unknown. The following descriptions are based on published illustrations and descriptions studied by Becker. Dunn (1894), in discussing the Italian appliances, stated that, by the early 1890s, five examples were "lost," presumably indicating that they had been sold or given to private collections. If this was Dunn's meaning, the three examples of appliances sent to Dr. Barrett might account for most of those described as "lost."

1 Barrett I

Present location: unknown (lost with Barrett Collection; Buffalo, New York).

Type: simple band (3 spaces).

Origin: Etruria.

Jaw part: maxilla, II plus (II, I2?).

Date: if Van Marter's (1886: 59) description of grave goods is correct, possibly 500–475 BCE.

Sex: female? (based on size of teeth/jaw).

Dimensions: L. 20, W. 1.2–1.6 mm.

Description: narrow band to stabilize three teeth. A simple, very narrow gold-band appliance, perhaps 0.2 to 0.3 mm thick, varying from 1.2–1.6 mm wide. It was designed to loop around no more than 3 anterior teeth, probably the 3 central incisors and a lateral incisor. Length as illustrated approx. 20 mm, with a buccal-lingual

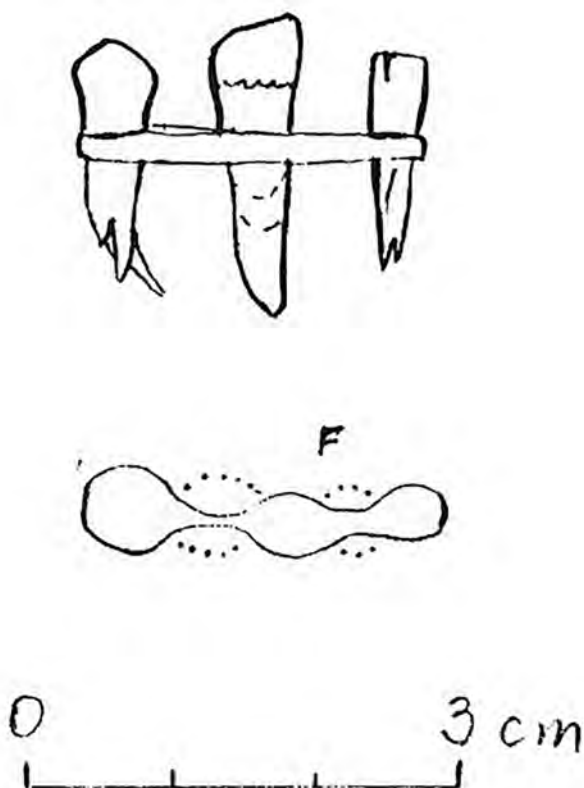


Figure 5.1 The Barrett I appliance, two views drawn after Farrar 1888: figs 6 and 8. View with dotted lines suggests the locations of the teeth originally bound by the appliance.

distance approx. 6 mm. The total length of the gold strip is calculated at 53 mm, after the overlapping ends had been welded. Published information suggests an elongated oval, with a constricted or pinched center that reflects post-interment affects. When in place it may have been ornamental, but may have been intended to stabilize a loose tooth at the center of the band. The tooth crown, from which the root has decayed, may be from the same tomb, even from the owner, but this cannot be confirmed.

References

- Becker 1999a.
Edlund 1981.
Farrar 1888, I: 33, figs 6, 8.
Naso 2011: 151, no. 10.
Van Marter 1885a, 1886: 59.

History and context

Van Marter (1886: 59) describes the source of this appliance as a tomb of the sixth century BCE (but see no. 2) “or about one hundred years prior to the dates of the oldest partial denture which *I sent you last year*” (emphasis ours). Van Marter’s comment suggests that around 1885, the year before he sent the “entire” collection of artifacts recovered from an Etruscan tomb to Barrett (see Barrett II), Van Marter had already sent him the first gold dental appliance. Van Marter’s published statement, quoted in full under Barrett II (no. 2), is clarified by reading Farrar’s 1888 publication, which describes the two Etruscan appliances sent to Barrett at his address in Buffalo, New York.

The Barrett I piece was excavated prior to Barrett II that has been more extensively described. Our only description of the Barrett I example comes from John Nutting Farrar. Farrar practiced dentistry in New York City but, like Barrett, came from upstate New York. Farrar may have seen both Barrett pieces in Buffalo; however, Barrett appears to have carried these artifacts to various other cities to show his colleagues. Lacking data on the other appliances then known in Italy, and intrigued by these early examples of restorative dentistry, Farrar illustrated his brief text on ancient dental gold-working technology with the two examples that were near at hand. This decision is quite fortunate for us, as today this record provides the only information that exists for this one of the two Barrett pieces.

J. N. Farrar was one of the pioneers of orthodontic practice, concerned with both the cosmetic and functional aspects of bridgework used in orthodontics (Farrar 1905). The Etruscan pontics that he examined had only cosmetic purposes, but may have had at least incidental orthodontic functions since the replacement tooth served to hold the living teeth in place. Simple band appliances could have served the ancillary function of sustaining normal dental

articulation by stabilizing loose teeth and/or maintaining normal tooth positions. In no case could they have served to shift the position of teeth, a complex aspect of orthodontics sometimes claimed for Etruscan appliances by recent commentators (Corruccini and Pacciani 1989; Torino *et al.* 1996).

Description

Van Marter's description of the Barrett I appliance (1885a; cf. Van Marter 1885b and c) is not very clear, nor is his illustration of much use in delineating its form. Van Marter's vague description (1885a) was used by Helbig (1886: 25), and provided Bliquez (1996) with an indication of the type of appliance that was once, and still may be, in Buffalo (see also Sterpellone and Elsheikh 1990: 139).

2 Barrett II

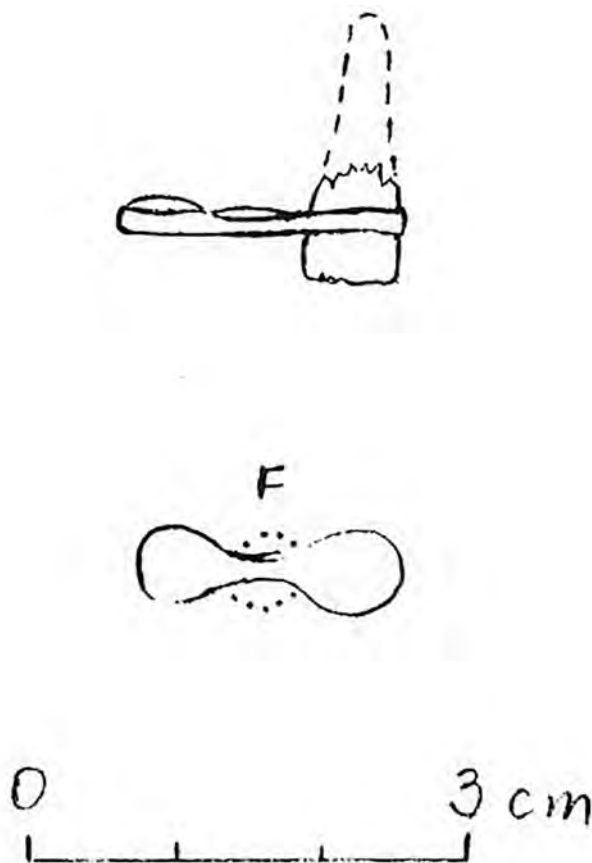


Figure 5.2 The Barrett II appliance, two views drawn after Farrar 1888: figs 5 and 7. The dotted lines suggest the locations of the teeth originally bound by the appliance.

Present location: unknown (lost with Barrett Collection; Buffalo, New York).

Type: simple band (4 spaces).

Origin: Bisenzio (ancient Visentium); necropolis at Capodimonte on western shore of Lake Bolsena.

Jaw part: maxilla (possibly 2I–I2).

Date: fifth century BCE? (Helbig 1886; see also Edlund 1981).

Sex: female.

Dimensions: 25 by 2.0–2.5 mm

Description: simple, narrow gold-band appliance, probably about 0.2 to 0.3 mm thick and varying from 2.0–2.5 mm wide. The length of the band appears designed to loop around the crowns of no more than 4 anterior teeth; most likely the 4 incisors. Length as illustrated approx. 25 mm, with a buccal-lingual distance of about 6–7 mm. The total length of the gold strip is calculated at 60 mm, after the overlapping ends had been welded. Published information suggests an elongated oval that, as originally illustrated with two constricted or pinched areas, held three teeth in place. These constrictions appear to hold in place three teeth introduced post-excavation. This post-interment alteration depicts the band as binding these teeth, probably not original to the appliance, at the neck or lower on the root, which would have been impossible. When in place it may have been only ornamental, but it may have been intended to stabilize one or two loose teeth at the center of the band.

References

Becker 1999a.

Bliquez 1996: 2654. Bliquez's appliance "R," fig. 28.

Cigrand 1893: 60n (cites Van Marter 1886).

Edlund 1981.

Farrar 1888: 33, figs 5, 7.

Helbig 1886: 25–26.

Koch 1909: fig. 5.

Naso 2011: 150, no. 1.

Sudhoff 1926: 86, fig. 52 ("*nach Guerini*," but clearly a photograph of a poor copy).

Van Marter 1886: 59–60, fig. 3.

Waarsenburg 1991a: 242, n. 5 (no. 14).

Weege 1913, Vol. II: 371.

Weinberger 1948: 125, fig. 41: 4 (from Sudhoff).

History and identification of context

Van Marter stated:

The most recently opened and the oldest Etruscan tomb yet discovered in Italy was lately excavated at Capadimonti [sic], near the Lake of Bolsena. The entire contents of this tomb, including three teeth bound together with a band of pure gold, gold spiral rings for the side hair, silver finger

ring, necklace of amber and glass, arm band, bronzes, vases, etc., etc., I take pleasure in sending you by first express. The part of the find of interest to our profession is the three teeth, a drawing of which I send you herewith. (See fig. 3.) This tomb belongs to the VIth Century BC, or about one hundred years prior to the dates of the oldest partial denture which I sent you last year.

(Van Marter 1886: 59)

From Farrar's comments one cannot tell if both Barrett dental appliances came from this tomb or from two separate tombs. Van Marter cites only one appliance. Other information indicates that two separate tombs were the sources.

Farrar (1888: 33) also says that the excavation took place in 1886, and that the entire contents of the tomb were bought by the American dentist Dr. William Carr. Perhaps Farrar simply made an error in the name, but it is more likely that he was noting the name of another intermediary in this transaction. Farrar notes that the tomb contents subsequently were presented to "W. C. Barrett, and now constitute a portion of his private museum. The specimens in this collection are more especially interesting to the dentist [including carious molars]." He continues:

Besides these teeth [the molars], there were three that were bound together with a gold band; also the crown of another (central), bound with a similar device. Figs. 5 and 6 illustrate the two specimens, and show the relation of the bands to the teeth in side view. In Figs. 7 and 8 the plain lines show the top view of the bands detached [sic] from the teeth as they appeared when I saw them. When in use, however, these bands were undoubtedly bent as shown by the dotted lines. These bands were probably bound to the teeth with gold wire as shown in Fig. 9 [where Farrar illustrates the Sidon I example, see no. W-1]. The tomb contained other antiquities, a list of which is given below, as evidences of the high rank held by the occupants of the tomb, and to show that consequently these specimens of denture were probably of the best that the period afforded.

(Farrar 1888: 33)

Farrar lists artifacts supposedly recovered with this piece (also Helbig 1886; Edlund 1981):

- two spiral gold rings;
- incised hair ornaments;
- one silver finger ring;
- necklace of lapis lazuli and pure amber;

bronze plaque nearly 2 feet in diameter, with lions' feet and sea horses in full relief;

one large bronze vase with ear pieces representing the Taurian Jove;

bronze ornament, evidently an armour-piece for the head;

bronze wine-strainer;

silver "fibrilla" [sic];

two bronze cups;

four earthen jugs (two finely molded);

two vase handles with carved head and tail pieces.

Weege (1913, II: 371) lists a gold dental appliance recovered from a tomb in Capodimonte, but whether it is the Barrett II piece or yet another remains uncertain. Also of importance in Weege's text is the reference to a piece that Van Marter says was sent to Barrett during the previous year, about 1885. This "partial denture which I sent you last year" (Barrett I), as we now know, was an actual appliance, not simply information concerning that appliance.

The Barrett II appliance must have been recovered late in 1885 or in 1886, because the associated tomb goods were described by Helbig in an 1886 publication (see also Edlund 1981). Helbig (1886: 25) suggests that the appliance is from one of the "*tombe a fossa che furono scoperte in uno strato inferiore a quello teste' descritto*," ("trench tombs discovered in a lower stratum from that described") Helbig (1886: 26) dated the earlier tomb to the sixth century, based on comparison with Attic vases from a tomb at Tarquinia. This made the Barrett I appliance the earliest dental appliance then known. Edlund (1981) identifies three surviving artifacts as dating c.500 BCE and probably from the tomb with the appliance. The list of tomb contents sounds like a mixture of types of the sixth through fifth or fourth centuries, as expected for a family tomb with multiple depositions. Only in 1898 was an earlier dental appliance discovered at Satricum (Satricum, no. 18).

Barrett's estate supposedly was divided between The University at Buffalo and the Buffalo and Erie County Historical Society (see Edlund 1981: 82). Although he died in 1903, the acquisition is reported to have become official only in 1930, possibly being considered as a loan until that time. Since Barrett's widow died about 1925, these vases may have remained in her possession until her death. If all the Barrett materials remained with his widow, the gold appliance or appliances, as well as other valuable items, may have been sold or otherwise lost from the collection between 1903 and 1925.

Aside from the three vases noted by Edlund, all other pieces from this tomb including the dental appliance, remain missing. Mrs. Mildred F. Hallowitz, History of Medicine Librarian of the Health Sciences Library (SUNY-Buffalo) provided Edlund with information regarding the bequest of artifacts to the School of Dentistry. One item, a bronze situla, has been noted as possibly being sent to Dresden, but (according to Becker) there may be some confusion between the Barrett tomb bronzes and other bronzes from this tomb that Helbig (1886) notes went to Dresden.

The Albright-Knox Museum in Buffalo, New York, which focuses on modern art, also holds a small collection of ancient artifacts. It is located across the street from the Erie County Historical Society, to which some of the Barrett collection was bequeathed. The enormous warehouse of the Erie County Historical Society, opened in 1990, contains vast numbers of boxes awaiting inventory. We may hope that three Etruscan gold dental appliances are among these holdings.

Description/identification

The earliest illustrations and descriptions leave much to be desired; each commentator offered a slightly different location and description of the teeth. Farrar illustrates this second Barratt appliance as if it were a mandibular prosthesis. Although this could be the case, it would be very unlikely (but, see no. 21). The teeth depicted appear to be a second premolar, canine, and central incisor, with one space between each where a first premolar and lateral incisor would be expected. Farrar's illustrations of Barrett II show a simple gold band that extended over five tooth spaces. This band may have been used to stabilize loose teeth rather than hold a false tooth in position. At least one of the 4 teeth surrounded by this appliance was lost after death, assuming that the remaining 3 teeth depicted by Farrar actually were original to this band (which Becker doubts).

Helbig suggests that this prosthesis, encompassing the upper left lateral incisor-canine-premolar, was worn by a "young woman," a "*giovinetta*." He clearly, and correctly, indicates that the band might only have served to hold a naturally loosened canine in place. The canine is less likely to be loosened by a blow or periodontal disease than either of the adjacent teeth, but Helbig is essentially correct. Koch (1909: 16, fig. 5) offers a very rough sketch, possibly taken from Van Marter, and erroneously illustrates it as mandibular.

Weinberger (1948: 125) accepts Farrar's conclusion that this appliance is mandibular (I1, C., and PM2) but completely errs in suggesting that it has rings with spacing bars where the PM1 and I2 are missing. The very narrow gold band appears to have been pinched together in the two spaces where no teeth were present, probably after recovery from the tomb. This appliance should be compared with Bruschi I (no. 8), a thin band from Tarquinia that has been pinched

together (probably recently) in the center, where it once embraced a tooth. (Note that Bruschi I is also described incorrectly as having a spacer “bar” separating “two rings.”)

3 Van Marter

Present location: unknown (perhaps in Buffalo, New York?).

Type: ring series (3), with central false tooth.

Origin: Lake of Valseno (Bolsena?), near Rome.

Date: 600 BC (?) according to Van Marter.

Jaw Part: maxilla?

Sex: female ??

Dimensions: 25 by 4 mm (estimated from published drawing).

Description: constructed of gold rings welded together to replace right central incisor with reworked human tooth (not cut from a central incisor). This three-ring appliance, with the central loop altered to a more rectangular configuration, is approximately 5 mm deep. The squared outline helps prevent the modified, or false, central incisor from rotating in this socket. The lateral rings as depicted in the drawing, the only extant record, appear similar but one may have been designed to surround a central incisor and the other for a lateral incisor, the teeth that served as anchors for a postulated replacement incisor.

References

Becker 1999a.

Naso 2011: 150–151 no. 4.

Van Marter 1889: 261.

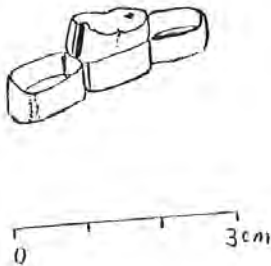


Figure 5.3 The Van Marter appliance (drawn after Van Marter 1889: 261).

History and identification

This dental prosthesis had been “buried” in the literature for over 100 years until recovered in the course of this study (Becker 1999a). Only a drawing and brief description, published by Dr. James Gilbert Van Marter (1889), preserve it for us. It is only fitting that this example be named for him.

In 1889 or shortly before, while Van Marter resided in Rome, he recorded that this dental appliance had been “taken from an Etruscan tomb lately opened not far from Rome on the lake of Valseno” (Lake Bolsena; Marter 1889: 261). This appears to be the third and final Etruscan dental prosthesis sent to Dr. Barrett in Buffalo, assuming that the publication date of 1889 accurately reflects the date of discovery. The two earlier examples sent to Barrett derive from tombs excavated around 1885–1886. The discovery and *purchase* of three gold appliances by Van Marter in this five- year period reflects the considerable extent of excavation of rich tombs in Etruria during the 1880s.

Fortunately, Barrett appears to have arranged publication of the note written by Van Marter, and illustrated this brief notice with a drawing slightly larger than life size. In many aspects this piece is similar to the Copenhagen example (no. 4), but here the replacement tooth appears to have survived while the post teeth did not, exactly the reverse of the situation with the Copenhagen example (Becker 1992a).

Description

Van Marter states that the rings of this specimen are welded, with no joints evident, as would be the case were the rings soldered together. While not explicit in his description, Van Marter certainly meant welding, which is exactly how the Copenhagen example and all of the other ring-series appliances appear to have been assembled. The anchor (or post) teeth to which this bridge was attached are depicted as missing, with only the central ring holding a “tooth.” Van Marter (1889: 261) describes this replacement tooth as “a bicuspid, turned one-fourth round on its axis.” This may have been meant to replace a large central incisor. Although Van Marter’s illustration shows this appliance in a “mandibular” position, the “tooth” appears to be a large right maxillary example or, more likely, a carved false tooth. Becker suggests that this may have been intended to replace a maxillary central incisor and not the PM1 (possibly bridging the spaces I1–I2, rather than C–PM1–PM2). This latter series would be very unlikely (replacing a first premolar).

From the illustration it appears that the replacement tooth is held in position like a gemstone, in a fashion similar to that of the Copenhagen example (Becker 1992a, 1994b; no. 4). That is, the “ring” is rectilinear and holds the replacement by pressing the gold band to conform to the contours of the tooth. In its construction the Van Marter piece is closest in type to the Copenhagen bridge.

4 Copenhagen



Figure 5.4a The Copenhagen appliance, frontal view.

Source: courtesy of the National Museum of Denmark, Copenhagen.

Present location: National Museum, Copenhagen (Department of Near Eastern and Classical Antiquities), inventory no. 8319.

Type: three welded loops, no rivet.

Origin: Orvieto (via Riis 1941: 161).

Jaw part: maxilla, 11–12.

Date: c.500–490 BCE or slightly later, based on ceramic associations.

Sex: female.

Condition: deteriorated somewhat since photographs were published.

Dimensions: 21.6 by 5.5 to 6.2 mm.

Description: “three-ring” appliance with lateral loops carefully fitted to the original and surviving dental crowns (left central incisor and right lateral incisor) and a sharply rectangular central gold unit holding a false tooth, not recovered. The three units have buccal-lingual distances that range from 5.1 to 6.7 mm.

References

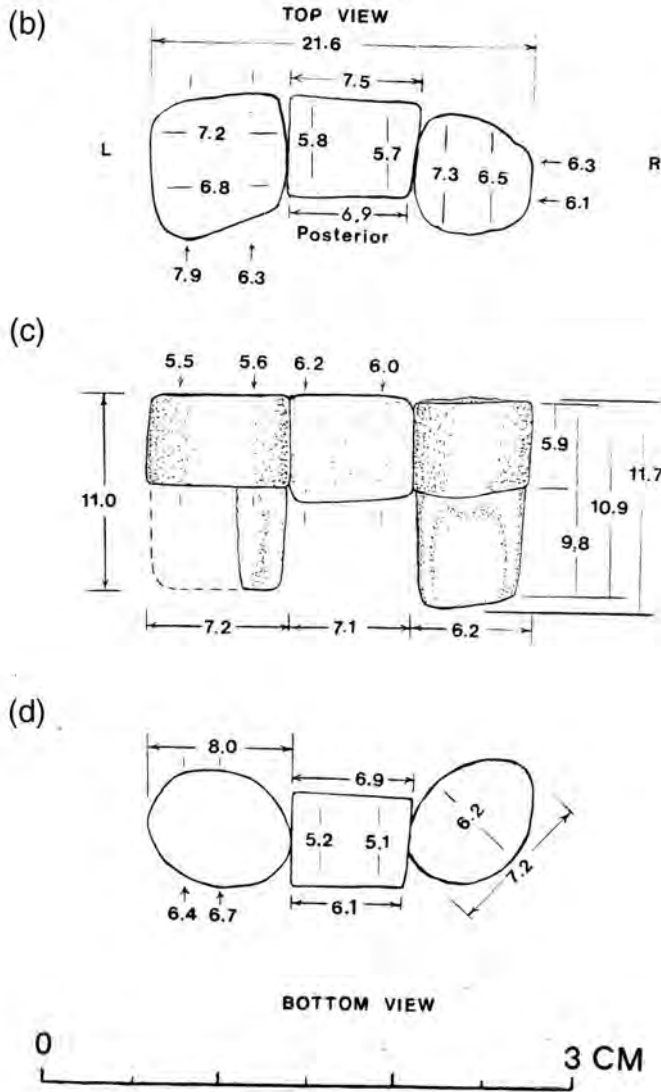
Baggieri (1998: 69) gives provenance “Volsinii” (modern Orvieto).

Becker 1992a, 1994b, 1995a, 1997, 1999b: fig. 3.

Bliquez 1996: 2656–2657, fig. 56. Bliquez appliance “W.”

Johnstone 1932a: 132n, pl. 94:17–18.

Marvitz 1982: 49.
Naso 2011: 151, no. 6.
Pot 1985: 38–39.
Poulsen 1927: 47.
Riis 1941: 161.
Villamil 1994.



Figures 5.4b–d Views of the Copenhagen appliance, drawn by Becker.

History and provenance

The National Museum of Denmark has an excellent photograph of this appliance (Negative no. M 194; FOI 132). The catalogue card states that it was received as a gift, in 1924, from the Ny Carlsberg Foundation, and that it comes from Orvieto. Riis (1941: 161) believed this because he thought it was one of “a number of gold objects—said to have come from the same grave as [a] censer” and other items. The censer is listed by Riis (1941: 79, pl. 15.3) as item B4, which came to the museum from the same source as several Attic vases, “the amphora of the Affector, the Oltus cup, the hydria of the Berlin painter,” etc. Riis believed that these items formed a mixed group, probably from different tombs rather than a single context, at Orvieto. Thus Riis clarifies Frederik Poulsen’s (1927: 47) statement that the Copenhagen dental appliance came from a grave (no site named) that included these famous vases plus five bronze vessels; the last batch of materials includes “*Einige Goldsachen, die sich jetzt im Nationalmuseum von Kopenhagen befinden*,” “some gold objects,” which must include the gold-band prosthesis. (The Affector painted between c.550 and 530 BCE; Oltos worked c.525–500 BCE, and the Berlin Painter from 500–460 BCE, leaving a wide chronological range—if we accept that these Attic (Athenian) vases were deposited with the person wearing the gold bridge.)

Description

Three gold rings held a central, false tooth, and two living “anchor” teeth. The surviving central incisor crown had been complete, but was damaged while on exhibit and the fragment lost before this study began (Figure 5.4a shows the appliance prior to damage.). The lateral incisor crown is largely intact. No portion of the false or replacement tooth was recovered.

Fashioned to be worn in the upper jaw of an adult female, this prosthesis was made using a complex variation of the simple band technique, welding a series of individual “rings” or small loops, each of which is fitted to a single tooth. This is only one of at least four construction variations now known (see Van Marter, no. 1).

The size of the teeth can be estimated from the empty gold rings, and must have been made for a woman of distinctly small stature.

Construction

To make this appliance, three separate straps or bands were fashioned and welded into three small loops. Then these three rings were welded into a series. Alternatively, each seamless ring may have been produced by expanding a small piece of gold by rolling and hammering. The left loop was fitted to the upper left central incisor (I1), the loop on the other end fitted to the right lateral incisor (I2). These sound teeth served as the anchor or “post” teeth to which the bridge was attached to hold it in place. The rectangular central loop held a false tooth

(now missing). No rivet is needed in this design since the false tooth was held by the same method by which a gemstone is fixed in its setting. A circular band was fitted with the false tooth in the same fashion that a goldsmith would make any bezel setting. The rectangular setting would prevent rotation and facilitate a good fit, then the false tooth would be secured in place by pressing the gold tightly, as with the gold of the lateral loops.

Each loop is seamless, and all are joined at their adjacent surfaces by invisible welds made after each had been shaped. Each loop was custom designed (form fitted) to encircle one dental element. The lateral loops were curved in such a way as to conform to the base of each anchor tooth's crown, with specific fitting done after the false tooth had been set in place and the appliance was ready to be inserted.

The replacement tooth in the Copenhagen appliance would have reproduced only the "crown," its "base" cut into a rectangular shape, which would remain stable in its collar and unlikely to slip. This series then forms the bridge, of a design remarkably similar to modern examples. The lateral bands were fitted to the living teeth by simply pressing the soft gold band securely around them. The construction of this appliance indicates that it was meant to be a permanent fixture. (The roots of the anchor teeth have dissolved, probably during burial.)

The widths of the three gold bands vary significantly, the variation exaggerated by the bending and fitting needed to adjust each band to a specific tooth, either natural or false. This resulted in the appliance being much more effective than those bridges that incorporate only a single long band. Viewing this unit from the rear, the mesial divider (the midline of the skull) would pass between the left and center dental elements. The left element, surrounding the central incisor, is a band worked to a thickness of about 0.3 to 0.4 mm. The thickness of the central element is closer to 0.3 mm, and that of the lateral incisor is under 0.3 mm, possibly 0.2 mm. These variations indicate a gradual thinning of these bands, perhaps because a single long strip was the source of all three units and this source was thicker at one end than the other. This might only reflect random variation, slight bends or disturbances that occurred after death, or attenuation that occurred during the fitting process.

The breadth of the gold bands also shows minor variations. The center of the anterior surface of the left unit measures 5.8 mm high along the entire front (labial) exposure. An encrustation of modern glue obscures the rear surface. The height of the band forming the central unit is 6.1 mm along the lingual surface, continuing to the front (labial) surface on the medial side. This strip broadens toward the midline where it is some 6.5 mm high. The right unit is formed from a band slightly less than 6.0 mm high at the lingual-distal corner but widens to 6.1 mm at the labial-distal corner; it continues to broaden across the front of the tooth to 6.3 mm at the labial-mesial corner. Variations may be due to changes effected in the gold band during close fitting rather than from variations in the original gold strip(s).

The left unit is relatively oval in plan at the lower margin, but nearly square at the top where it surrounded the widest aspect of the tooth. The central unit is nearly a perfect rectangle in plan at top and bottom, with the lower aspect slightly smaller than the upper. The right unit is a very irregular square at the upper side and an irregular oval at the lower edge.

The probability is that the gum would recede away from contact with the replacement “tooth” and that a new element would have to be fitted from time to time. This could be done with relative ease, or at least as easily as the removal of the entire device from the adjacent teeth.

Human remains with the appliance: dentition

Marvitz (1982: 49) illustrated the crowns of the anchor teeth when they still appeared to be intact within the lateral loops of the bridge. Both roots and most of the dentine, however, probably had disintegrated in the tomb. When examined in 1989 the tooth on the left (I1) in the Marvitz photograph was badly fractured, and what appears to be translucent glue covered much of the rear of these teeth as well as the gold band. The damage and glueing may have resulted from its display in an exhibition held about the time that Marvitz (1982) published. Much of the neck area of the I1 is preserved within the gold band; the exposed area is now largely missing except for the labial-mesial aspect of the enamel.

Both anchor teeth are worn down into the dentine along their articular or cutting edges, but there is no trace of secondary enamel replacement in the dentine area. The lateral tooth (the I2, which shows very little recent damage) has some tiny chips along this surface reflecting continued biting of hard materials or coarse elements. The entire I2 crown is intact; a trace of shoveling² (1 on a 0–4 scale) is evident. A major question regarding the permanence of this appliance involves the probability of its trapping decay-causing particles. The adverse long-term effects, and the probability that the gold bands would be distorted and loosened in days if not months, leads us to wonder if appliances of this design were intended to be worn continuously or if they were inserted only on special occasions.

Maximum diameters of these teeth cannot be measured directly but may be estimated from the inner dimensions of the enclosing gold-work. (Approximate crown heights: right lateral 10.9 mm, left central 11.5 mm.) Shoveling appears along the left lateral margin of the lateral incisor, and may exist as a trace along the damaged medial margin. The extent of the damage to the labial surface of the central incisor is such that no observations regarding shoveling can be made. The relatively small size of these teeth and the gap bridged by this appliance implies that the wearer was female.

Age determination by degree of wear requires a comparative population, and is less effectively applied to anterior teeth. The individual represented by these teeth appears to be over 30 years of age, but probably under 50 years of age.

5 Poggio Gaiella

Present location: Museo Archeologico Etrusco, Florence (inv. no. 11782).

Type: single complex “band,” now in two pieces, spanning eight tooth spaces.

Origin: Poggio Gaiella in Città della Pieve (7 km south of Chiusi, 50 km south-west of Perugia).

Jaw part: originally probably maxillary, 1PM–PM1. This appliance was not original to the mandible in which it was displayed where it surrounds the mandibular 1PM–PM2 (nine tooth spaces) of a jaw from which the 2I was lost, possibly during recovery.

Date: uncertain (see Rastrelli 1998). Menconi and Fornaciari (1985: 93, 94) offer a date of the fourth–third century BCE, but without supporting archaeological evidence.

Sex: female, age 25–40 years.

Dimensions: L. 42; W. 2.82 to 3.3 mm.

Description: This elaborate appliance was crafted to surround eight teeth, from around both first premolars and all teeth in between. Two strips of gold were used, varying in width from 2.82 to 3.29 mm, with a median width of 3.16 mm. The thickness of the strips ranges from 0.12 to 0.17 mm, with a median of 0.14 mm.

Two separate strips of gold are shown in Figure 5.5a. The sequence by which the form of the original piece was reconstructed is shown in two illustrations (Figures 5.5b4) providing two interpretations of its construction. One band some 90 mm long passes along the labial side of the central six teeth, then curves into complete circles at each end to form loops surrounding the premolars. The buccal aspect is 42 mm long, welded at either end to the loops formed from the labial element. This could have been done at the same time that the lateral loops were welded. A center weld stabilized the appliance between the central incisors.

After recovery this appliance was much manipulated in order to mount it into a skull for display, resulting in some separations of welds and minor distortions. The alveolar margins of the skull, believed to be a relatively modern young adult, reveal no areas that would require bracing from a dental appliance. This may be another case of the modern mounting of appliances in more photogenic skulls.

References

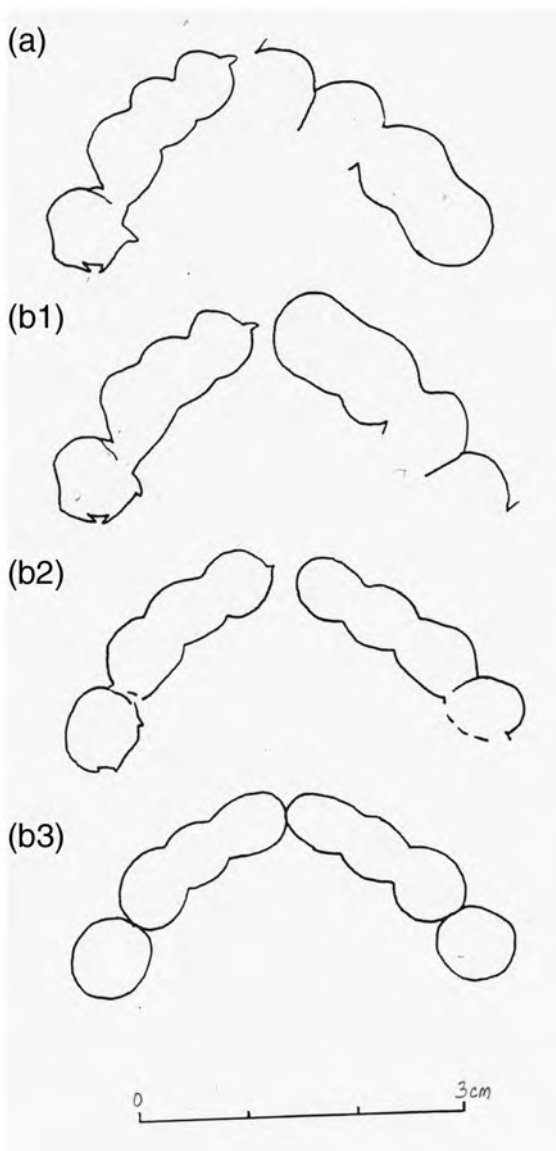
Baggieri 1998: 67 (the popular basis for his 1999 essay)

Baggieri 1999 (an error-filled essay)

Baggieri 2001: 323, pl. xxxviii (repeats his 1999 essay)

Becker 1995a, 1996, 1999b: 44, fig. 1.

Bliquez 1996: 2646–2648, “D,” also “E,” and possibly “F” (see Bliquez 1996: 2647, n. 5, also figs 5–9.)



Figures 5.5a–b3 Views of the Poggio Gaiella appliance: (a) schematic drawing of the superior view of the configuration of Poggio Gaiella as it now rests in the display mandible; (b1) Becker's reconstruction of the Poggio Gaiella appliance in its correct orientation, (right half inverted and reversed); (b2) Poggio Gaiella, damage in course of "repair"; (b3) Poggio Gaiella, as reconstructed by Becker.

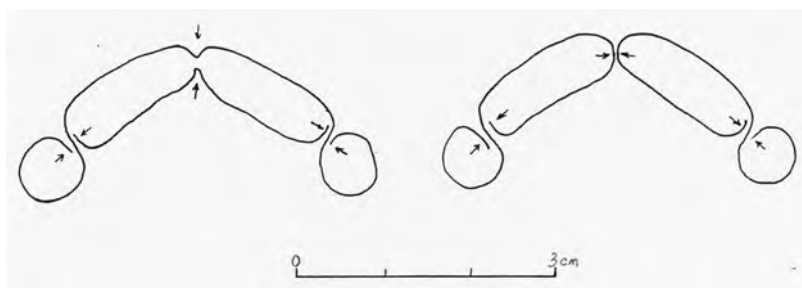


Figure 5.5b4 Variant possibilities of fabricating Poggio Gaiella, if not made using four-ring technique.



Figures 5.5c–e Photographic views of the Poggio Gaiella appliance, as displayed in a modern skull.

Source: courtesy Soprintendenza Archeologica per la Toscana and Lawrence J. Bliquez. (Florence Museo Archeologico neg. nos. 23530, 23528, and L. J. Bliquez.)

Bobbio 1958: fig. 14 (after Casoti 1957).

Capasso 1986: 52–55 (with 3 figures).

Cassoti 1927: 627 (illustrates copies only); 1947: 671–672; 1957:105, fig. 5; and 1958.

Corruccini and Pacciani 1989: 61–62, figs 1, 2 (text derives from photographs, not direct examination).

Dunn 1894: 4 (fig.).

Emptoz 1987: 557, 558, fig. 20 (no. XXIV).

Frassetto 1906: 156.

Ghinst 1930: 407 (after Dunn 1894).

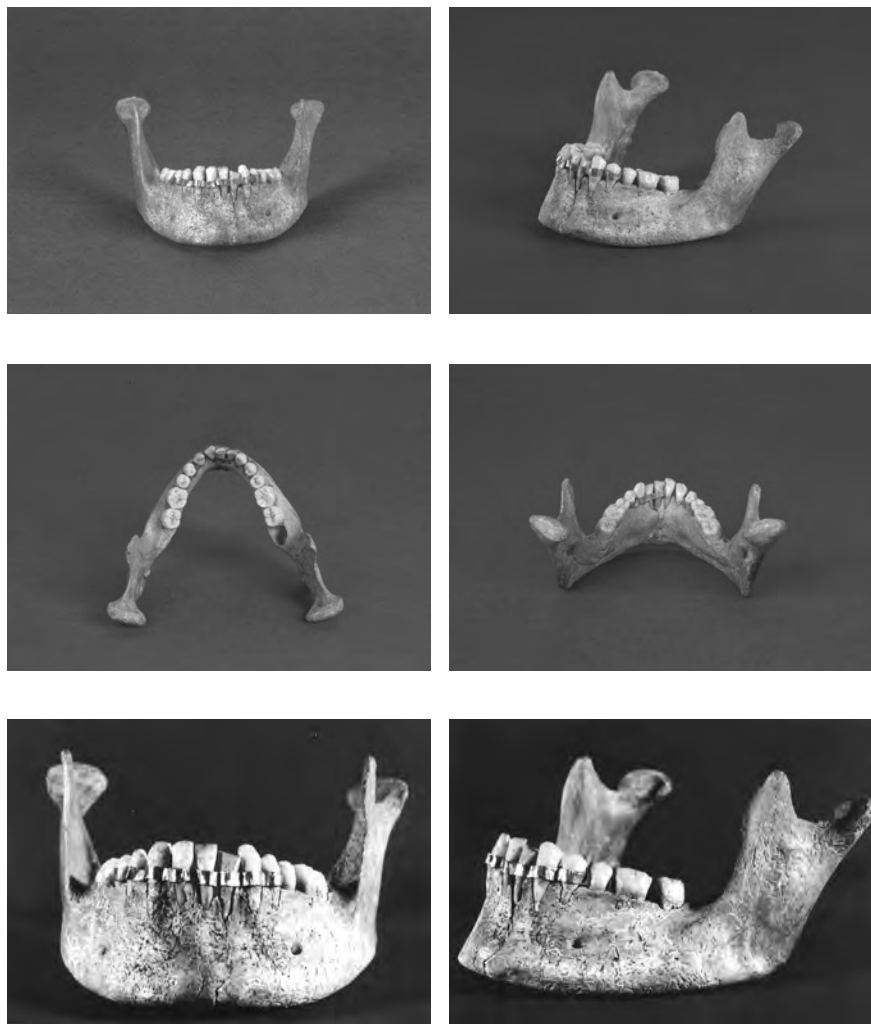
Hoffmann-Axthelm 1985: 78, fig. 65 (not in 1981 edition).

Laviosa *et al.* 1993: 131, figs V4 and V5. The two color plates from the Soprintendenza Archeologica della Toscana (Firenze) are the best now available.

Menconi and Fornaciari 1985: 93: figs 3, 4. (Location called “Chiusi,” p. 94.).

Naso 2011: 150, no. 3.

Penso 1984: figs 143–145?



Figures 5.5f–k Views of mandible of Poggio Gaiella appliance.

Source: courtesy Soprintendenza Archeologica per la Toscana and Lawrence J. Bliquez. (Florence Museo Archeologico neg. nos. 23523, 23524, 23526, 23525, and L. J. Bliquez.)

Platschick 1904–1905: 239, fig. 2 (from “Giojella”).

Sterpellone and Elsheikh 1990: 140, fig. 68.

Torino *et al.* 1996: 120–121.

Torino *et al.* 1997: 536–538.

History and provenance

The ancient necropolis at Poggio Gaiella, a hill south of ancient Chiusi (Etruscan *Clevsin*, Latin *Clusium*, *Camars*), was the city's largest, with tombs of the Orientalizing and Archaic periods. Unfortunately the exact context from which this appliance was recovered remains unknown. Anna Rastrelli (1998) presents an impressive summation of the history of excavations in Chiusine necropoleis.

A report on the excavations at Poggio Gaiella appeared only a year after the last of the Casuccini "excavations" at nearby Città della Pieve (Braun 1841). The earliest known account of the Poggio Gaiella appliance was written by its first known owner, then resident in Florence (Dunn 1894: 4). Platschick (1904–1905) says that Dunn's skull and mandible were found "*a Giojella, presso* ["near"] *Chiusi*." Dunn donated this appliance and the skull on which it had been mounted to the museum in Florence (Corruccini and Pacciani 1989: 61); the exact date of this transfer remains unknown. The Poggio Gaiella appliance and associated skull were in Florence in 1957, where Casotti (1957: 105, fig. 5) saw them in Sala XLIII of the Archaeological Museum.

The breakage into two pieces may have been a result of trying to fit it around the teeth that are present in the mandible. Frassetto (1906: 156) studied 15 Etruscan skulls, 2 of which came from Poggio Gaiella. His data may enable us to tell if the skull that now has this dental appliance in it is one of these 15. In January of 1994 both the Poggio Gaiella and the Valsiarosa appliances were on display at an exhibition developed by Professor Luigi Capasso (Capasso 1993). Subsequently the Poggio Gaiella appliance and the skull were returned to the storage area of the Museo Archeologico Etrusco in Firenze, where Becker made this study in March of 1994.

Description: the skull and mandible

This is the skull of a female (cf. Becker 1993a), but not the person for whom this appliance was made. Only a few measurements (in mm) were taken to confirm this sex evaluation. Mandibular molar dimensions also provide evidence: see Becker 2014; cf. Kieser 1990. Dental wear is minimal, but complete closure of the basilar suture confirms the age at more than 21 years (cf. Baggieri 1998: 67). Externally the cranial sutures appear to indicate an even greater age, perhaps about 40 years (cf. Corruccini and Pacciani 1989). Studies of the Etruscan population at Tarquinia (Becker 1990, 1993a) suggest that external closure of the cranial sutures commonly occurs at a relatively young age (see also Becker and Salvadei 1992). The minimal evidence of periodontal disease plus the configuration of cranial sutures indicates that she was about 25 years of age, if not older, at her death. An extensive review of the dentition confirms the estimated age at death (Becker 1995a).

The condition of the mandibular dentition yields evidence that the appliance was not made for this mandible. This mandible now has only three incisors in place. The modern restoration of the jaw failed to leave an empty socket for the 2I, which was lost post-mortem, (see also Sterpellone and Elsheikh 1990: 126).³ A central incisor (1I) now “fills” both left incisor spaces, but this 1I appears discolored or darker than the surrounding teeth and may not be original to this mandible. The shifting of the mandibular teeth during the conservation process has produced a closing of the gap left by the “missing” tooth, causing the opening of spaces between the other teeth.

The Poggio Gaiella appliance spanned a maxillary 1PM to PM1 in the person for whom it was designed. After recovery from a tomb it had been placed in a mandible for which it was not made. Dunn, implying that this “appliance” was found *in situ* around eight teeth, indicates that “*i bicuspidati*” were included within the appliance. Dunn’s reference cannot be interpreted as indicating that all 4 premolars were bound (10 teeth), but may have meant that only the premolars on one side of the jaw were held in the appliance. The two pieces or parts of the band are now placed so that the 2PM of this mandible is not encircled and since an incisor is missing, probably only recently, the appliance now extends to include the PM2. Dunn’s observations suggest that the appliance had been set into this mandible before he acquired it. The “excavator” may have rudely set this appliance into a skull obtained (from a modern context? see Becker 1993a) specifically as a showpiece.

Becker’s examination of the configuration and sizes of the extremely distorted pieces of this appliance suggests that it originally extended from a maxillary left first premolar across the front of the mouth to the right first premolar. This interpretation would include spanning two premolars. Thus Dunn may have evaluated this appliance correctly, with the problems of the present placement arising only after he had donated the Poggio Gaiella appliance to the museum. As to Dunn’s belief that this prosthesis was worn in a mandible, we suggest the following scenario. The original skull did not survive and Dunn made his evaluation either on the basis of the configuration of the appliance or perhaps the location in a mandible as he first saw it before it was purchased. The distortions now evident in the appliance (extreme bending and folding, plus breaks), and the damage that renders an easy evaluation impossible, suggests that the appliance was forced into the mandible of an intact skull derived from an unknown source, and after some teeth were lost.

Clearly, at some time before Poggio Gaiella came into Dunn’s possession a great deal of damage had been done to this delicate gold appliance. In addition to the bending and springing of the long sections of the band, the band appears to have been torn, and several welds have separated. Among the breaks may be one at a point where these two parts had been joined. All the evidence indicates that this appliance had not been associated with a mandible articulated with a complete skull at the time of discovery.

Description of the appliance

Detailed examination of the two pieces of this dental appliance suggests that the two elements were joined as shown in the illustration. Both pieces of the appliance are so ill-fitted to the teeth of this display mandible and so badly bent that there is no possibility that they match the teeth of this restored mandible. More significantly, there has been no ante-mortem loss of teeth, nor any evidence of loosening of the teeth, which might require the use of a dental appliance, nor evidence for a blow, which might have had a temporary effect of loosening the teeth.

Beginning at the right end of the right unit (cf. Platschick 1904–1905: fig. 2) is a sprung part of the appliance that was located behind the premolars of the display mandible, then pressed between the PM2 and M1. The unit is broken off after a sharp bend. A small “tail” under 2 mm long does not appear to have been connected with the loose end (width 2.82 mm), which extends back from the labial aspect of the right unit. This entire unit appears reversed, with the aspect now shown as mesial originally around a PM with the “tail” passing around the maxillary PM1. The portion of the band passing in front of this PM is continuous with the “divider,” which would have been the piece of a small band that passed between the canine and the PM1. The most distal aspect of the right unit, as now seen, was welded onto the larger band of the left unit. Visual observation suggests that this “divider” was part of the PM1 ring that had been welded to the larger band, as is the case with the left unit.

At its closest point to the left unit this part of the right unit has a very sharp bend ending in another “tail.” There is no indication where this may have attached. These two “tails” plus the divider seemingly welded to the front all may have connected to a piece of this unit now missing. It appears that three simple welds held four separate bands together.

Beginning at the most distal part of the left unit we find an extremely bent and folded “ring” now surrounding the 1PM, probably originally fitted to a canine. As we now see it this “ring” has a very peculiar double internal fold at the distal-lingual area, an equally peculiar “bump” at the distal-mesial portion, and a mesial part clearly formed by a very thin (0.13 mm) band, which had been welded in place. The weld on the labial aspect still holds, but the distal weld has separated. Microscopic examination might determine if this weld had once attached at the distal end of the “bump.” The remainder of the left unit appears to be a simple ring (band) that probably spanned three teeth, the left canine to the left central incisor.

The Poggio Gaiella appliance presents an interesting variation of a simple four-ring prosthesis. Four rings would also require only three simple welds to fasten the units together in a series. One of the many reasons for doubting that this appliance was a single, long braced band is the fact that if it were one appliance it would be unusually long and be unique in being made in two parts. Braced bands loop over sound “anchor” teeth at either end in order to support loose teeth that

are between the sound teeth. An effective means of securing the entire band would be to place braces (cross pieces) at either end to hold the band more firmly to the “anchor” teeth. With an unusually long appliance such as this, braces at either end would be essential. Stability could have been achieved by making two units, each carefully fitted with a braced area that would loop around a sound tooth (PM1’s in this case) at the distal ends, and welding these two units together at the center. The center weld would reduce unwanted flexibility created by a long band, and would increase stability. We can “construct” a normal prosthesis if we reorient the various parts seen as they were mounted for display. Assuming that the mesial area of the right unit of Poggio Gaiella had been the distal portion, and if it originally had looped around a sound right first premolar (PM1) as an anchor tooth, the result would be a reasonably ornamental and probably functional appliance.

Two possible variant techniques of manufacture, using long bands, are possible, but unlikely. Although only three welds might be needed in either case, two of these welds would be “complex”; that is, they would have to weld two different sets of units together at the same time. Close fitting is much more difficult when long bands are used. Given all these possible variations, the Poggio Gaiella appliance is likely a variation of a simple four-ring prosthesis.

Ghinst (1930: 407) believed that the Poggio Gaiella appliance was intended to support teeth loosened by periodontal disease. This is a reasonable assumption, but Ghinst says that he based his conclusion on alveolar resorption evident in the mandible (also Emptoz 1987). Either the Poggio Gaiella appliance was in a different mandible prior to 1930, or Ghinst seriously erred in his evaluation of the health of the alveolar margin of this skull. Quite possibly the person for whom the Poggio Gaiella appliance was made did have periodontal difficulties, but there remains the basic problem of whether this appliance was actually found in this skull. Unfortunately, “*Les photos que nous présentons ici illustrent nos hypothèses*”—do not appear in Ghinst’s publication. His statement may refer only to photographs that he had shown as illustrations at the congress where his paper was read. Unfortunately these are not known to have survived. Capasso’s (1986: 53) illustrations (reversed?) indicate that the appliance has recently been in two pieces, as does a reverse print shown by Corruccini and Pacciani (1989: fig. 2).

Measurements

Detailed measurements may be of some aid in determining if these two elements are from a single dental appliance. Dr. Jacopo Moggi-Cecchi (Università degli Studi, Firenze) provided the use of an extremely accurate, illuminated sliding dial caliper (Mitutoyo Digimatic 500–110 electric). The two pieces of this appliance, as they are mounted in this mandible from PM1 to PM2, have a maximum “length” of 41.76 mm. If we draw a straight line as a tangent to the curves of the two most distal parts and then a line perpendicular from the most anterior aspect, that

perpendicular line measures 20.41 mm long. The two elements may be considered individually as the right and left, with reference to their present position as displayed in the unrelated mandible.

The widths of the “right” band and its parts vary from 2.82 mm to 3.23 mm. The thickness of this ribbon of gold varies from 0.12 to 0.15 mm. The “left” unit varies between 3.02 and 3.33 mm wide, while the thickness of the gold strip varies from 0.12 to 0.17 mm. The more narrow aspects of the right unit are under 3 mm wide and are found at the mesial part of the appliance. The more narrow aspects of the left unit, while wider than the right, also appear toward the mesial aspect. Poggio Gaiella may have been constructed in a similar fashion to the GHENT appliance (no. 7), as a single complex band, somewhat more elaborate and longer than the GHENT appliance. These detailed measurements indicate that Ghinst (1930) was the most accurate in his appraisal of the width of this appliance (see also Casotti 1957). Less specific estimates are provided by Menconi and Fornaciari (1985: 94 = 0.20 to 0.25 mm) and by Tabanelli (1962: 92 and pl. 49). Emptoz (1987) offers only the vague estimate of 4.5 mm.

The very length of this band suggests that it was not purely decorative, implying that the anterior teeth of the person for whom it was made had actually been loosened by periodontal disease, or a blow. Periodontal disease would be expected in a relatively mature person of at least 30 to 40 years of age. Also, the uniform width of the band reinforces the conclusion that this was a functional prosthesis. It also might have been designed to support teeth that had been loosened by a blow, in which case a mandibular placement also would be possible. If true, this would be the best evidence from Etruria to suggest that dental appliances were used as a therapy for teeth loosened by accidental trauma as well as for cosmetic purposes.

Casotti (1947: 672, 1957: 105) offered a complex, but fanciful, description of how this appliance was held in place, concluding that the two pieces had once been “wired” together and otherwise manipulated to serve as an orthodontic corrective device to close “gaps” between the teeth (also Capasso 1986: 54). Not only is this a fanciful observation, but the fictitious goals of such a device would be counterproductive. Moving the teeth only would have disrupted their dental alignment and loosened them all. Casotti (1957) later noted which teeth were present when he examined the supposed Poggio Gaiella skull. He did not recognize the post-mortem loss of a mandibular incisor, leading him to speculate that the appliance was meant to close a “gap” supposedly created by the loss of the maxillary I2 (also Pacciani and Corruccini 1986: 55, fig.). However, no tension can be provided by any of these ancient appliances, and therefore there is no possibility that any prostheses could have altered the position of teeth (cf. Torino *et al.* 1996). Most devices served only inadvertently to maintain spaces between natural teeth.

This appliance is commonly confused with the more simply constructed band that had been in the Florence collections (Cat. no. 84467 and listed below as Populonia). Waarsenburg (1991a) points out that Ghinst’s (1930) description of the two “pieces” in Florence actually describes the very same example (see also Penso 1984: figs 143–145).

**6 Populonia: buried at Populonia (missing:
lost in Florence Flood 1966)**



Figures 5.6a–c Views of Populonia appliance, set in a human mandible.

Source: courtesy Soprintendenza Archeologica per la Toscana (Florence Museo Archeologico neg. nos. 20055, 20056.20057.)

Present location: Museo Archeologico Etrusco, Florence, inventory no. 84467.

Said to have been lost in the 1966 flood.

Type: four-“ring” series, with one rivet (at I1?), four spaces. Cf. no. 15, Valsiarosa.

Origin: Populonia, San Cerbone necropolis, part of “*membra disiecta*” from damaged tombs.

Jaw part: probably maxillary 2I–I2.

Date: fourth century BCE.

Sex: probably female.

Dimensions: approx. 25–30 mm long.

Description: generally described as a four-ring appliance, with a rivet through one of the (central?) rings. Original reports (Guerini 1894a, b) are insufficiently clear to distinguish this item from the Valsiarosa appliance (see no. 15, below), with which it has been often confused. Bliquez (1996) first revealed this and other problems created by authors writing without any primary study of the appliance (see also Waarsenburg 1991a).

References

Bliquez 1996: 2648, 2652, “F” and probably “O,” figs 10–11.

Capasso 1985: 53. Capasso conflates the Populonia example with the Poggio Gaiella appliance (no. 5).

Casotti 1935: fig. 1 (reproduces Guerini 1894a: fig. 7).

Casotti 1947: 676 (668, 674–675: 3 cm long and 0.6 cm wide).

Emptoz 1987: no. XI?

Ghinst 1930: 406–407.

Guerini 1894a: 393, figs 1 and 2 (these illustrate the Valsiarosa example).

Guerini 1894b: figs 1 and 2 (show the Valsiarosa example).

Guerini 1909 (once again depicts the Valsiarosa example).

Hoffmann-Axthelm 1985: 78–79, Abb. 65.

Lufkin 1948, fig. 17 (photograph of a poor copy?).

Minto 1943, 183; pl. 48 fig. 1 (photo of mandible, but no appliance is visible).

Naso 2011: 151, no. 7.

Platschick 1904–1905: clearly confuses Populonia and Valsiarosa.

Tabanelli 1962: 92, 94–96, pl. 48, 52.

Torino *et al.* 1996: 120–121 (from Casotti 1947; cf. Becker 1994d).

Waarsenburg 1991a: n. 5, no. 1.

Weinberger 1948: 148, fig. 53 (Weinberger’s fig. 41, no. 7 is either a copy, or the Valsiarosa example). Weinberger suggests that this is for mandibular I2–PM2.

History and provenance

The necropoleis of Populonia held numerous rich seventh–sixth-century tombs, many plundered during the nineteenth–twentieth centuries CE. Minto’s brief publication (1943: 183) of items excavated in the vicinity of the Tomb of the Funeral Beds (1908–1909 and 1920–1921) lists several of the more spectacular sporadic finds, including:

quella singulare legatura dei denti, ottenuta mediante una sottile fettuccia d'oro, ritrovata in una mandibola di cranio (pl. XLVIII, 1), proveniente da una tomba ad inumazione distrutta, che richiama ad un esempio consimile riscontrato nella mandibola di un cranio, proveniente da una tomba etrusca di Città della Pieve, con dentiera legata in oro (R. Museo Archeologico di Firenze, Sezione topografica).

this unusual binding of the teeth, achieved by the use of a thin strip of gold, [was] found in a mandible of a skull (pl. 48, 1) that came from a destroyed inhumation tomb, which recalls a similar example, found in the jaw of a skull that came from an Etruscan tomb at Città della Pieve, with a denture bound in gold (Royal Archaeological Museum of Florence, Topographic Section).

Minto was referring to the Poggio Gaiella appliance, also stored in the Museo Nazionale Archeologico di Firenze. This may explain why subsequent authors have confounded the two items. The difficulty of seeing any appliance in the mandible illustrated in his plate 48 further compounds the problem. Tomb of the Funeral Beds at Populonia is an Archaic structure, but the gold jewelry from the same area of sporadic finds in the necropolis includes a number of pieces in fourth-century style, so the date of the appliance could be from the seventh to the fourth century BCE—the latter date was suggested by Ghinst (1930).

Analysis of past accounts

Ghinst says that this appliance is gracile (presumably meaning that it was found or displayed in a gracile mandible), involves 6 mandibular teeth, and that the mandible held 14 teeth. This appears to confuse the appliance with the skull in which it is mounted. Ghinst says that the appliance is held by “*un mince ruban d'or*” and measures “*larg. 2 1/2 mm env épais 1/3 cm,*” passing from tooth to tooth “*sous forme de spire.*” He also says that the alveolus is strongly resorbed, suggesting “*un cas de pyorrhée traumatique*” but that this is difficult to see since there is earth still adhering to the neck of the teeth? (“*au colet*”). In many respects this description sounds like that for Poggio Gaiella. We are not at all certain that Ghinst actually saw either of these appliances, and suspect that his data also are derivative.

Regarding the confusion between Populonia and Valsiarosa, Weinberger's (1948: 125–126, fig. 41: 7) illustration for Valsiarosa (no. 15) looks more like a copy of the Populonia piece. Weinberger cites Deneffe (1899: pl. 3, no. 1), Sudhoff (1926: fig. 45), and Farrar (1888: figs 1 and 2) as his sources. Weinberger's illustration appears to show a copy that may also be illustrated by Lufkin; it resembles an industrial casting rather than any known appliance.

Weinberger's photographs of the Populonia appliance clearly indicate a very common type that bound four maxillary incisors. However, for reasons unknown he states that it is a mandibular prosthesis (I2–PM2), and his illustration and text

(1948: 148, fig. 53, right) incorrectly suggest that the PM1 was replaced by a false tooth, held in place by a rivet. Weinberger's caption for the copy (fig. 41: 7) also says it held a premolar. The chamber pierced by the rivet appears to be square, while the other three are more ring-like (although somewhat distorted by their archaeological past). Three rings were formed, one each for the two left teeth and for the right anchor tooth. The chamber for the false tooth appears to have been formed by connecting the flanking rings with two short strips of gold welded in place.

Later in his book Weinberger (1948: 148, fig. 53) describes the original of what is supposed to be the Populonia example, and includes two "original photographs" said to be in the "author's collection." These appear to show the original appliance in Florence. These may be photographs of the piece which Guerini (1894a: 393) noted during his visit in Rome: "*je trouvai dans celui de Pape Luigi un appareil en or . . .*"

Guerini's poor drawing of a gold appliance of four "rings" with the central pair somewhat quadrangular, indicates that one central ring has a rivet. Although his fig. 1 depicts the central pair of rings as quadrangular (cf. Copenhagen appliance), in fig. 2 they are circular. Guerini's drawings also make these four rings appear to be a graded series, small to large, with the riveted example the second largest.

Guerini's (1894a) "Papa Luigi" museum in Rome is, of course, the "Museum of Pope Julius," the Villa Giulia Museum, devoted to Etruscan antiquities. Guerini (1909: 70) offers a correction to his earlier (1894a) identification.

Waarsenberg (1990) suggests that 4 or 5 specimens often said to be in Florence are "probably only one specimen." Many of the phantom pieces have been created from the photographs of the (apparently) two different inventoried examples once identified at the Museo Nazionale Archeologico in Florence. In addition, *copies* of at least one piece may be creating some of the confusion, as Waarsenburg explains. Other specious "specimens" of dental appliances have emerged in Florence (see Appendix III, no. 6, below) as the result of misidentification of copper-stained human teeth as "gold crowns."

7 Ghent (lost)

Present location: unknown: lost during German occupation of Ghent in 1940s.

Type: complex band forming three "loops," the central of which is double size (and surrounded two teeth). Spans four dental spaces; no rivets.

Origin: Orvieto (ancient *Volsinii*).

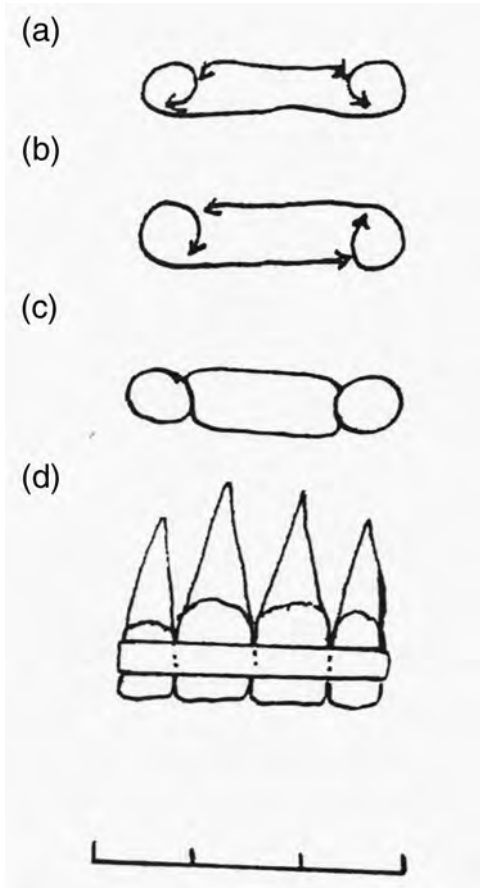
Jaw part: maxilla, probably 2I–I2, rather than the commonly believed 1I–C (now attached to a suspect or added canine).

Date: sixth century BCE or later (estimated).

Sex: female?

Dimensions: L. 27, W. 2.8 mm.

Description: This appliance surrounded four teeth, almost certainly the maxillary central incisors. It may have been used as an ornament, or perhaps to stabilize



Figures 5.7a–d Drawings of the Ghent appliance (said to be from Orvieto, now lost), by Becker: (a) and (b) construction technique; (c) plan of the appliance, to scale; (d) labial surface of the appliance showing how it would brace the four incisors.

one or more loose teeth. The width of the band is unusually uniform. An unusual manufacturing technique using two separate strips of gold, a pattern similar only to the longer Poggio Gaiella appliance (no. 5). Ghent, like Poggio Gaiella, may have been constructed in one of two ways. A strip approx. 39 mm long could have been rolled at both ends and tacked to form the lateral loops, with a second strip only 17 mm long then welded to form the back portion of the appliance. Alternately, two identical strips, about 28 mm in length, could each have been rolled at one end, then that “loop” welded to form a “9” (each loop to surround one anchor tooth) before having the long end of each of the pair welded to the loop of the other. This final welding created an elongate opening to surround the two central teeth.

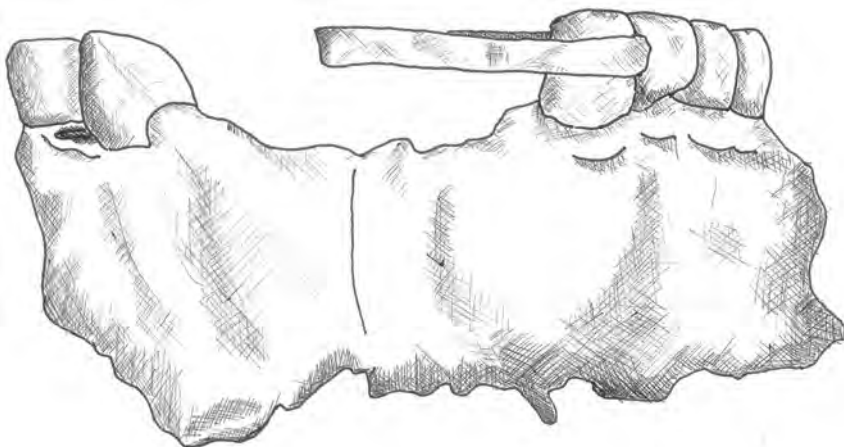


Figure 5.7e Appliance from Orvieto, now lost, side view, as illustrated by Guerini 1909.

Source: drawing by Tim Downs (timdownslocations@verizon.net).



Figure 5.7f Appliance from Orvieto, now lost, view from below, as illustrated by Guerini 1909.

Source: drawing by Tim Downs (timdownslocations@verizon.net).

References

- Bliquez 1996: 2646, “C,” figs 3–4.
Boissier 1927: 29, fig. 5.
Casotti 1947: 673–674, fig. 3 (from Guerini 1909: fig. 21).
Cavenago 1933: fig. 1.
Dechaume and Huard 1988: fig. 4B3, lower. Photograph may show a copy (see Appendix II.D).
Deneffe 1899: 59, 63–67, fig. 1 (probably the original appliance).
Emptoz 1987: 557, fig. 19 (no. XXII).
Guerini 1909: 74–75, figs 21, 22.
Lufkin 1948: Fig. 21.
Sudhoff 1926: 84–85, figs 48–49.
Tabanelli 1962: 93 (not at all useful).
Waarsenberg 1991a: n. 5, no. 12.
Weinberger 1948: 123–125, fig. 41: 1.

History and provenance

Emptoz (1987), in a review of the early literature pertaining to the Ghent piece, concludes that this appliance together with an associated maxilla come from Orvieto. Boissier (1927: 29) clearly errs when he says that the Ghent appliance came “from Tarquinia, near Orvieto.” Deneffe’s (1899: 59, 63) estimate of date, 600–500 BC, is based on the dates of the famous stone masonry tombs and their original depositions at the Crocifisso del Tufo necropolis of Orvieto (also Sudhoff 1926: 84).

Soon after its discovery this appliance was purchased by Ghent University, Belgium, and by 1899 it was in the Ghent University Museum. It *may* be the only one recovered with a portion of the original associated skull. Skull and appliance were together in 1899, although a more photogenic skull easily could have been chosen for display, as happened with the Poggio Gaiella and Valsiarosa appliances.

Description

Deneffe (1899) offers the most complete description of the Ghent appliance, derived from the original, rather than a copy as appears to be the case with his other descriptions. It disappeared during the German occupation of Ghent in the 1940s.

Much of the maxilla in which the Ghent appliance was seated had been lost by the time of Deneffe’s photograph (1899: pl. 1). Assuming that this facial fragment is actually part of the skull of the owner of this appliance, much can be determined from this photo. Possibly all 16 teeth of this maxilla were in place at the time of death, but the 3M, 1PM, 2I–I2, M2 and M3 appear missing in the photograph, probably lost *post mortem*. If this is a retention band for loose teeth, as we believe, there need not have been any loss of teeth before death.

The Ghent band appeared attached to the right canine as a post, and would then have attached to the left central incisor in order to support loose incisors.

However, its configuration suggests that it actually spanned the four incisors (2I–I2) rather than following an asymmetrical pattern. The incisors commonly are lost post mortem, and the Ghent appliance must have been placed on this canine to hold it in place for display (see Cavenago 1933: fig. 1).

Deneffe's (1899: 64) description is both complex and contradictory. First he says that this appliance is a single oval, with three rings each built from two lengths of band attached at each end to post teeth. He gives the total length of the appliance as 26 mm, noting that the *central* band is 1.5 cm long and 0.6 mm wide (thick). This indicates that the central opening, formed by the long arms of the ribbons, is 26 mm long, or just the width expected of two broad, female central incisors, with the smaller openings at the ends conforming to expectation for the size of lateral incisors. Deneffe describes the breadth of the gold band as 3 mm, but where it encircles the post teeth, at the ends, it is 3 mm broad. However, Deneffe notes that the appliance is 5 mm broad in the rear, and notes this variation of 2 mm (1899: 67). He later states that the band and the "partitions" are a bit more than 0.5 mm thick.

Sudhoff (1926) estimates the width of the band at 4 mm, a figure copied by Emptoz, who also incorrectly states that it has rivets. Weinberger (1948) describes the gold along the mesial surfaces of the anchor teeth as "partitions." Only discovery of the actual prosthesis could confirm this evaluation (see also "The Ghent copies" in Appendix II).

The Tarquinia series (nos. 8–12): history and provenance

In the village of Corneto, today recognized as ancient Tarquinia (Etruscan *Tarchna*, Roman *Tarquinius*), during the end of the nineteenth century, two separate collections of archaeological materials included Etruscan dental appliances. One belonged to Count Bruschi and included three Etruscan dental appliances. Dasti (1878: 386) indicates that the private "Museo Etrusco nel Palazzo Bruschi Falgari," was open to anyone requesting admission. This rich collection was assembled by the Contessa Giustina Quaglia Bruschi Falgari, one of the important women antiquarians/archaeologists of the nineteenth century. Dasti (1878: 386) says "*sia smalti oggetti in oro, de' quali molti eseguiti artisticamente con rara perfezione*" ("there are enameled [sc. shiny] objects in gold, of which many were artistically executed with rare perfection").

Two other dental appliances found their way into the Corneto Museo Civico, which opened in 1878. Excavation of tombs at Tarquinia was at its height throughout the later nineteenth century into the twentieth (see Leighton 2004: 10–24). The number of dental appliances in Corneto is reported differently by early authors, depending on when they saw these growing collections, which were merged into the new national archaeological museum (today Museo Archeologico Nazionale Tarquiniense). In the move, some pieces either were not actually transferred or subsequently disappeared. The quality of early descriptions and illustrations is uniformly poor, which prevents us from determining with certainty which pieces came into the two original collections, and in what years. We also cannot

determine when some items were last seen. The general confusion even renders difficult an exact count of these examples.⁴

In 1992 only one appliance was on display in the Museo Archeologico Nazionale Tarquiniese (Bruschi I); another was in storage (Bruschi III); these two are described from Becker's observations. Among these five examples, the similar Bruschi III and Corneto I are frequently confused, and they continue to be difficult to distinguish in most reports. Van Marter (1885a) saw Corneto I and II in the Corneto Civic Museum, and says that he received certificates from Dasti attesting to the dates when these tombs were opened. Why Van Marter sought such documents is not known; his papers might furnish a trove of information regarding this matter as well as other aspects of Etruscan archaeology.

An example of the problems created by the early reports is an anonymous note in the February 1885 issue of the *Journal of the British Dental Association* (p. 123), referring to Van Marter's publication of the previous month (1885a) and describing:

An example made of rings of gold [which include] the first bicuspid "and two centrals" said to be carved from animal teeth [probably Corneto II]; a second example which has 2 lower incisors, both said to be natural [Corneto I?].

This second example, said to be Roman, is dated to 400 BCE without explanation. Waite (1885a: 316–317) subsequently, and unfortunately, stated that the two examples in Liverpool are so similar to the pieces (from Corneto) described by Van Marter (1885a), and repeated in the *Journal of the British Dental Association*, that they need not be illustrated. All of this is further confused by Waite's (1885a: 508, 512) relation of his then recent trip to the United States, during which he may have seen the two appliances then owned by Barrett.

Guerini (1894a: 393) also notes that he saw (only?) two appliances in Corneto, presumably the pair then in the Civic Museum. Dunn (1894) describes two appliances in each museum. Deneffe (1899: 77–78) suggests that two pieces were in the Civic Museum at Corneto in 1899, and that three were still in Count Bruschi's possession. This implies that Count Bruschi acquired the third appliance between 1894 and 1899. Deneffe, however, describes only two of the three pieces that he said were in the Bruschi collection (Bruschi I, and a second less clearly identified). The fifth Tarquinia appliance now is believed to have come into Count Bruschi's collection about 1894. Platschick (1904–1905: 239), following Dunn (1894) says that only four pieces were in the two museums in Corneto, three of which were maxillary and one mandibular. Platschick does not list Bruschi I, probably because he did not see any examples, and only repeats what Dunn reported.

Deneffe (1899) implies that Corneto I as well as a second prosthesis from Tarquinia had been among the eight appliances copied (see Appendix II.A) for the Ghent University Museum. Deneffe also thanks Helbig, as the "*Directeur du Musée Comunal de Corneto*," for permission to copy two appliances, and then (1899: 82) thanks Count Bruschi for his permission to copy, for the Ghent

collections, the three prostheses in his possession. In fact, copies of all five of the Tarquinia pieces were made for the Ghent collections (Ghent copies nos. 4–8, see Appendix II.A). Moltesen (1987: 242) documents the fact that Helbig was, for some time, Honorary Inspector of Excavations at Tarquinia. Helbig certainly facilitated the copying of these prostheses.

Waarsenburg (1991a: 243, n. 11) suggests that it was Helbig (1877: 64) who provided a fifth-century BCE date for one of the Tarquinia appliances. We have very little other information useful in establishing specific dates for these pieces.

Today we can account for only two of the five examples that had been in the two earlier collections. In June of 1991, when Becker studied the two examples now in the Museo Archeologico Nazionale Tarquiniense, staff suggested that the three pieces that now cannot be located were lost during the transfer of items to the then-new national museum system collections around 1916.

8 Bruschi I

Present location: displayed in Museo Archeologico Nazionale Tarquiniense, formerly in association with two teeth. Previous collection of Count Bruschi-Falgari, Corneto.

Type: simple, narrow band spanning four spaces (not three spaces as commonly reported).

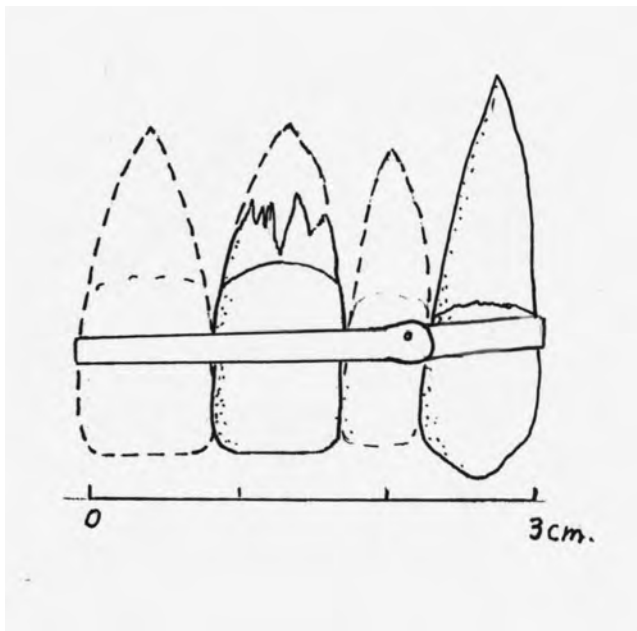


Figure 5.8 Tarquinia appliance: drawing of Bruschi I appliance, by Becker.

Origin: region of Tarquinia (?)—also said to have been found at Cerveteri (Guerini 1909: 74).

Jaw part: maxilla, four spaces. Probably spanned all four incisors, but Guerini and Deneffe took the presence of a left canine (not original to the appliance) to infer its use as in Figure 5.8).

Date: possibly fifth century BCE (see Waarsenburg 1991a: 243, n. 11).

Sex: female.

Dimensions: L. 32; W. 2.0 mm. Gold band: W. 1.9 to 2.2 mm; Th. less than 0.20 to approx. 0.25 mm; total L. of gold ribbon estimated 74 mm. Weight 2.5 g.

Description: simple band, formed from a strip of relatively uniform width, with slight variations from 1.9 to 2.1 mm, and about 74 mm long. The ends of the strip, which is 0.20 thick, were overlapped for a length of about 3 mm and welded. The location of the weld is, unusually, quite evident as indicated by a slight expansion of the width.

References

- Baggieri 1998: 68 (popular basis for his 1999 essay).
Baggieri 1999: 38–39, fig. 7, top pair (numerous errors, details omitted).
Baggieri 2001: 324–325, pl. xxxviii and graph 1.
Bliquez 1996: 2649–2650, 2654, figs 15, 16, 37 #7 (prosthesis “I” and also “S” figs 29–30).
Brothwell and Carr 1962, fig. 3 (illustrate only a copy).
Casotti 1947: pl. 1 (copy of Guerini 1894a: fig. 8).
Cataldi 1993: 125 (mentioned, not illustrated).
Deneffe 1899: 81–82, fig. 7 (from a copy).
Emptoz 1987: 556, fig. 18 (no. XXI). See also p. 557, no. XXIII.
Guerini 1894a: 394, fig. 8 (of a copy?).
Guerini 1894b: fig. 8 (see 1894a).
Guerini 1909: 74, fig. 20 (looks like a copy; see figs 23, 24 for possible original).
Hoffmann-Axthelm 1981: fig. 65 (not in 1985 edition).
Lufkin 1948: fig. 22 (cf. fig 23); data copied from Guerini 1894a, 1894b and 1909.
Menconi and Fornaciari 1985: 94, 97, fig. 10 (perhaps the best published photograph).
Naso 2011: 151, no. 8.
Rath 1958: fig. 3. Illustration secured by Rath from “Foto Istituto Scala, Firenze.”
Sterpellone and Elsheikh 1990: 136, fig. 64 (two teeth in place, but depicted as mandibular).
Sudhoff 1926: 84, fig. 47 (his #8, on p. 85 with figs 51a and b, also appear to be Bruschi 1).
Tabanelli 1962: 93.
Torino *et al.* 1996: 120.
Torino *et al.* 1997: 536.
Waarsenburg 1994.
Weinberger 1948: 125, fig. 41: 6 (from Sudhoff 1926).
Weiss 1989: fig. 29 (mislabelled). His suggestion that it is in the “Civic Museum of Corneto” indicates that his research, showing the appliance as mandibular, was long out of date.

History and archaeology

Guerini (1909: fig. 23) says that the Bruschi I appliance was “found in 1865 in a tomb by Cerveteri [sic]” and that it then belonged to “Castellani’s collection, Rome.” (Much of the Castellani collection did in fact come from the Caeretan region and included gold jewelry.) Possibly the appliance was recovered from Cerveteri, located between Tarquinia and Rome, in 1865 and only later, before or around 1894, sold to Count Bruschi. No number is written on this piece, but the number “160” appears on a small white circular tag on the velvet backing in this display, possibly the velvet in use when this piece was in Count Bruschi’s collection. It was in the collection of Count Bruschi-Falgari when Guerini (1894a: 394) supposedly first saw it, but was not noted by Dunn (1894), who lists only two examples of gold appliances in the Bruschi collection. Casotti’s (1927: 628) note that only two appliances in this collection appears to be copied from Dunn.

The Bruschi I appliance may have been acquired by Count Bruschi about 1894, or after Dunn had seen and published the two examples in the Bruschi collection. Deneffe’s publication (1899) reveals that by 1899 Count Bruschi owned *three* examples, which apparently included this piece. This apparent increase in the number of appliances in Count Bruschi’s collection reinforces the belief that Bruschi I was added to the Count’s inventory *after* Dunn had examined Count Bruschi’s holdings.

Description: the Bruschi I appliance and associated teeth

Becker first examined this appliance in 1991, in a laboratory in the storage area of the Museo Archeologico Nazionale Tarquiniense, in Palazzo Vitelleschi, Tarquinia. At that time the two teeth within this appliance were found to have a slight calcareous coating, not dental calculus but a deposit acquired during the centuries of burial underground. (Such deposits are not uncommon on inside chamber tombs, especially in the region of Tarquinia and Vulci.) The teeth themselves reflected the pattern of deterioration typical for teeth recovered from Etruscan tombs in this area (see Becker 1990). The roots are typically warped and cracked, with considerable damage in a pattern commonly seen on teeth from other “open” tomb contexts (i.e. chamber tombs where air circulates around the bodies). Unlike our findings with other dental appliances with which intact teeth may be associated, those few examples with teeth from which the roots have deteriorated are most likely to have been with the prostheses when they were found. The teeth with Bruschi I appear to be a maxillary left canine and a left central incisor, suggesting that the appliance may have spanned at least the left canine to the left central incisor. This placement is exactly what Guerini (1894a) and Deneffe (1899) both concluded, and suggests that the teeth now with the appliance were the ones seen by these two authors. This does not prove that these are the original teeth, nor that they are in their original positions. The wear on the teeth suggests that the owner was a mature female, perhaps 45–55 years of age, but we cannot be certain that the teeth represent the person who actually wore the appliance.

Examination confirmed that this appliance was worn in the maxilla, but the Bruschi I band probably spanned four teeth rather than three. The canine now in Bruschi I, rotated in its location, has a maximum length of 25.0 mm, with M-D breadth 7.2 mm and B-L width 8.3 mm. The surface is worn, but obscured by calcareous deposits. The left central incisor now within this band has lost much of its root, with the remainder measuring 17.6 mm long. The remainder of the root has fractured and was recently repaired. The incisor crown breadth is 8.6 mm and the maximum B-L diameter is 7.3 mm. This incisor has been worn evenly into the dentine, and no trace of shoveling is evident.

The thin gold band is a simple construction, for most of its length just under 2.0 mm wide varying from 1.9 to 2.1 mm (2.2 in one location, see below). Baggieri (2001: 324) gives the weight of the band as 2.5 grams. This is an unusually narrow band for Etruscan dental appliances, particularly if it had been intended to stabilize loose teeth. Its thickness varies from under 0.20 mm to about 0.25 mm at the point where it is welded. This wider section, near the medial aspect of the canine now placed within it, appears more globular (slightly wider and thicker), suggesting that a weld was made at this location by overlapping the ends of a long band. The total length of the original ribbon from which this band was formed is estimated at 74 mm. Baggieri (2001: 324) claims that microscopic examination of the exterior surface at 170 magnification reveals a line marking the junction of the welding and trauma to the band, which he attributes to damage from mastication. Becker notes that considerable damage is evident even without high magnification, but what is visible probably reflects excavation and subsequent handling. At the point of overlap on this gold band there also appears a tiny raised dot of gold, the function or origins of which are unknown. This “dot,” of insignificant size and barely visible to a casual observer, would have been on the right labial side of the mouth. The maximum length of this appliance as it now exists, after distortion, is 24.5 mm, with the breadth at the inserted canine 6.1 mm, and at the incisor, 8.0 mm. The distance across the buccal and lingual aspects of the band, at the present compressed center, is about 1.6 mm. The actual maximum length of the oval band is shown in Figure 5.8, and demonstrates that the band had spanned four teeth.

This band certainly could not have held any false teeth, but could only have been used as a decoration, or to stabilize teeth loosened by a blow or by periodontal disease. Yet it appears too narrow to have served a purely decorative purpose, unless a discrete hint of gold was intended or the owner could not afford a more elaborate example. Severe periodontal disease in the area of the incisors has not yet been demonstrated as common among the Etruscans (cf. Becker 1990), but the incisor area is often involved in periodontal disease in other populations.

The loss of some of the teeth within this band (or perhaps they simply were not recovered in the excavation) led to its central area being deliberately pinched together to hold the remaining (or added) teeth “in place” within the band.

Critique of other published observations

Hoffmann-Axthelm (1981: fig. 65, but not illustrated in 1985 edition) shows what appears to be the Bruschi I appliance—a very thin band around an I1 and canine, with the center of the band (recently) bent to hold these teeth at the ends. Illustrations of this appliance generally depict two teeth, with intact roots, at the lateral ends of the appliance. Other descriptions of these teeth vary, probably because they derive from one or more copies that have included human teeth (incisors and/or canines).

The most common error associated with the Bruschi I appliance is to describe the pinched central area of this simple oval band as a “spacer bar” connecting two loops. This error obviously derives from naive evaluations of photographs, or possibly from flawed reproductions. Deneffe’s description is roughly accurate. At first he notes that the gold band was pressed together in the center, but later Deneffe (1899: 82) describes this as a “*barre*” and a bar-like stricture is exactly what he illustrates. Becker suggests that Deneffe’s 1899 illustration derives from a Guerini copy (see Appendix II). Emptoz (1987) repeats this error when he describes this piece as having “*une tige*” (“a shank”) of gold. Emptoz even suggests that this “*tige*” may have supported a false tooth. Other authors repeat this erroneous “observation” when they state that a bar connecting two rings acted as a spacer bar, or by various other descriptions revealing that they have never examined the piece. The use of the term “wire” for this thin band has led other evaluators to suppose it resembles the Phoenician examples (e.g. the two from Sidon, nos. W-1 and W-2). Perhaps this is why Emptoz (1987: 557) infers that a sixth appliance could be found at Tarquinia; his example XXIII that he believed to be made of wire. Emptoz says that the example he assigns to number XXIII had been part of the Bruschi Collection, and mistakenly notes it as a gold-wire retention band. Clearly this confusion results from Emptoz’s misreading of earlier descriptions and photographs, in which the unusually narrow band looks like wire.

Sudhoff, normally a careful observer, also created two examples from Bruschi I through the use of two different photographs, both shown without crediting their sources. Sudhoff (1921: fig. 47; 1926: 84, fig. 47) describes Bruschi I as coming from Tarquinia, and describes the maxillary left central incisor and canine in place. Later, Sudhoff (1921: figs 51a and 51b; 1926: 85, figs 51a and b) describes a supposedly different piece said to be from Caere, although he suggests that it might have been made in Tarquinia. Both illustrations seem to be of the same piece.

Weinberger (1948: 125, fig. 42: 2 and 6), following Guerini (1909) and Sudhoff (1921, 1926), also created two phantom appliances from this single example. One is derived from Guerini’s figs 23–24 (Sudhoff’s fig. 51) and the other from Guerini’s fig. 20 (Sudhoff’s fig. 47). The value of conducting primary research is nowhere more clearly seen than in these examples.

Guerini (1909: fig. 20) illustrates what is probably a very poor copy of the Bruschi I appliance, perhaps his own replica. He incorrectly shows it as if it were

a “bar” extending between two rings. Guerini’s (1909) description could be of another thin band (see Weinberger fig. 41: 2). Guerini’s (1909) fig. 24 depicts another reproduction of this piece (possibly the Wellcome Museum copy, see Appendix II), further confusing the matter.

Most recently Baggieri (2001: 324–325) has presented data supposedly derived from an X-ray diffraction analysis of the Bruschi I appliance, which he identifies as “T-A,” but a number of errors recur.

9 Bruschi II

Present location: missing since c.1916–1925. May have been transferred from

Count Bruschi’s collection to the Museo Archeologico Nazionale Tarquiniense.

Type: four rings, with a rivet in the only empty ring (1I?).

Origin: area of Tarquinia?

Jaw part: maxillary(?), probably 2I–I2.

Date: not known.

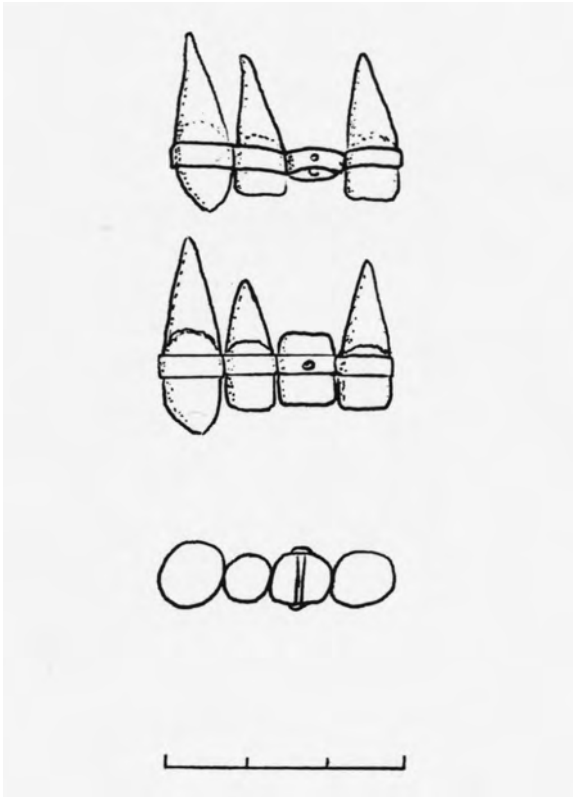


Figure 5.9 Tarquinia appliance: Drawing of Bruschi II appliance, three views, by Becker.

Sex: female ???

Dimensions: L. 29, W. 2.1 mm.

Description: This now missing piece was a four-ring appliance with a single rivet though one of the center loops similar to the Populonia pontic (no. 6). Early illustrations suggest a 7.0 mm rivet with a shaft length of 5.9 mm and a diameter of ca 0.75 mm. The peened heads of the rivet have diameters of 1.5 to 2.0 mm. Generally believed to have surrounded a canine (right) and three incisors. If correct, the replaced tooth would have been the right central incisor.

References

- Bliquez 1996: 2652, “N,” figs 21 and 37 (no. 6).
Bobbio 1958: fig. 11 (taken from the copy illustrated by Rivault 1953: fig. 4).
Brown 1934: fig. 4b.
Casotti 1947: 671, pl. 1, 7 (from Guerini 1894a: fig. 7).
Deneffe 1899: 80–81, fig. 6 (shown as maxillary).
Dunn 1894: 9, with figure.
Emptoz 1987: 555, fig. 16 (no. XVIII).
Guerini 1894a: fig. 7; 1894b: fig. 7.
Guerini 1904: 394, fig. 7.
Guerini 1909: 73–74, fig. 19 (sketch) notes it as maxillary.
Platschick 1904–1905: 241, fig. 8 (shown as mandibular).
Rivault 1953: fig. 4 (from a copy in Paris).
Sterpellone and Elsheikh 1990: 134, fig. 62 bottom (rough drawing copied from unknown source).
Sudhoff 1926: 81, fig. 44.
Weinberger 1948: 125, fig. 41: 3.
Woodforde 1967: fig. on p. 11 (sketch of rear view).

History

When the appliance was first described (Guerini 1904) in the collection of Count Bruschi-Falgari at Corneto (Tarquinia), the Bruschi II appliance presumably was excavated in the territory of Tarquinia, on land owned by the Bruschi-Falgari (which encompassed the famous Hellenistic Tomba Bruschi: see Vincenti 2009). Sudhoff (1926: 81) specifically says that it comes from the “*Gräberstadt von Tarquinii*,” “the Tarquinia necropolis.”

Description

The three earliest depictions of the Bruschi II appliance are in remarkable agreement as to details of its configuration, each depicting a four-ring appliance that seems to be welded together, with a single false tooth held in place by a rivet. As always one should not assume that the teeth depicted were original to the appliance. By 1894 the false tooth originally riveted in place had been lost, but

three natural human teeth were then present. The replacement tooth may have been made of perishable material, although it is just as likely that all the original teeth as well as the replacement tooth dissolved in the tomb. Our depiction of this appliance is based *entirely* on the assumption that the canine shown in early illustrations was original to the appliance, which is not a likely pattern of manufacture. More likely this appliance spanned all four incisors, but that could only be confirmed by precise measurements, which are no longer possible.

Dunn's drawing (1894: 9) appears to show the lingual surface of Bruschi II in an inverted position, with the tooth size as depicted suggestive of the four incisors, and with the rivet for a false tooth at 1I. Dunn depicts rather wide rings, perhaps 5 mm wide (later copied by Platschick). Guerini's drawing (1894a: 395, fig. 7; also 1894b: fig. 7) suggests a buccal view of a maxillary appliance, with the tooth on the left being a canine. This would indicate that the replaced tooth was an 1I, as depicted in Figure 5.9.

Deneffe (1899: pl. III, 6) says that Bruschi II is a maxillary appliance, 3 cm long, a measurement only 1 mm longer than Becker's calculation from the photograph. However, Deneffe states that the band is 4 to 5 mm wide; in agreement with Dunn but twice the 2.1 mm estimated by Becker. Deneffe's photograph appears to show the lingual surface. Although the configuration of the tooth on the right is unclear, Deneffe specifies that it is a canine and that the appliance spans the 1I–C. Guerini (1909) depicts it thus. Sudhoff (1926: 44) copies Deneffe exactly. Deneffe's photograph, however, shows much more slender rings than those drawn by Dunn. The Deneffe illustration may be a copy and not the original. The width of the individual rings probably did vary, however, as each ring of Bruschi II may have been made separately before all 4 were welded together.

Based on an evaluation of the three teeth "in place" one might assume that the Bruschi II appliance was mandibular. These, however, could be modern replacements rather than originals. More likely this piece was worn in the maxilla, as suggested by most of the early observers. A medial tooth more recently depicted as present in this appliance does not appear to be the correct tooth for that location, and should thus be omitted (see photograph in Weinberger 1948: fig. 41:3; also Brown 1934).

The Bruschi II piece also is often confused with the Valsiarosa example. A copy of an appliance bridging four spaces (1I–C.), which appears to be a copy of Bruschi II, is in the Musée de l'École Dentaire de Paris (Rivault 1953: fig. 4; see copies in Appendix II).

10 Bruschi III (often confused with Corneto I)

Present location: Museo Archeologico Nazionale Tarquiniense.

Type: stabilizing band with four braces welded within to form five spaces (no rivets). Two spaces are now empty, three others held human teeth when Becker examined it, but all appear to be recent replacements.

Origin: Tarquinia area? (from collection of Count Bruschi-Falgari).

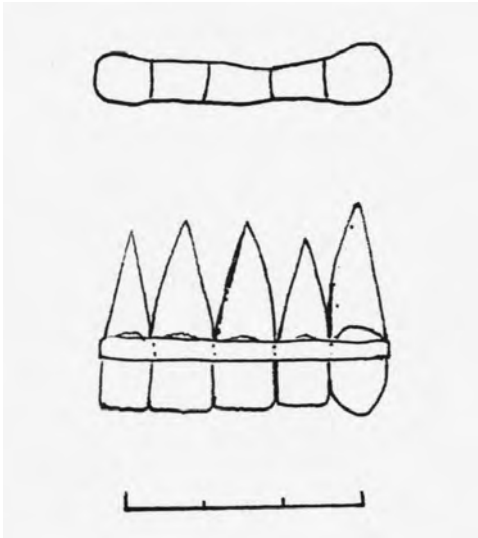


Figure 5.10 Tarquinia appliance: drawing of Bruschi III, a braced band for stabilizing loose teeth, two views, by Becker.

Jaw part: maxilla? Possibly left canine to I2. Menconi and Fornaciari (1985) suggest that this spanned the maxillary 2PM–I1.

Date: Menconi and Fornaciari (1985) imply a second-century BCE date, but this is unconfirmed.

Sex: female.

Dimensions: L. 36.8, W. 2.6–2.8 mm. See Table 5.1 below for additional measurements.

Description: band apparently fashioned from a single long (about 83 mm) strip of gold in an oval into which four braces (dividing strips) have been fitted and welded to create five tooth spaces. The elongate “rectangles” of these openings suggest that this appliance had been flattened, creating its unusual length (cf. Corneto I). The sizes of these openings clearly represent one canine and all four incisors. There is no indication of a replacement tooth. The gold dividing strips are slightly narrower than the minimum diameter of the surrounding oval. The teeth displayed with this appliance are not those of the original owner.

References

- Baggieri 1998: 68.
- Baggieri 1999: 38–39, fig. 7 lower (useful illustrations).
- Baggieri 2001: 325, pl. xxxviid.

- Bliquez 1996: "H," figs 13, 14 and 37 (no. 5).
Bobbio 1958: fig. 11 (copy, photo taken from Rivault 1953).
Casotti 1947: 673 (max. 2I–C.) with three teeth *in situ* (from Deneffe?).
Deneffe 1899: 79–80, fig. 5 (maxillary). Apparently a photograph of a copy with random teeth inserted.
Dunn 1894: 8–9, fig. on bottom of p. 8 (very poor drawing).
Emptoz 1987: 553? fig. 13?, no. XIII?
Guerini 1909: 70, fig. 15 (a copy that appears to be a "casting").
Menconi and Fornaciari 1985: 92, figs 1, 2 (fig. 1 depicts this with two teeth; p. 94, fig. 2 is an excellent "plan" view with no teeth in place) (Maxillary 2PM–I1?).
Micheloni 1976. Appliance belonging "*all'arcata inferiore*" thus mandibular, according to Torino *et al.* (1996: 119–120).
Naso 2011: 151, no. 9.
Platschick 1904–1905: 241, fig. 7.
Rath 1958: 12, fig. 3 (illustrations only).
Rivault 1953 (from a copy in Paris, see Appendix II).
Sterpellone and Elsheikh 1990: 135, fig. 63 (photograph has only two teeth in place).
Sudhoff 1926: 85, fig. 50.
Tabanelli 1962: fig. 50 (q.v. for teeth in place) (Mandibular C–I2).
Torino *et al.* 1996: 119–120. These authors accept this as a mandibular appliance, supposedly following Micheloni 1976.
Torino *et al.* 1997: 536.

Critical observations on the Bruschi III appliance and associated teeth

This appliance often is confused with Corneto I, which also spans five spaces, but has two rivets. Bruschi III appears to be a maxillary appliance, which spanned (included) five dental spaces, probably from the left canine to the right I2. This location is inferred from the size and relative positions of the "rings." The premolar seen within the band in 1991 certainly is a "display replacement" tooth not original to the appliance, and the other two teeth seen in 1991 also are recent additions. When Menconi and Fornaciari (1985) saw this appliance it had only two human teeth associated with it. Based on the two teeth that he saw and identified as mandibular, Baggieri (1998: 68, "T-B") believes it was worn in a mandible (Baggieri 2001: 325).

An appliance of this type would be unlikely to span 1PM–I1: probably it was placed around a maxillary canine and across all four incisors, as suggested by Tabanelli (1962). It was seen by Menconi and Fornaciari in 1985. This description is based only on Becker's provisional examination of this appliance, since Bruschi III then (1991 to 1995) was securely glued to a plastic sheet and displayed in inverted position; it could not be detached.

Bruschi III appears to have been formed by constructing a simple gold band into which a series of four braces, or internal cross-pieces, have been inserted by welding. The four braces partition off five separate spaces. The exact manner of construction is not clear. Becker (11 June 1991) identified all

three teeth as maxillary, although they are not original to the appliance (cf. Dunn 1894: 8, fig.). All three are loose in the gold sockets: the teeth would fall out if placed right side up on the vertical display surface. The measurements listed below are approximate, but adequate for comparative purposes. The inference that two false teeth had been held in place by this appliance (Casotti 1947; Tabanelli 1962) is not supported by the sizes of the individual spaces, the shapes of the sockets or the breadth of the band (cf. Copenhagen). We also must agree with Deneffe (1899) and Bliquez (1996) that no false teeth were in this appliance. The three teeth “traditionally” displayed in this appliance might reflect only the remaining fragments in place when Bruschi III was first seen.

Description

The total length of this narrow band is 36.8 mm, spanning five tooth spaces from the left canine-I2). Each of the enclosing units formed within the band, also referred to as “rings,” is somewhat warped or distorted from its original shape, but some are much more bent than others. Therefore, the estimates of the size of individual units must be inferred from the few measurements that could be made. All measurements are presented in Table 5.1.

The largest chamber appears to be the “ring” that would have held the suspected left lateral incisor (2I). This considerable mesial-distal length, however, has been created by a compression of the B-L diameter. The lingual surface is a single low curve, but the buccal surface has a bend toward one side that gives the appearance of two curves. The present B-L width is about 4.5 mm, but varies due to the bend noted above. The B-L dimension may even have been greater than the M-D, but what is presented here are only those lengths that could be measured and described.

The “ring” for I2 appears to be the second largest, and would be expected to be about the same size as that for the left lateral incisor (2I).

Table 5.1 Dimensions (in mm) for Bruschi III (measured while it was affixed to the display mount noted above)

<i>Tooth space</i>	<i>M-D length*</i>	<i>B-L breadth*</i>	<i>B-L breadth**</i>	<i>Height of band ***</i>	<i>Thickness of band ***</i>
I2	7.2	6.9	7.3 (est.)	2.7	?
I1	6.1	5.9	6.3	2.75	0.2
1I	5.3	5.3 approx.	6.2 (est.)	2.6	
2I	7.9	4.5 (bent)		2.8	
C	5.9	9.3 (est.)		2.7	0.25

* Measurement across the interior surfaces of the “ring.”

** Measurement across the interior surfaces of the “ring.”

*** Taken at the center of the lingual surface.

Table 5.2 Measurements of exterior surfaces of band Bruschi III

Between Teeth	12/11	11/11	11/21	21C
Exterior	8.1	—	—	—
Interior Length	3.9(?)	3.8*	3.8**	4.1***
Thickness	—	0.2	—	—

* The brace is narrower than the band (by approximately 0.1 or 0.15 mm only) on the “inferior” surface.

** This brace is wider than the encircling band to which it attaches, on the buccal surface, by at least 0.5 mm, and approximately 0.2 mm wider on the lingual surface (all these are on the inferior surface of the band, which is uppermost as it is displayed).

*** The brace between 21/C is detached from the surrounding band at the buccal surface. An imperfect weld is suggested. At the lingual attachment the brace is about 0.1 mm higher than the band to which it is attached.

Measurements (B-L) were taken to the external surfaces of the band at the points where the four “braces” separate the anterior from posterior surfaces (includes the length of the brace and the thickness of the band. The thickness of only one brace could be measured as per Table 5.2.

The central socket or “ring” of this appliance, which would have surrounded the 11, clearly is the smallest, as expected for a mandibular appliance. The socket for the 11 appears to have been circular, but also appears to have been bent outward on the buccal surface for attachment to the present mounting, resulting in the distance between the braces on the buccal surface being less than on the lingual surface. (All lengths in the table are from lingual surface.) The “circular” form suggests that this socket was meant to surround a living “peg”-shaped incisor, rather than a false tooth, which would have been made rectangular in cross section (cf. Copenhagen). The narrow width of the band of Bruschi III supports Deneffe’s and Bliquez’s conclusion that it was designed only to stabilize loose teeth.

The “ring” in the position of the 11 seems to be second smallest. This very small size, if it held a lateral incisor, is also consonant with a mandibular prosthesis. The socket formed for the left canine appears to be in the middle-range of sizes in this series of five. The I2 space within the band, however, has a M-D interior diameter of 7.2 mm. This distance seems distorted (elongated) through flattening of the band. This may explain why the tooth located in this socket was found, in 1991, to be rotated in its socket within the band.

The width of the “lingual” surface of the band, measured in five places, varies from 2.6 to 2.8 mm (average 2.71 mm). Menconi and Fornaciari (1985: 94) give a range in width from 0.23 to 0.25 cm. The variations suggest that four individual units were joined by welding. Also, the buccal widths of the band appear much narrower than the lingual, but these could not be measured while the appliance was fixed to its display mount.

The three human teeth in Bruschi III

Although Becker believes that none of the associated teeth are original to this appliance, a history and description may be useful. Dunn (1894: 8) originally

thought that this appliance encompassed five teeth or tooth spaces, believing that the three teeth were natural and served as anchors. Dunn's suggestion that there were two false teeth is indicated by the depiction of two rivets in his *drawing* of this appliance, although no rivets exist nor does Dunn indicate their presence in his text. At least one of the "false" teeth was said to be missing at an early date, but a later account states that one of the surviving false teeth was split. By the time that Menconi and Fornaciari (1985) examined this piece, at least one of the three human teeth that had been earlier in place appears to have been lost. This lost tooth had been replaced by another tooth before 1991, the premolar now seen in more recent illustrations.

This possible scenario, that lost teeth have been replaced, or did not survive with this appliance, brings us back to the problem of the location of the Bruschi III appliance in the jaws. It is doubtful whether *any* of the teeth now with the appliance were the owner's original teeth, serving as anchors for artificial teeth: we should infer that all have been replaced. All three teeth found with this appliance in 1991 appeared to be rotated in the sockets. The premolar appears rotated a full 90 degrees and an incisor is rotated about 45 degrees. We cannot even be certain which surface of the appliance is being viewed, but possibly what can be seen is the lingual surface, and the buccal is affixed to the plastic display panel. All observations and measurements were made with the supposition that we are examining the lingual surface.

No teeth were found in the sockets made for either left incisor; the tooth now found in the socket made to fit the I2 is actually a maxillary left canine. The tooth wear pattern on this I2 suggests that this person had an overbite, down in a left distal direction, indicating that this is a maxillary tooth. The extent of wear is considerable, suggesting that it comes from a mature individual who may have lost teeth or at least have had loose teeth. Some tartar remains on the tooth.

The left maxillary first premolar is so worn that both cusps are flattened almost to the level of the dividing groove, but not quite to a single flat surface. This premolar, which appears to be within the socket designed for a canine, appears to have had a great deal of tartar on it, but most of it has become detached, as indicated by the clean enamel showing through patches among the remaining tartar. The root has grooves on both surfaces but is not actually bifurcate, and also has some bumps at about one-third of its length up from its apical end. The articular surface of the tooth is worn down towards the lingual margins, again suggesting

Table 5.3 Dimensions of extant teeth in Bruschi III (believed to be modern inserts, of unknown sex)

<i>Tooth</i>	<i>Height</i>	<i>A-P diameter</i>	<i>M-D diameter</i>
Left canine	25.3	7.9	6.4
Left first PM	17.6	9.3	5.9
Right central I	?	6.9	7.2

an overbite such as is indicated by the canine, and suggesting that both derive from the same individual.

Critique of the literature relating to the Bruschi III appliance

Dunn (1894) describes Bruschi III as one of two examples that he saw in Count Bruschi's collection. This indicates that the third example owned by Count Bruschi (here identified as Bruschi I) came into his collection after 1894, but before 1899 when Deneffe indicated that Count Bruschi owned three examples.

Dunn suggests that the Bruschi III appliance was fashioned to replace the two front teeth (central incisors). His text does not mention rivets, but his illustration depicts rivets in both empty "central incisor" positions. The three "real" teeth shown by Dunn appear to be the maxillary left canine, a left lateral incisor (which in the drawing looks like a broad central incisor) and a right lateral incisor.

Deneffe (1899) describes this as one of three dental appliances then in Count Bruschi's collection. The accompanying illustration shows the left canine and both right incisors in place (differing from Dunn's illustration). Deneffe's description, however, is confused since he states that "*la canine droite*" is in place, while his fig. 5 suggests that it is the left canine in place. Casotti (1947) also says that three teeth are in place.

Deneffe (1899) describes the appliance as approx. 5 mm high, nearly twice the actual height (see also his evaluation of width for Bruschi II). His drawing, however, depicts a band much closer to the correct width. Deneffe's estimate of the band's length (40 mm) is reasonably accurate. Menconi and Fornaciari (1985) give the width of the band as 2.5 mm, but others tend to estimate the width as being much greater than it is, probably following Deneffe's published measurement. All descriptions supposed to be of Bruschi III, but including rivets, are probably of Corneto I, or derived from Dunn's (1894) early error. All the difficulties with the five examples supposed to be in Tarquinia (see Deneffe 1899: 59) appear to derive from these early contradictory descriptions.

11 Corneto I



Figure 5.11 Tarquinia appliance: Corneto I Appliance, drawing by Becker after Van Marter 1885a: figs 3–4.

Present location: missing (at least since c.1916–1925) from Museo Archeologico Nazionale Tarquiniense.

Type: braced band (?) for five spaces, one of which has a rivet (one space still had a replacement tooth; no other teeth remain). It is likely that the modern copies of this appliance are “braced,” while the real example may have been formed by a series of welded rings.

Origin: excavated from a tomb in the necropolis of Corneto-Tarquinia, in contrada Montarozzi northwest of Secondi Archi, by Luigi Dasti *et al.* (season October 16, 1876 through Spring, 1877). The project was financed by the Comune and the *Arte agrariane* of Corneto-Tarquinia.

Jaw part: maxillary (?), left C–I2.

Date: sixth–fourth century BCE (on the basis of Helbig 1877: 64).

Sex: ?

Dimensions: L. 28, W.5–6 mm.

Description: following Deneffe’s description, the five spaces of this “braced” oval band are believed to have surrounded all four maxillary incisors and the left canine. The construction appears remarkably like Bruschi III, but with a single rivet. This rivet would have passed through a replacement or cut-down central right incisor. Illustrations suggest a rivet 7.5 mm long, with a shaft length of approx. 6.7 mm. The peened heads of the rivet probably had dimensions similar to Bruschi II.

References

- Bliquez 1996: 2651–2652, “M,” figs 20 and 37 (no. 4).
 Boissier 1927: 22, figs 3 and 4.
 Brown 1934: fig. 4a.
 Casotti 1947: pl. I (from Guerini 1894a, see below); and also Casotti’s pl. II (from Platschick 1904–1905: fig. 6).
 Dasti 1878: 17, 367–368.
 Deneffe 1899: 77–78, fig. 4 (2.6 cm long, 0.5 cm wide) (Weinberger says p. 75, pl. II: fig. 4).
 Dunn 1894: 8 (fig).
 Emptoz 1987: 552, fig. 11? (no. XII?).
 Farrar 1888: 33, figs 3–4.
 Guerini 1894a: 394, figs 5–6; 1894b: figs 5, 6.
 Guerini 1909: 73, fig. 18 (“very old Etruscan tomb”).
 Helbig 1877: 59–64 (see also Waarsenburg 1991a: 243, n. 11).
 Koch 1909: 9, 16 figs 3–4.
 Lufkin 1948: fig. 20.
 Platschick 1904–1905: 241, fig. 6.
 Ring 1985: 45, fig. 36 (from a copy? maxillary?).
 Sudhoff 1926: 80–81, fig. 43.
 Tabanelli 1962: 93.
 Van Marter 1885a: 3, figs 3–4.
 Waarsenburg 1991a: 243, n. 11.
 Weinberger 1947: fig. 8 (supposedly from Sudhoff 1926: 90 [sic], but looks like a copy).
 Weinberger 1948: 125, figs 41: 5 and 41: 9.
 Woodforde 1967: fig. on p. 11 (inverted).

History/possible origins of Corneto I

From Helbig's early account (1877: 56–57) it appears that this piece was discovered during the 1876–1877 excavation season. Dasti (1878: 359) notes that the museum was inaugurated in 1878, on the ground and first floors of Palazzo Demaniale, Via dell'Ospedale, n. 15. This appliance was then displayed in room 3 on the ground floor. Van Marter (1885a: 2), who saw it in the Corneto Museum about 1885, says that he has a certificate from Dasti attesting to the dates during which this tomb and the tomb containing the Corneto II appliance were opened.

Dasti (1878: 367–368, Chapter IV, “*Oggetti d'Arte Etruschi di recente trovati*”) notes that “this” appliance, along with smaller objects, was located in the “3^a Stanza” of the Museo Etrusco, Tarquiniese, suggesting that a piece in the Civic Museum is meant, very probably Corneto I.

Una dentiera formata di [sic. “da” in Helbig's version] una lamina d'oro, lunga 0.03, con sei fori, uno dei quali, il quarto, contando dalla sinistra, contiene un dente. I due fori estremi, alquanto più larghi degli altri, erano destinati ad essere infissi in due denti effettivi della persona che doveva portare la dentiera.

A denture formed from a gold plate, 0.03 long, with six holes, one of which, the fourth counting from the left, contains a tooth. The two end holes, quite a bit wider than the others, were destined to be affixed to the two actual teeth of the person who was supposed to wear the denture.

This is the same description that appears, with slight orthographic variation, in Dasti's letter to Helbig (Helbig 1877: 64). Although no rivets are noted in this description, at least one appears to be present. Dasti may have counted the spaces on either side of the visible rivet (that actually would be halves of a single “tooth” location) as two separate “*fori*” (“holes,” “openings”) thus creating “*sei fori*” or six holes by his count.

Helbig's (1877: 59–64) publication of Dasti's letter contains the listing of the objects from all these chamber tombs combined, and distinguished only as having held Greek black- or red-figured vases (thus probably Attic imports of the sixth to fourth centuries BCE). There is no mention of finds of Etruscan bucchero pottery, often ignored in old excavations. Helbig (1877: 64; also Dasti) notes a sardonyx scarab and gold, granulation-decorated grape-cluster earrings, which sound typical of fourth-century Etruscan art, but there is no way to determine which burial produced them or the gold appliance.

Description, based on past publications

Van Marter (1885a) was the first to illustrate and describe this appliance in a clear and useful fashion, as an oval band with cross braces, an unlikely but not impossible construction. Usually it is described as made of five rings, as in Dunn's drawing (1894: 8; copied by Platschick 1904–1905), but elsewhere it appears to

be a single band with four cross braces forming five sockets. Dunn's drawing and others showing five sockets appear to result from confusion with the Bruschi III appliance. Dunn (1894) illustrates in the central hole a single tooth with a portion of the root intact. This suggests a natural tooth and not an artificial or re-placed crown riveted into the appliance. In turn, the presence of a human tooth at this location implies that the appliance had but one rivet.

Van Marter (1885a: 3, figs 3–4) provides a description together with its most widely known illustrations, including what appears to be a hand-applied gilt overlay representing the appliance, approximately 5 mm broad. Van Marter says that the Corneto I appliance came from a “Roman” tomb, dated to 400 BCE: he gave no explanation for the statement, but many authors assumed that the more “complex” appliances are Roman rather than Etruscan in origin. This information, including Van Marter's illustration, was among the material copied by Koch (1909: 9, 16, figs 3–4).

Sudhoff (1926: 81) depicts a mandibular prosthesis (2I–C.), as indicated from its positioning and from the shape of the teeth, of which four are present, three being real tooth crowns and one apparently a replacement or false tooth. He shows a braced band with two rivets, stating (1926: 80) that the band is 5 mm wide on both the labial and lingual sides. By 1926 this piece may have been missing, so it would be useful to know if Sudhoff actually saw the appliance or instead derived his information from earlier sources or a copy. Other illustrations show but one tooth, the false tooth riveted in the central space in the band. Possibly Sudhoff's illustration was made before the loss of the natural tooth crowns that had been found with the appliance.

Deneffe describes this appliance, or a copy of it, as 26 mm long, 5 mm high, holding a well-preserved tooth crown riveted in the central opening, but lacking a root. The fourth hole from left also bears a rivet, but the false tooth is missing. Deneffe's description appears to have been taken from Guerini (1894a: figs 5 and 6). Deneffe (1899: 78) also credits Helbig with facilitating arrangements for making copies for the museum in Ghent of the two appliances then in the Corneto Civic Museum (see Appendix II.A).

Van Marter's figure illustrates this piece with what appears to be the crown of a small incisor in the central opening. Guerini (1894a: figs 5–6) also shows but one central incisor crown in place. However, Weinberger (1948: 125, figs 41: 5 and 41: 9) provides two interesting photographs. One, labeled 41: 5, appears to show this piece situated in a mandible, with all teeth present except for the top of the I1. This is probably a photograph of a copy set in a mandible for display. Weinberger's illustration 41: 9 shows the appliance without a jaw, apparently the illustration used by Guerini (1894a) and Van Marter (1885a). Most interesting is Weinberger's failure to make any reference to his own fig. 41: 9 depicting this appliance.

Commonly published illustrations show but one remaining tooth and it is positioned as if it were a mandibular I1. If this tooth were a mandibular I1, then the chamber next to it (to the left when viewed from the rear; and the one with a rivet)

would have held a replacement for the I1. The openings or spaces in this appliance are relatively square in plan, and similar in shape to the central chamber in the Copenhagen appliance that was clearly meant to hold a false tooth. Possibly the I2 here was not false and we must conclude that the C and 2I served as anchor teeth. The exterior of the band appears perfectly flat.

Dunn's note on an appliance that has been confused with this prosthesis says that it is in the "*Stanza Ottava*" ("Eighth Room") of the museum. Platschick (1904–1905: 241) says that it is in the "*ottava sala*," and shows the false tooth as projecting below the appliance, but clearly most of the root is missing. Dunn suggests that only the fourth tooth was false, having a rivet located in that ring, but that when he saw it the tooth was missing. Clearly what he described is not the same piece as Corneto I, but which of the others (at Tarquinia or elsewhere) it might be is not certain.

12 Corneto II

Present location: missing (at least since c.1916–1925) from Museo Archeologico Nazionale Tarquiniense.

Type: compound: seven rings bridging eight spaces. Two of the five human teeth are in place as are "two" of the three false teeth. The "two" false teeth are fashioned from one piece of bone (possibly ivory). The false teeth were held in place by three rivets.

Origin: Necropolis of Tarquinia.

Jaw part: maxilla? Possibly 2PM–right canine.

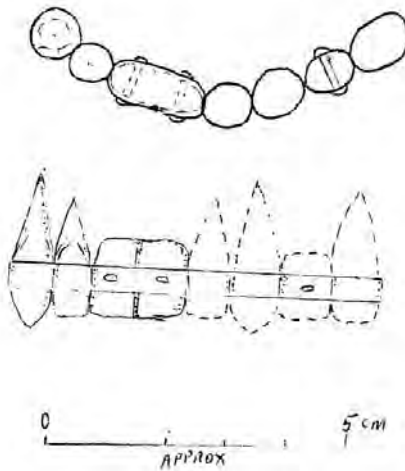


Figure 5.12 Tarquinia appliance: Corneto II, two sketch views. Becker's reconstruction, combining Clawson (1933: fig. 8a) and Van Marter (1885a: figs 1–2).

Date: 500 BCE (per Van Marter 1885a, citing Helbig).

Sex: Female??

Dimensions: L. 62.5, W. 4.0–4.3 mm.

Description: This impressive seven-ring appliance, spanning eight tooth spaces, is perhaps the most complex of Etruscan bridges. Three teeth are replaced, using three gold rivets. Both central incisors are replaced by the major feature, an elongate “ring” in which an animal tooth or piece of ivory was held in place by two rivets. The right lateral incisor and canine served as anchor teeth. This appliance also provided a replacement for the left first premolar, also riveted in place but not surviving. The second premolar served as the anchor tooth for this end of the bridge. Loss to decay of a first premolar prior to the loss of the molars and the second premolar is rare. The possibility that this tooth, like the incisors, was deliberately removed must be considered.

References

- Bliquez 1996: 2653–2654, “Q,” figs 26, 27 and 37 (no. 3).
 Bobbio 1958: fig. 13 (figure credited to Van Marter 1885a, but actually from Rivault 1953: fig. 1, who took his illustration from a copy of the appliance in Paris).
 Boissier 1927: fig. 8.
 Brown 1934: fig. 5.
 Casotti 1947: 669, pl. 1 (from Guerini 1894a: figs 3 and 4).
 Cavenago 1933: fig. 3.
 Clawson 1933: fig. 8a; also 1934: 24, fig. 1.
 Dunn 1894: 6–8, fig. p. 6.
 Deneffe 1899: 73–76, fig. 3.
 Emptoz 1987: 556, fig. 17 (no. XIX). Dates it fourth century BCE.
 Farrar 1888, I: 32–33, figs 1–2.
 Guerini 1894a: 393–394, figs 3–4.
 Guerini 1894b: figs 3–4.
 Guerini 1909: 71–73, figs 16, 17 (16 is printed in reverse).
 Hoffmann-Axthelm 1985: 68, fig. 63 (also fig. 63 in 1981 edition). The “appliance” illustrated here is a copy.
 Koch 1909: 9, 16, figs 1–2.
 Lufkin 1948: figs 18, 19.
 Platschick 1904–1905: 240, fig. 5.
 Rivault 1953: fig. 1.
 Sterpellone and Elsheikh 1990: 134, fig. 62 top (a drawing), 138–139.
 Strömberg 1919: figs 2–3 (from Guerini 1909).
 Sudhoff 1926: 82–83, fig. 46.
 Tabanelli 1962: 93.
 Van Marter 1885a: 3, figs 1–2 (best illustrations).
 Weege 1913: 371.
 Weinberger 1948: 126, fig. 41: 8 (error in transcribing date).
 Woodforde 1967: fig. on p. 10.
 Weiss 1989: fig. 28 (but mislabeled, with two views shown. Said to be in the “Civic Museum of Corneto,” suggesting that his data comes from Guerini 1909).

History

This dental appliance appears to have been excavated around 1885, or during Van Marter's residence in Rome. First described by Van Marter (1885a: 2) when it was in the Museo Comunale di Corneto. He suggested a date based on Helbig's estimate of 500–400 BCE, presumably founded on the style of artifacts found with this appliance. Van Marter specifies that he received a certificate from Dasti attesting to the date on which the tomb was opened. While it is remotely possible that this unusual seven-ring appliance was excavated before 1885, the fact that it was located in the Museo Civico suggests that Corneto II was a local find.

Van Marter's report later (1909: fig. 16) indicates the "Civic Museum of Corneto," where it had been since at least 1885 (see Van Marter 1885a). Koch (1909) also copied Van Marter's data and illustrations. Platschick's reference (1904: 240) to "*otto anelli*" ("eight rings") probably indicates that this appliance extended over eight tooth spaces, bridged by an appliance of seven rings. Strömberg (1919: 24, figs 2–3) provides lower case letter designations for these rings (a–g), an extremely useful means of identification, a system unfortunately not followed by subsequent authors. Strömberg's letter "g" indicates the largest ring in this appliance, which may have circled the left 2PM.

Description, based on past publications

This appliance, if maxillary, appears to replace at least three teeth (1PM and I1–I1). Deneffe gives the dimensions as 6 cm long, with a band 0.5 cm broad. Van Marter's illustration (1885a: fig. 1) suggests that the various rings are only 2.5 mm broad. Deneffe's illustration (also Platschick 1904–1905: fig. 5) shows one rivet for each of the three replaced teeth. The false incisor unit (the one-piece replacement for two teeth) has one rivet for each "tooth." The space for the 1PM has a rivet, but no false tooth remains. Details of construction techniques cannot be confirmed without direct inspection.

In place in this appliance are an intact, slender right canine and right lateral incisor. Next to them is a false tooth, replacing the central incisors, that appears to derive from an animal tooth with root filed down and otherwise sculpted. Platschick (1904–1905: 240) covered all possibilities when he noted that it might have been made "*di terra cotta o di pietra, oppure fatti da un pezzo di conchiglia*" ("of terracotta or of stone, or perhaps made of a piece of seashell") or from the tooth of a large animal. This "observation" clearly indicates that Platschick had not seen the appliance. Guerini (1909: 71–72) also suggests that one rooted anchor tooth was a canine. Guerini also suggested that either one of the owner's incisors never erupted (dental agenesis), or that shifting after the loss of a tooth may have allowed the teeth to close over this position, basing his inference on his belief that the large ring on one end was meant to fit over a molar (also Weinberger 1948: 126). In addition, Weinberger (1948: 126) believed that a "lug" (spacer bar?) was inserted between the last two empty rings, as suggested earlier by Guerini. No such "lug" existed, (a parallel is the fictional "spacer" often cited for the Bruschi I appliance).

The attachment of an Etruscan dental bridge to a molar is extremely unlikely. Van Marter is correct in indicating that the large empty ring probably fitted over a large left second premolar, which also reinforces the conclusion that this is a maxillary appliance. The cleverly carved replaced teeth make this is a sophisticated example of these appliances.

Guerini (1909: 72) also notes that “the large [false] tooth employed . . . does not show any signs of being worn by mastication” and suggests that it was made from an unerupted maxillary incisor taken from a calf (see also Deneffe 1899: 75, who cites Guerini). The replacement is cleverly carved with a vertical groove on the anterior surface to represent two incisors, complete with two rivets. Guerini, in English translation (Guerini 1909: 72) uses the British term “pivot” (in British dentistry the term used to describe a false tooth mounted on a pin set into a tooth root) rather than the term “rivet.”

Guerini (1909: 72) states that when making copies of “all the ancient prosthetic pieces existing in the Italian museums, I met with special difficulty in the reproduction of . . .” this example. The copy of the Corneto II appliance illustrated by Hoffmann-Axthelm (1985: fig. 63) is not accurately made, having but one rivet securing the large, carved false tooth representing the original “pair” of incisors.

The two appliances in the Liverpool World Museum

In 1867 Joseph Mayer, a goldsmith of Liverpool, England, donated his entire collection of antiquities, including two Etruscan gold dental appliances, to the Public Museum of his city, then known as The William Brown Street Museum (Waite 1885a: 508). Mayer’s Etruscan pieces had been purchased by or for him in Italy or in sales of collections formed in Italy. Many objects, especially bronzes and gold ornaments, can be traced to nineteenth-century collections gathered after what we today would consider the plundering of Vulci. Some items still bear labels linking them to finds sold by the estate of Prince Canino, one of the Bonapartes who had a country home in the vicinity of the city plain near the necropolis of ancient Vulci. Unfortunately, there is no documentation of the exact sources of the two dental appliances that were offered to Mayer because of his interest in gold-working. (None of Mayer’s Etruscan gold jewels has a recognized provenance either.)

We know from Purland (1857–1858) that by 1857 Mayer already had “some examples” of Etruscan gold dental appliances in his collection (also Weinberger 1948: 79). Waite (1885a: 508) believes that Mayer had actually bought these pieces during the year 1857, but this is uncertain. Waite mistakenly states that Hockley (1858: 121) had published a note on at least one of these appliances in Mayer’s collection (M 10334).

Perine (1883: 163) clearly knew of the existence of these two prostheses but his descriptions bear no relationship to the actual appliances. Quite possibly they derive from a verbal account provided to Perine by a third party. Arthur Weinberger (1948: 79) states that both of the Liverpool examples were reported in an article written by Henry H. Burchard, but this has not been confirmed. Emptoz (1987) includes in his inventory only one of the two examples from the Mayer Collection.

Both Liverpool dental appliances were studied (1980–1981) for incorporation in the catalogue of Etruscan objects being prepared by Turfa; they were then studied by Becker in 1988. After a theft occurred in 1989, the appliances were eventually recovered unscathed along with most other gold items. They were again examined by Turfa in 2011 with the assistance of Margarita Gleba and Cynthia Reed (goldsmith), and again examined by Turfa in May 2015 and May 2016. The descriptions that appear below are based on those direct observations, and the *Catalogue of the Etruscan and Italic Collection in the Liverpool World Museum* (Turfa, in press). What is most interesting about these two appliances, given that they became part of a single collection, is that both are bands enclosing four tooth spaces yet one has only what appear to be the natural lateral teeth in place and the other bears only the artificial central teeth (cf. Johnstone 1932b: 449). The gold bands are somewhat distorted, perhaps through post-excavation factors.

One of the copies of these Liverpool appliances in the Wellcome collection of the Science Museum, London (see Appendix II.B) came with the following note: “two pieces of Etruscan bridge work acquired Liverpool Museum 1915 ex Mayer collection there. WR 2/26/4 WR 2/25/28 (3).” Possible meanings of these numbers are discussed in Appendix II.B.

Bibliography for both Liverpool M 10334 and M 10335

These dental appliances have been cited in numerous general publications, with or (very often) without credit, often using the sketches published by Johnstone (1932b) or Macnamara (1973), or the photos of Tabanelli 1962: pl. 51. The main references are cited below:

- Bonfante, L. 1986. “Daily Life and Afterlife,” in L. Bonfante, ed., *Etruscan Life and Afterlife* [232–278] 251 fig. VIII–28. Detroit, IL: Wayne State University Press.
- Cohen, R. A. 1959. “Methods and Materials Used for Artificial Teeth [Abridged],” *Proceedings of the Royal Society of Medicine* 52: 775–786, fig. 1. (Incorrect identification as Egyptian.)
- Emptoz, F. 1987. “La Prothèse Dentaire dans la Civilisation Étrusque,” in *Archéologie et Médecine: VII Rencontre Internationale d’Archéologie et d’Histoire (Antibes 1986)* [545–560] 554–555 no. XVII (inventory includes only one Liverpool appliance, M 10334). Juan-les-Pins: Editions A.P.D.C.A.
- Grant, Michael, 1980. *The Etruscans*, 240–241, fig. London: Weidenfeld and Nicholson.
- Hockley, A. 1858: 121 (M 10334 only). Waite (1885a: 508) gives this incorrect reference: “Mr. Hockley on Mechanical Dentistry.” *Quarterly Journal of Dental Science* (April 1857–January 1858) I: 393–403.
- Hoffmann-Axthelm, W. 1970. “The History of Tooth Replacement,” *Dental Science and Research* 8 (9): 81–87, fig. 4.
- Hoffmann-Axthelm, W. 1985. *Die Geschichte der Zahnheilkunde*, second edition, 78–79, fig. 64. Berlin: Quintessenz Verlags-GmbH.
- Lloyd-Morgan, Glenys and S.P. Girardon, 1988. “The Etruscan Collection,” in M. Gibson and S. M. Wright, eds, *Joseph Mayer of Liverpool, 1803–1886*, 77–85. London: Society of Antiquaries, Occasional Papers N.S. XI.

- Macnamara, E. 1973. *Everyday Life of the Etruscans*, 170, fig. 104. London: Batsford.
- Mayer, Joseph, 1852. *Egyptian Museum: No. VIII, Colquitt-Street, Liverpool*. 1852: 20, no. 222A (April 15). (Booklet/guide to Mayer's museum in Colquitt Street, Liverpool. Archival copy in the World Museum Liverpool is annotated with inventory numbers by Charles T. Gatty.)
- Moorehead, T. F. 1983. "Chirurgia, Medicina, Dentifricium," *British Dental Journal* 154 (10): 340–342 (May 21): 341, fig.
- Perine, G. H. 1883. "A History of Dentistry from the Earliest Period to the Present Time," *New England Journal of Dentistry* 2 (6): 161–166, 199–205, 269–273, 343–345 (descriptions inaccurate).
- Purand, T. J., Jr., 1857–1858. "Dental Memoranda," *Quarterly Journal of Dental Science* I: 63–65, 201–204, 342–346, 460–463; II: 121–123, 242–243, 353–355, 490–493; see 1857: 63.
- Waite, W. H. 1885. "Association Intelligence, Western Counties' Branch" (Letter; Discussion), *Journal of the British Dental Association* (now *British Dental Journal*) 6: 316–317; 499–512, and 2 figures.
- Weinberger, B. H. 1981 (1948). *An Introduction to the History of Dentistry*, Volume 1, 79. Saint Louis, MO: C. V. Mosby, 1948. Special edition by Gryphon Editions in Birmingham, Alabama, 1981.
- Scarborough, J. 1969. *Roman Medicine*, pl. 8. Ithaca, NY: Cornell University Press.
- Sterpellone, L. 2002. *La medicina etrusca. Demoiaatria di un'antica Civiltà*, 139. Noceto: Edizioni Esemblieme.
- Tabanelli, M. N. 1962. *La Medicina nel Mondo degli Etruschi*, pl. 51. Florence: Leo S. Olshki.

Numerous other popular and general publications mention or illustrate (often from the nineteenth-century drawing) one or both appliances (for instance, Recke and Wamser-Krasznai 2008: 139 fig. 58). The Wellcome Collection of the Science Museum, London, holds copies of these prostheses, said to have been acquired in 1915.

For additional discussion of Etruscan gold dental appliances, see:

- Becker, M. J. 1999a. "Etruscan Gold Dental Appliances: Three Newly 'Discovered' Examples in America," *American Journal of Archaeology* 103: 103–111.
- Becker, M. J. 1999b. "The Valsiarosa Gold Dental Appliance: Etruscan Origins for Dental Prostheses," *Etruscan Studies* 6: 43–73.
- Turfa, J. M. in press, 2016. *Catalogue of the Etruscan Collection in the World Museum Liverpool*, ed. Georgina Muskett, entries no. J29 and J30.
- Turfa, J. M. and M. J. Becker 2013. "Chapter 47. Health and Medicine in Etruria," in J. M. Turfa, ed., *The Etruscan World*, 871 figs 47.8 and 47.9. London: Routledge.

13 Liverpool I (Cat. no. M. 10334)

Present location: World Museum Liverpool, Mayer Collection, inventory no. M 10334.

Type: band with two false teeth, which survive, each held in place with one rivet. Origin: unknown, but surely from an excavation before or during 1857.



Figures 5.13a1–a2 Buccal views of appliance.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

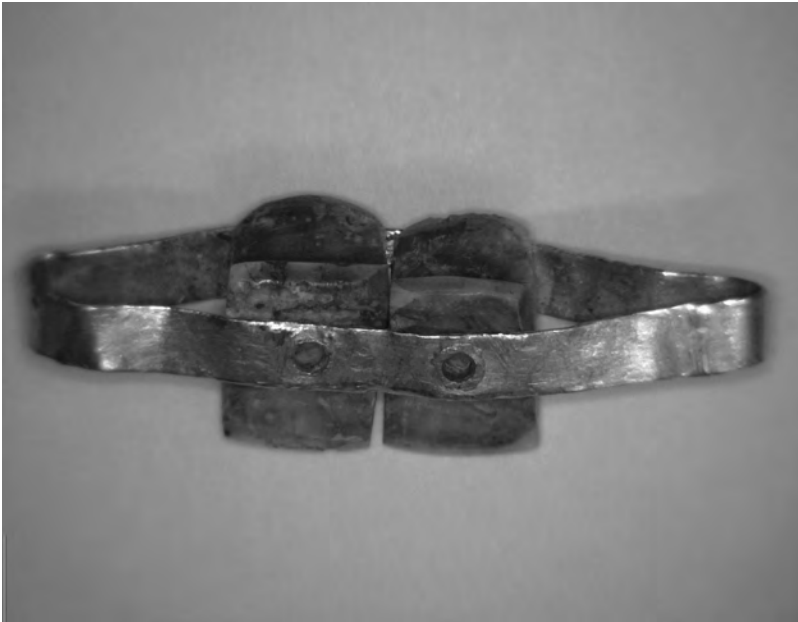


Figure 5.13b Lingual view.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

Jaw part: maxillary 2I–I2.

Date: not known.

Sex: female?

Dimensions: L. 30.9, W. 3.0–3.1 mm. Rivets: L. 7.0 mm, diam. 1.1 mm.



Figure 5.13c View of trimmed bases of inserted teeth.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

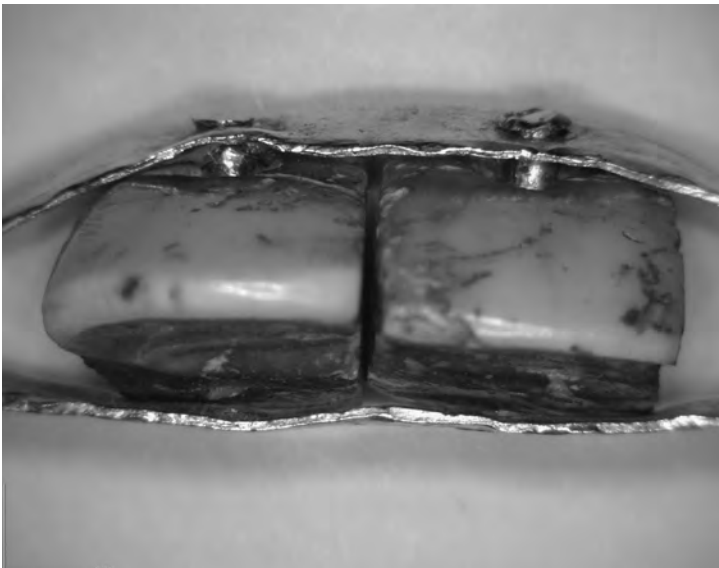
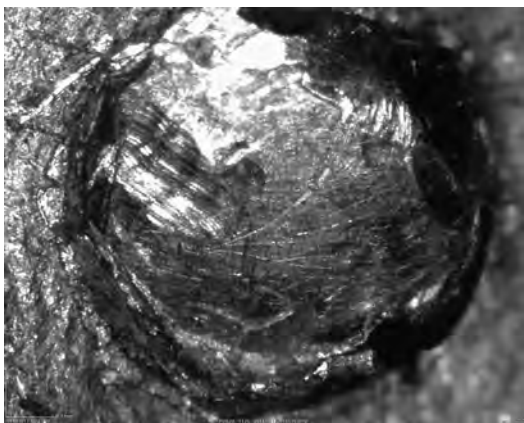
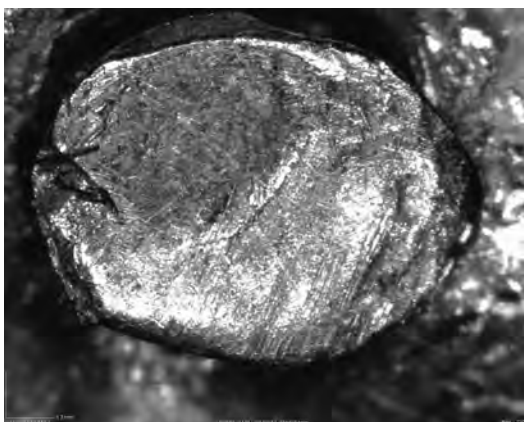
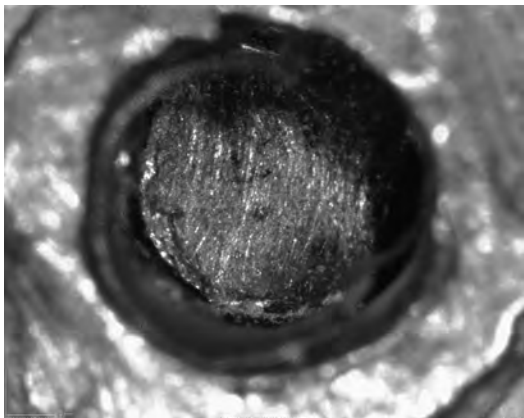


Figure 5.13d View of cutting edges of inserted teeth.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.



Figures 5.13e–g Details of the preserved three rivets.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

Description: This single band that surrounded all four incisors is 0.3mm thick. Since the location of a weld cannot be visually identified it is possible that this seamless oval was formed from a piece of gold that was expanded to shape. The appliance has two cut-down human central incisors riveted into the center. These may have been the teeth removed from the owner. The open ends of the band were fitted over the owner's lateral incisors. The rivets holding the replaced teeth are 7.0 mm long and have heads peened so close to the surface of the band that in some locations they have pulled free.

References (main dental references for identification; see also preceding discussion)

See Turfa in press for additional nineteenth-century catalogues etc.

- Bliquez 1996: 2650–2651, “K,” fig. 17 upper.
 Bonfante 1986: fig. VIII–28 (on p. 251: shown inverted).
 Cohen 1959: fig. 1.
 Emptoz 1987: 554–555, fig. 15 (no. XVII).
 Hockley 1858: 121: a mistaken reference, see below.
 Hoffmann-Axthelm 1970: 83, fig. 4 (upper).
 Hoffmann-Axthelm 1985: 78–79, fig. 64 (upper).
 Johnstone 1932a: 132–134, pl. 94 (17).
 Johnstone 1932b: 448–449.
 Macnamara 1973: 170, fig. 104.
 Mayer 1852: 20 no. 222A.
 Moorehead 1983: 341, fig.
 Naso 2011: 151 no. 12.
 Purland 1857–1858: 63.
 Scarborough 1969: pl. 8 (shown upside down).
 Tabanelli 1962: pl. 51.
 Waite 1885a: 316–317, 507–512, fig.

Description

Dental prosthesis made of gold sheet and rivets, covering the four upper incisors of a human jaw, preserves two central replacement teeth (the anchoring teeth are missing). The two “false teeth,” held in place with one rivet each, are cut down from two human teeth, probably recycled teeth originally removed from this person (Johnstone 1932a: 448–449). They were heavy, robust teeth, with traces of large pulp cavities, and the back surfaces seem eroded or stained from burial (front surfaces are shiny white), making them resemble the teeth of a larger mammal. Recent examination (2012) by experts at the Liverpool Museum has affirmed that they are human. The replacement incisors are sharply trimmed with straight vertical edges, and are notched at the top to create a ledge that would have been seated against the gums. The band is a narrow strip of sheet gold worked to fit

closely to the teeth, and band and rivets were carefully hammered and smoothed, to be worn comfortably. Small size of teeth implies a female owner.

Analysis

This band appears to have enclosed the spaces for the four maxillary incisors. The lateral incisors, on which this bridge was fixed, are not present. The space for the central incisors is occupied by two false “teeth” cut down from the incisors (almost certainly) of the human wearer, not from “some hard ivory” as Waite (1885a: 316) suggested. Both false teeth are notched at their uppermost surface (above the crown; see Figure 5.13c), suggesting that they were placed over the gums. The lateral edges appear to have been filed to ensure a close fit between the false teeth and the adjacent anchor teeth, (not extant). The pulp cavities of the crowns of these teeth are now exposed by the trimming, which created a ledge resting against the gum.

The length across the oval of the Liverpool I band is 3.09cm. The band has been effectively fashioned to form a seamless gold ribbon 0.3 mm thick, formed by expanding a single piece of gold. (No joint or alteration in the surface of the band could be detected in a series of examinations under magnification, confirmed by Turfa and staff at Liverpool Museums including Drs. Georgina Muskett and Chrissy Partheni, 5/31/2016. In May, 2016, on the lingual side of the proper right tooth there were visible several very short, shallow, parallel diagonal scratches that continue onto the gold band: it was not possible to determine if they resulted from ancient activity or nineteenth-century cleaning.) The band is 3.0 mm wide at the left side, as seen from the rear, and 3.1 mm wide at the right side, and molded to fit closely to both sides of the teeth. Each “tooth” has been drilled (B-L) and fixed within the band by a single rivet of gold, both of which are 7.0 mm long. Each rivet is estimated to be 1.1 mm in diameter, and passes through the tooth and band on each side. Both rivets have been hammered (peened) flush with the outer surfaces of the band. On the exterior or labial surface there is a slight bulge where the head of the rivet overlaps the band, the left being an oval 1.9 by 1.2 mm and the right almost circular, with diameters of 1.95 by 2.0 mm. On the lingual surface, where both rivets have been peened nearly flat, with less material overlapping the band, both have pulled free from the band. Dimensions of the cut-down replacement teeth are listed below.

Table 5.4 Dimensions of the false teeth in Liverpool I (in millimeters)

	<i>Left</i>	<i>Right</i>
B-L depth at the top	7.5	7.2
Height along buccal edge	11.4	11.0
M-D width at top	6.9	6.2
M-D width at cutting edge	6.5	7.0

14 Liverpool II



Figures 5.14a1–a2 Buccal views of appliance

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.



Figure 5.14b Lingual view.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

Present location: World Museum Liverpool, Mayer Collection, inventory no. M 10335.

Type: band with 2 false teeth (now missing) riveted in place.

Origin: unknown, but surely from an excavation before or during 1857.



Figure 5.14c View of roots and rivets.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

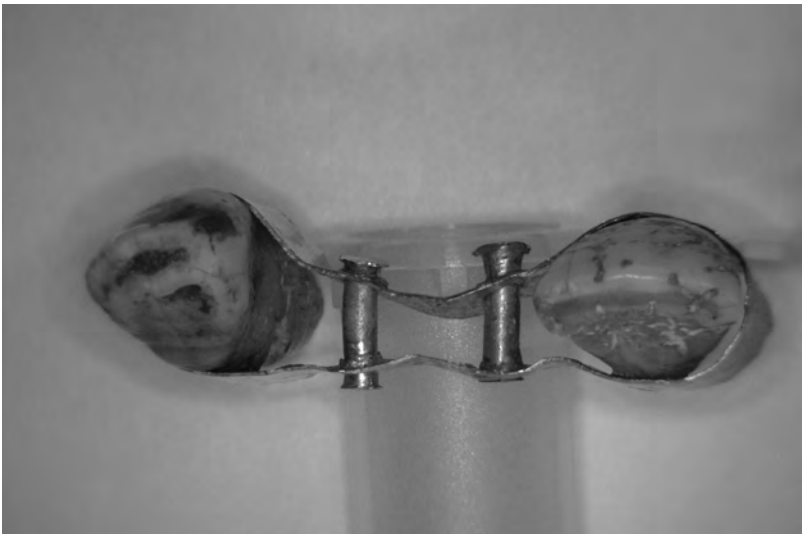


Figure 5.14d View of rivets.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

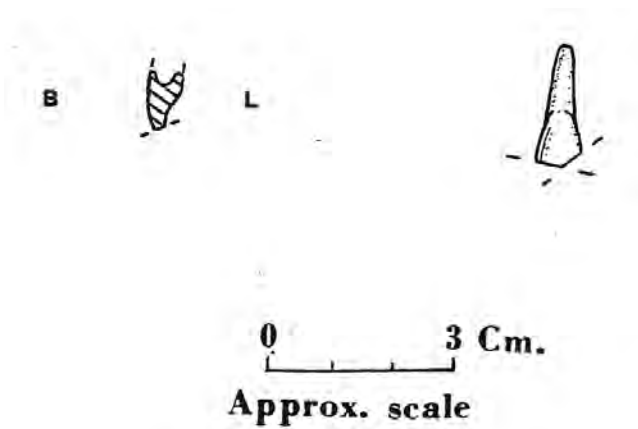


Figure 5.14e Diagram of cutting surface of rooted tooth.

Source: sketch by M. J. Becker.

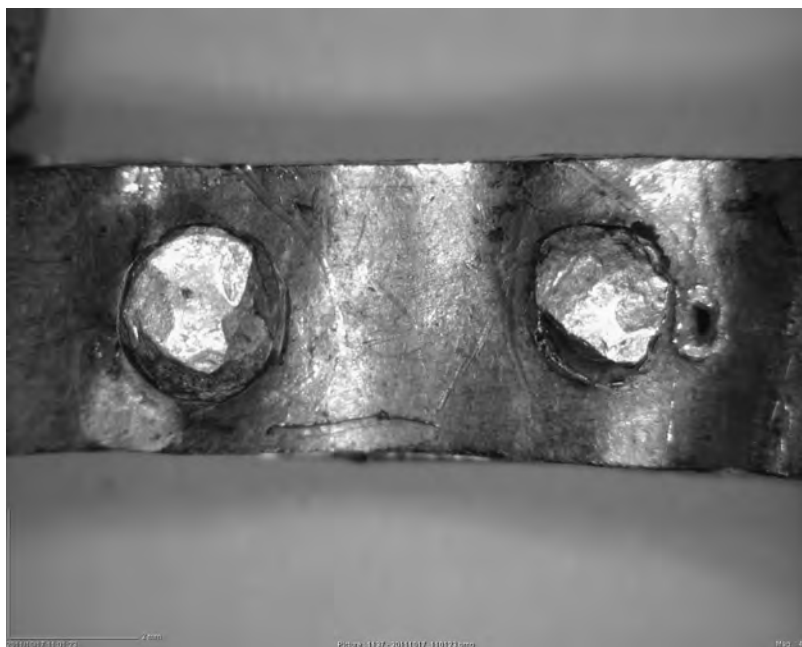


Figure 5.14f View of rivet surface.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.



Figure 5.14g View of rivet surface.

Source: courtesy of the National Museums Liverpool and Georgina Muskett, also by Margarita Gleba.

Jaw part: maxilla, 2I–I2.

Date: not known.

Sex: female?, age 45+ years.

Dimensions: L. 28.2, W. 4.9–5.2 mm.

Description: this single band, once surrounding all four incisors, is ca 0.2 mm thick, considerably more gracile than the band of Liverpool I. Construction appears much the same, with two rivets where the centers of the central incisors would have been. The replacements, however, have not survived. Since we do not know the tomb conditions that would have influenced survival of artifacts, we cannot know if human teeth or some other organic materials had been used in forming the replacements. The rivets measured 6.0 and 6.2 mm long and have been peened nearly flat as with Liverpool I.

References

- Bliquez 1996: 2650, “J,” fig. 17 lower.
Bonfante 1986: fig. VIII–28 (p. 251, shown inverted).
Cohen 1959: fig. 1.

- Hoffmann-Axthelm 1970: 83, fig. 4 (lower).
Hoffmann-Axthelm 1985: 78–79, fig. 64 (lower) (text in 1981 edition was unchanged).
Johnstone 1932a: 132–134, pl. 94 (18).
Johnstone 1932b: 448–449.
Macnamara 1973: 160, fig. 104.
Moorehead 1983: 341, fig.
Naso 2011: 151 no. 11.
Purland 1857: 63.
Scarborough 1969: fig. 8.
Tabanelli 1962: 94, pl. 51.
Waite 1885a: 316–317, 507–512, fig.

Description

Dental prosthesis made of gold sheet, surrounding four teeth of a human upper jaw, has spaces for two false teeth (missing) and currently preserves in the position at either end, two complete human teeth, although they may not have been the original anchoring teeth. With roots preserved, they resemble two canines but are actually, according to Becker and recent examination by Cynthia Reed (October 18, 2011), a left central incisor and a right canine (of the upper jaw). The original central incisors were replaced by this bridge, but the teeth currently installed are not original. Two rivets remain to show the central teeth were replacements. Becker and goldsmith Reed have noted a pin-hole punched through the gold band, with its rough edges smoothed, as if there had been a slight mistake in drilling and fitting the replacement tooth, or perhaps an adjustment to the appliance after it had been made and worn. Small size of teeth clearly implies a female owner. The rivets are slightly larger than those used in the other appliance. As with Liverpool I, there is no visible sign of the joining of two ends of the band: it seems possible that the band was worked from a pierced solid of gold into an oval ribbon-like band, using the techniques common in making gold and silver hollow wares.

Analysis

The Liverpool II dental appliance (examined by Becker when earlier on display in Case 44 at the Liverpool Public Museums, now Liverpool World Museum), is a band forming an elongate (3 cm long) oval 5 mm wide, which bridged four maxillary spaces. Most likely this example conforms to the general pattern of spanning the four maxillary incisors, with the present teeth placed within the appliance at a more recent date. The two false teeth, originally held in place by rivets, now are missing. Neither of the present human “anchor” teeth now in this appliance are securely fixed into the positions they occupy, at the extreme ends of the oval gold band. It is not at all certain that these are the original anchor teeth. Both appear to be from a female, as indicated by their size. The appliance originally may not have been well seated, and quite possibly, if these are indeed the original teeth, this was a removable cosmetic appliance, a possibility reinforced by the extremely high

placement of the band on the teeth. The anchor teeth appear to be a left central incisor and a right canine, indicating that both right incisors were bridged.

The following comments assume that the teeth are the original ones (Figure 5.14e). The canine shows slight shoveling exhibited along the posterior mesial margin. Shovel shaped canines now have been found at several central Italian sites (e.g. Osteria del'Osa: Becker and Salvadei 1992) and this tooth is well within the expected range of configurations for canines. The right canine is 22.8 mm long, and has a maximum breadth of 7.2 mm. The wear pattern suggests the loss of one or more opposing teeth. The replacement central incisors would have a total M-D length of less than 12.5 mm. This suggests that the tooth now in place at the left is not original to this appliance.

The wide left central incisor is 20.9 mm long and 8.7 mm broad along the articular edge and is clearly shoveled. The degree of shoveling (2 on a 0–4 scale) is not great, which is also common with central incisors as compared with the more pronounced shoveling usually found on lateral incisors in central Italy (Becker and Salvadei 1992). Central incisor shape and width also are genetic characteristics.

This band, at approximately 5 mm wide (range 4.5 to 5.2), is wider than Liverpool I but has been made by the same technique, suggesting a related time and/or place of origin. The ribbon is particularly thin at c.0.2 mm and has been formed into a seamless band. The longest diameter of this oval band measured 28.2 mm. The width of the band varies from 4.9 mm in the center to 5.2 mm at the ends. Two holes were made (punched?) in the rear of the band and the rivets were driven through from that surface. The rivets were flattened at the rear, as in the other Liverpool example, and peened over on the anterior (buccal) surface. The left rivet is 6.2 mm long and has a head peened “flat” to form a circle some 2.9 mm in diameter. The right rivet is only 6.0 mm long and has a head peened into an oval some 2.2 by 2.0 mm.

An interesting, and unexpected, feature of this appliance is the presence of a small hole on the anterior surface of this band, lateral to the rivet on the right side and less than 0.5 mm from the edge, as seen from the inside of the band. This pin-size hole appears to have been punched from the rear (the interior) of the gold band, with the raised surface around the hole where it exits on the anterior surface then being peened over. If this was meant to add a small stabilizing pin to better secure the false tooth one would expect that it would have entered from the exterior. The origins and function of this very tiny hole remain unknown, but it appears to be a feature unique to this appliance.

15 Valsiarosa

The archaeological context from which the Valsiarosa appliance derives is the best documented of all the known examples. The specific tomb location has been identified and published. Only the Satricum appliance is nearly as well documented.

Present location: Museo Archeologico dell'Agro Falisco, in Forte Sangallo, Civita Castellana. (Formerly Museo Nazionale di Villa Giulia inv. no. 1515.)

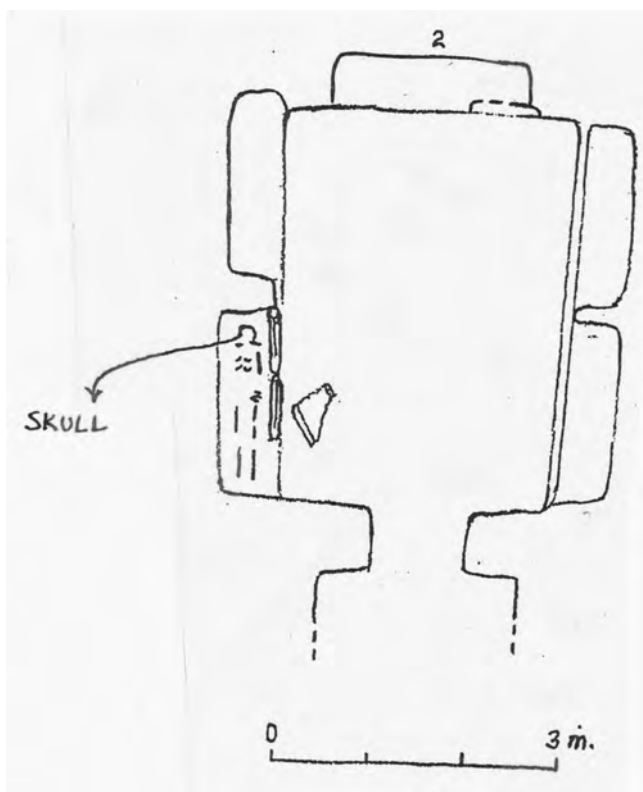


Figure 5.15a Valsiarosa Tomb 20 CIX (81). Plan, as rectified from Cozza and Pasqui (1981: 188, 201). They indicate that the chamber measures 2.50 m by 4.20 m. The arrow on the plan leading from the skull points to the text “Teschio colla dentiera”—“Skull with denture.”

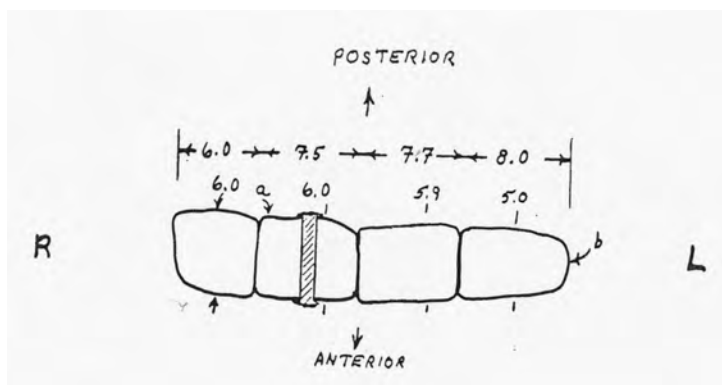
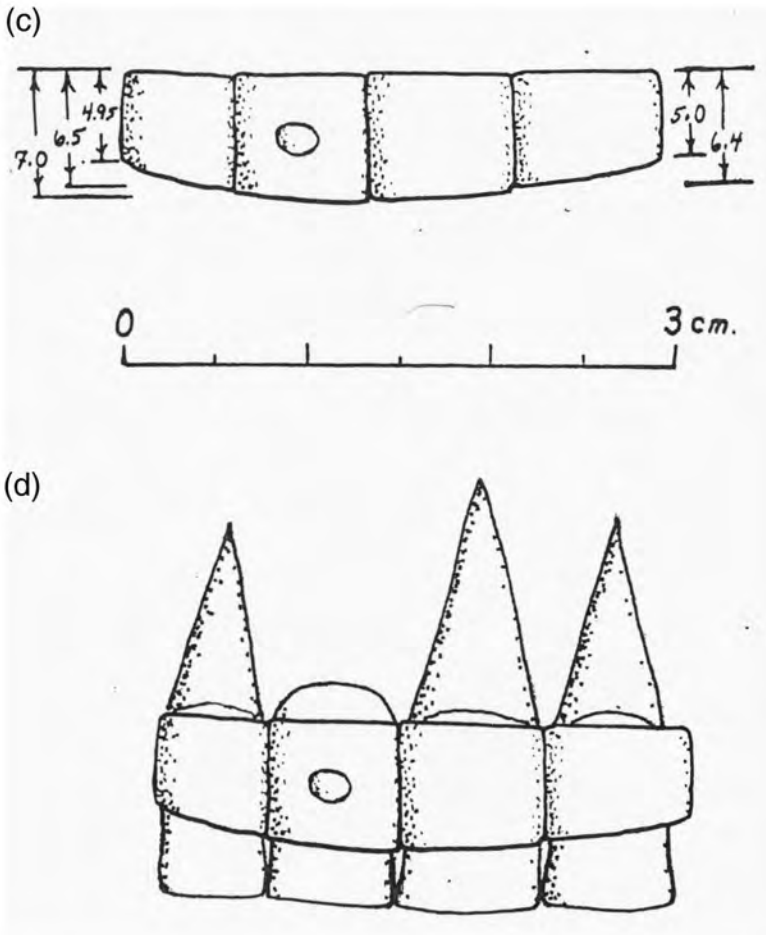


Figure 5.15b Drawing of the superior view, by Becker.



Figures 5.15c–d (c) Drawing of the anterior view, by Becker; (d) view with teeth, roots in place, drawing by Becker.

Type: four chambers (for four maxillary teeth), which appear to be welded loops, with one rivet at II (cf. Copenhagen). No teeth survive. (Often confused with Populonia no. 6.)

Origin: Valsiarosa Necropolis about 0.5 km from modern Civita Castellana (ancient *Falerii Veteres*), “Tomba a camera CIX” = Tomb 20, a chamber tomb with *loculi* for bodies in the walls.

Jaw part: originally fitted to a maxillary 2I–I2 (spanning four spaces), although not so displayed today.

Date: fourth–third century BCE (based on Waarsenburg’s evaluation of the grave goods).

Sex: female ?

Dimensions: L. 38, W. 5.0–7.0 mm.

Description: The Valsiarosa appliance sur rounded all four maxillary incisors, and provided a modified or false tooth for the right central incisor. No teeth survived, including any replacement tooth. The gold rivet, flattened at both ends, is now approximately 1 mm in diameter and 6.9 mm in length. The rectangular chamber for the left central incisor may reflect the shape of the actual tooth, but the buccal–lingual flattening of all four “rings” reflects post-mortem activities. At some time after its recovery this appliance had been mounted in a modern human skull for display.

References

- Baggieri 1998: 68 (an error-filled prelude to Baggieri 1999).
 Baggieri 1999: 39–40 (repeats errors in Baggieri 1998).
 Baggieri 2001: 327–328, pl. XXXVIII C and E, graph 3.
 Baggieri and Allegrezza 1994: 4–5. Includes a useful photograph of appliance as seen from below, but printed in reverse!
 Baggieri and De Lucia 1993: 15 (briefly noted).
 Becker 1994d.
 Becker 1999b: 45, 49–59, figs 2A–2C, 4A–6.
 Bliquez 1996: 2651, “L,” figs 18 and 19.
 Bobbio 1958: figs 9–10 (data taken from the modern copy in Paris (see Appendix II) as shown in Rivault 1953: figs 2–3), and fig. 12 (taken from Casotti 1947, who provides excellent views of this copy).
 Boissier 1927: 27, fig. 2.
 Cali 1901: 11.
 Casotti 1927: 627, fig. 4 (illustrating an edentulous skull and mandible in a plate with the appliance on the left side (cf. Dall’Osso 1907), plus a stylized, but fairly accurate view (Casotti’s Form B)).
 Casotti 1947: 668, pl. I. Waarsenburg (1991a: n. 23) points out that five of Casotti’s appliances (A, I, P, Q, R) are identical; see Bliquez 1996, and the text below.
 Casotti 1957: 105 and fig. 6.
 Cavenago 1933: fig. 2 (copy of Deneffe 1899).
 Cozza and Pasqui 1981: 187–201 (archaeological data).
 Dall’Osso 1907: fig. (illustrates this piece, incorrectly identified, in a skull resting on a plate).
 De Lucia Brolli 1991: 35, 36 fig. 10.
 Deneffe 1899: 59, 69–72, pl. 2 and 3. Plate 2 illustrates a copy of this appliance, mounted in a skull that has all of its teeth, also printed in reverse. Plate 3 shows what may be an original view of this appliance, seen as an unnumbered item in the upper left.
 De Vecchis 1929: 18f.
 Dunn 1894: 5 (Waarsenburg (1991a: 244) describes these drawings as “good”).
 Emptoz 1987: 553, figs 12, 14 (no. XIV, from “Falcini Veteres” [sic]; also XVI).
 Guerini 1894b: figs 1–2.
 Guerini 1909: 70–71, fig. 15 (probably a copy).

- Laviosa *et al.* 1993: 130, figs V2 and V3 (these two color plates, from the Soprintendenza Archeologica per l'Etruria Meridionale at the Villa Giulia Museum, are probably the two best now in print).
- Menconi and Fornaciari 1985: 95–97, figs 5, 6.
- Michelsoni 1976 (illustrates the Praeneste appliance, but says it comes from Civita Castellana).
- Naso 2011: 150 no. 2.
- Platschick 1904–1905: 239, figs 3–4.
- Quintarelli 1946: 104, fig. 2 (skull on plate plus view of appliance; cf Casotti 1927).
- Rath 1958: fig. 4. (illustration provided by “Foto Anchora, Roma,” shows only what appears to be a fragment of the mandible).
- Rivault 1953: figs 2–3 (taken from modern copies in Paris).
- Sterpellone and Elsheikh 1990: 138, fig. 67.
- Strömgren 1919: fig. 4 (skull on plate).
- Sudhoff 1926: 81–82, fig. 45.
- Tabanelli 1962: 93 (from Dall’Osso 1907?).
- Torino *et al.* 1997: 540–542.
- Waarsenburg 1991a: figs 1–2.
- Weege 1913: 371.
- Weinberger 1948: 125, fig. 52. Text repeats Casotti (1947), but illustration, courtesy of Curt Proskauer, said to have been taken at the Villa Giulia in Rome, appears to be the Poggio Gaiella example in Florence.

History and date of the Valsiarosa appliance

The data published by Cozza and Pasqui (1981: 187) on excavations at Valsiarosa provide the most extensive archaeological record for the recovery of any ancient dental appliance. They include a map of the excavations with Tomb 20 noted. The grave goods (*corredo*) were identified by the number CIX, and the “Catalogo Pasqui” assigned number 81. The full series of numbers for this tomb at Valsiarosa often appears as 20CIX(81). The lack of human skeletal data from these tombs severely restricts our ability to interpret the details of their use.

Cozza and Pasqui (1981: 187) indicate that the “*Proprietario della tomba*” was “*Mancinelli (scavatore)*,” the “year of excavation” 1888 (*anno di scavo* 9/7/1888/a). This information was not included elsewhere (“*Non è descritta in*” *Notizie degli Scavi* for 1887 “*ne dal catalogo Magliulo [10/3/1886–10/7/1890]*.” Baglieri and De Lucia (1993) provide background on Francesco Mancinelli Scotti, who conducted numerous excavations for Italian and foreign sponsors in the territories (necropoleis) of Falerii and Narce—see Mancinelli Scotti 1917; for background on the Faliscan culture, see De Lucia Brolli and Tabolli 2013.

The site plan (Cozza and Pasqui 1981:188) locates Tomb 20: CIX(81) as the third in a line of eight large tombs running southwest to northeast along an ancient road, “*fra il terreno Brunelli e la strada comunale*” (“between the Brunelli land and the municipal road”) (1981: 201). Entered from the southeast, it is trapezoidal in plan (2.50 by 4.20 m; see Figure 5.15a). *Loculi* (s. = *loculus*),

body-length niches, were cut horizontally into the walls; two more *loculi* were set above one another in the back wall of the tomb. All were sealed with large pan tiles or flat stone slabs, a design common in the Faliscan region by the fourth century BCE.

The published Tomb 20 plan (Figure 5.15a) includes the *loculi*, but only one has the position of the skeleton sketched on it. A line drawn from the skull of this skeleton leads to the left where it ends in an arrow pointing to a text that reads “*Teschio colla dentiera*” (“skull with the denture”). In a note Cozza and Pasqui (1981: 201) state that this was the “*Teschio di scheletro che conserva nella mascella inferiore la legatura in lamina d’oro di quattro molari.*” (“Skull of skeleton that preserves in the lower jaw the binding in gold sheet of four molars.”) They also cite page 75 of Della Seta’s Villa Giulia Museum catalogue, which provides the catalogue number of “1515” for this appliance (see Guerini 1894b: figs 1–2).

A kylix (cup, probably Athenian) was mistakenly assigned to this tomb: “*Il Pasqui aggiunge una kylix che invece, secondo la sua Relazione, fa parte della tomba di Cella CIX bis*” (Cozza and Pasqui 1981: 143, 201). According to Waarsenburg (1991a), inv. no. 1516 was an Attic skyphos (cup) from “tomb CIX^{bis}.” A partial list of the goods recovered from Tomb 16: CXI(82) is in Cozza and Pasqui 1981: 199–200; see also Waarsenburg 1991a.

Tomb 20: contents and dating

Waarsenburg (1991a: n. 25) extracts an excellent description of the tomb from a document that he diligently sought out in Rome (*Archivio Centrale 1888*). It indicates that the tomb had been looted prior to the official excavation, but a large number of remaining items were recovered, including a “*dentiera a tre denti in oro, oggetto rarissimo*” (“dental appliance of three teeth in gold, a very rare object”). Waarsenburg (1991a) suggests that “*tre denti*” refers to the number of actual teeth then found in place in the prosthesis, not the number of spaces bridged, which clearly is four. Waarsenburg also points out that the appliance comes from Tomb 81, while the artifacts are noted as from Tomb 82. Why Cali (1901:11) says that item 1515 came from Tomb 84 of Faleri Veteres is unknown.

The date for the artifacts in Tomb 82 is based on their description, as reconstructed from the record by Waarsenburg (1991a: 245 with n. 25). He notes especially “a thymiaterion with lion’s paws for stands and plastic doves along the plate, an Etrusco-Campanian jug and a bronze mirror with engraved Lasa.” The first items might be fourth–third century in style, but the Lasa mirror is surely a third-century creation, although we cannot be sure that it belonged to the woman with the dental appliance rather than one of her relatives. Although the Valsiarosa appliance has one of the best archaeological records associated with it, the ability to date the tomb to any specific decade through ceramics was lost when the tomb was robbed. Tabanelli and Casotti both suggested a fourth-century date, also used by Bliquez (1996).

The association between the “*dentiera*” and the skull and mandible in which it is displayed is not known. We strongly doubt that the skull comes from any of

the tombs noted by Cozza and Pasqui (1981). Very likely a convenient skull was selected in which to mount this appliance in order to create an attractive display.

This appliance was acquired by the Villa Giulia Museum soon after being excavated in 1888. Dennis (1888–1889: 164) notes that a gold dental appliance was present in a skull displayed among the Faliscan tomb groups at the Villa Giulia Museum when it opened in 1889 (see also Appendix I.B). Deneffe describes them there (1899: 69) when he says that a gentleman by the name of Barnabei (surely F. Barnabei), described as the conservator of the Villa Giulia Museum, gave him all possible help. Deneffe seems to have made a copy of the appliance. Platschick described it as at the Villa Giulia, as it was when Casotti (first?) saw it. The artifacts from Tomb 82/CXI were returned from Rome to Civita Castellana in 1955 with other parts of the Faliscan collection (Waarsenburg 1991a: 245–246). The skull and appliance followed in 1967 and have been on display in the Museo Archeologico since 1977.

Waarsenburg, following Casotti (1947), reviews the early, confusing literature relating to the Valsiarosa piece and its “multiple identities.” Citing Weege (1913: 371), Della Seta (1918: 75) and Leopold (1923: 208), Waarsenburg (1991a) attempts to clarify the history of the many “fabricated examples” (cf. Bliquez 1996). According to him, Cali (1901) also says these are the skull and mandible that he saw at the Villa Giulia Museum “*tra gli oggetti della tomba 84*” from *Falerii Veteres*.

Description of appliance and correction of past discrepancies

The Valsiarosa appliance, now in the Museo Archeologico dell’Agro Falisco in Civita Castellana, still retains the Villa Giulia inventory number “1515.” Photographic negatives of the Valsiarosa appliance are on file at the Villa Giulia Museum, Rome (negative nos. 4044 and 4045, taken during the 1950s; see also 88-567 through 88-572). In August of 1993, with the aid of Dr. Maria Anna De Lucia Brolli (Director of the Museo Archeologico dell’Agro Falisco), Becker was granted permission to remove the skull from its display case and make a detailed study. Direct observation immediately resolved most of the confusion that exists in the literature, including the lack of any relationship between the Valsiarosa appliance and the skull in which it is displayed.

Examination of the shape of the individual rings of this appliance confirms that it had been designed to span the four maxillary incisors, with a rivet clearly demonstrating that a false tooth had replaced the I1 (upper left central incisor). The squared shape of the “loop” or “ring” that is pierced by this rivet further confirms that a false tooth had occupied this space. The configuration of the loop for the living right central incisor (I1) of Valsiarosa reveals the delicate and careful shaping necessary to fit this gold band around a tooth that has a complex profile. The portion of the gold ring surrounding the crown of the I1 rapidly shifts from a nearly round shape near the gingival margin (gum line) to the chisel shape of the biting edge of this incisor. Although this natural tooth is now missing, as are all the natural teeth once surrounded by this appliance, the characteristic shape and

size of a central incisor is retained in this loop, providing conclusive evidence for the original location in the owner's *upper* jaw. The relative size plus the precise incisor-shaped configurations of the four units in the appliance clearly indicate that it surrounded all four incisors. Furthermore, the shape of the "false" tooth so closely matches that of the right central incisor that we conclude that it is the original tooth, recycled after evulsion (see Becker 1995c). The absolute size of these teeth, as indicated by the size of the once tightly fitting loops, is not clearly diagnostic of the sex of this person. However, in each case where sex has been determined the wearers of such appliances have been female. In these examples, where high status individuals are most likely the owners, greater stature and general robusticity might be expected.

The Valsiarosa appliance is displayed in the skull of a person who had all of his teeth at death but had lost all of the dental crowns and most of the roots post-mortem. Constructed to fit over the maxillary 2I–I2, the appliance as displayed rests in the left side of the face (area from 2M–2PM), and is attached to the mandible by wax or a similar material probably used in the conservation process in the 1880s. This display has led naive observers to assume that this was the original position of the bridge and assume that this is the skull for which it was intended. Menconi and Fornaciari (1985: 95, fig. 5) attribute a lesion in the *display* mandible as resulting from the wearing of this appliance during life.

The waxy material holding this appliance in place also fills several of the dental sockets, forming a coating that appears on superficial inspection to be like a second sheet of gold. This wax, or glue, reduced the opportunity to gather data from this appliance. Later in 1993 the Valsiarosa appliance was sent on loan to an exhibition in Rome (see Laviosa *et al.* 1993: 130). During its time away the material attaching it to the mandible was removed and both mandible and appliance were at least partially cleaned for photographing and study (Baggieri and Allegrezza 1994: 5; Baggieri 1998: 68; Baggieri 1999: 39). Baggieri (2001: 327) states that the weight of the appliance is 1.96 grams, apparently after cleaning.

The actual construction technique used in making this piece continues to be uncertain. The four units were fashioned as individual loops that then were welded to each other (not as a single long loop or braced band). Each ring now is roughly rectangular when viewed from "above" (looking at the superior margin, or into the appliance from the alveolus). The loop for the false tooth (the 1I) clearly has the most rectangular configuration. Thus the rectangularity of the superior margins of each of these loops reflects the configuration of the appliance when it was made, and affixed into the owner's mouth.

The original superior margin of the Valsiarosa appliance, which would have been close to the gums when worn, has been subjected to considerable bending and distortion by being battered by the maxilla of the display skull. The actual edges on both sides of each loop are distorted, bent and torn. The total length of the Valsiarosa appliance was approximately 29.2 mm, although it now has a somewhat greater maximum length (particularly along the exposed edge) due to distortion at the right lingual corner by the I2 and a flattening of the ring for the 2I.

Table 5.5 Tooth measurements (socket diameters, at centers), in millimeters

Tooth	A-P	M-D
2I	6.0	6.0
1I	6.0	7.5
I1	5.9	7.7
I2	5.0	8.0 (greatest distortion)

The peculiar “dihedral” form of the anterior view of this appliance appears to be unique. This shape is not evident in Waarsenburg’s illustrations (1991a), but it can be detected in the color plates provided by Laviosa *et al.* (1993). From an anterior view (see Figure 5.15d) one may note the differing heights of the appliance where it passes between the teeth. Thus at the far right margin of the appliance, where it separated the I2 and the right canine, the band is only a bit more than 4.9 mm high. Where the ring surrounding the I2 is attached to the I1, the height of the band is 6.5 mm. In the very center of the appliance, between the loops for the two central incisors, the height of the band is greatest (7.0 mm). The band narrows from the center down to 6.4 mm at a point between the 1I and the 2I, and ends at the distal side of the 2I with a height of only 5.0 mm. The thickness of the ribbon of the loop for the I2 is the only one that could be measured at that time due to a coating of glue, but it appears to be 0.2 mm thick.

The rivet pierces the anterior and posterior walls of the ring or chamber for the 1I, but the diameter of this pin could not be determined. Becker estimates a diameter of approximately 1.0 mm (cf. Menconi and Fornaciari 1985: 95). The loop pierced by this rivet has an anterior-posterior diameter of 5.9 mm at its inferior and superior margin. Since the rivet itself has been peened to a length of 6.9 mm, where it passes through the center of the ring, the false tooth must have been closer to 6.0 to 6.1 mm in diameter. Even accounting for the thickness of the heads on both ends of the rivet, plus the thickness of the loop on both sides, the false tooth at the point where the rivet passes through it clearly has a greater diameter than is seen at the inferior opening of the chamber, similar to Copenhagen. The rivet head on the lingual side has been peened to a rough oval about 1.9 by 2.1 mm in diameter. On the anterior surface the configuration of the head is nearly the same, with diameters of 2.0 by 1.8 mm.

Although we do not know what material was used for the false tooth in this appliance, only minimal functional use could be inferred. The soft gold used could bear only limited stress. The positioning of the appliance across all four incisors balances the decorative design, as seen in the Liverpool appliances, and was not an attempt to provide additional support by adding a loop around another tooth, as suggested by Baggieri and Alegrezza (1994: 5).

Analysis: the skull now associated with the “Valsiarosa” appliance

Two important publications contribute to our understanding of the archaeology and recent history of the Valsiarosa appliance. Francesco Mancinelli Scotti’s

nineteenth-century excavation data has been published by Cozza and Pasqui (1981). Waarsenburg (1991a) adds the archival records, but does not refer to Cozza and Pasqui (1981).

All of the photographs of the Valsiarosa appliance published prior to that of Baggieri and Allegrezza (1994: 4) show an edentulous skull and mandible, with the appliance resting between the left jaws, approximately in the location of 1M–C (see Waarsenburg 1991a; Laviosa *et al.* 1993). These illustrations plus all of the earlier descriptions suggest that Valsiarosa had been mounted in this skull soon after it was discovered, probably when it was first displayed at the Villa Giulia. The skull in which the appliance has been displayed is that of a male who died at about 40 years of age (Becker 1994d). All of his incisors were in place at the time of his death. Since none of these teeth had been lost, there would have been no need for this person to be fitted with a false incisor. As Baggieri and Allegrezza (1994) point out, the mandibular right second molar appears to be the only tooth lost before death; yet this observation does not prevent them from reaching the incongruous conclusion that this is the skull of the owner of the appliance.

In the skull now displayed, from a source yet to be determined, 29 of the 30 original teeth (two molars never erupted) were in place at death. The margins are in good condition, reflecting relatively good dental health. At least 13 fragmentary roots of these teeth remain in place (Baggieri and Allegrezza 1994). The partial survival of these dental roots reflects normal conditions within Etruscan tombs where the variably moist environments are conducive to hydration (water bonding) of bone. The bony portions of the skull in question have survived quite well, but the teeth have splintered (see Becker 1999d). Becker's age evaluation derives from the configurations of the cranial sutures as well as by the pattern of tooth loss. The Etruscan population from Tarquinia appears to have enjoyed generally good dental health, but molar loss due to developing dental caries commonly begins at about age 40–50 in that population (Becker 1993a). We infer that soon after its discovery this appliance was mounted in a skull from another context at Valsiarosa. Baggieri and Allegrezza (1994: 4) offer notes on pathologies associated with the skull, and recognize that the Valsiarosa prosthesis is a maxillary appliance, but Baggieri (1998: 68, 69) failed to recognize that this skull was not the one for which it was made.

Much of the confusion regarding the Valsiarosa prosthesis derives from earlier observers making fanciful evaluations from the original or from a number of poorly constructed replicas. The extent of confusion among these authors is addressed elsewhere (Becker 1994d; Becker 1999b: 45, 49–59, figs 2A–2C, 4A–6). These problems were exaggerated by reversing negatives or printing them upside down. One illustration commonly associated with the Valsiarosa appliance, the source of which remains unclear, depicts this skull resting in what looks like a dinner plate. Sudhoff (1926) shows the skull in this plate, but his print is probably a copy. This also is what Casotti (1927: fig. 4) illustrates. Casotti (1947: 674–675) says that this piece is about 30 mm long and 6 mm high, figures repeated by Weinberger (1948: 125) and Waarsenburg

(1991a: 243). In many respects the piece that Sudhoff and others depict resembles a model from which copies appear to have been *cast*, rather than fabricated using techniques like those employed in making the original. Casotti (1947: figs 4–6 and pl. II) provides illustrations of five “different” appliances, all of which appear to be created from the Valsiarosa example. Casotti realizes, from photographs, that at least five different identities had been created from this prosthesis, but then he creates at least two new examples in his own text (see Appendix I.B). Casotti’s problems multiplied over the years, as noted by Becker (1999b).

16 Teano

Present location: Institut für Geschichte der Medizin, Universitätsklinikum Charité, Humboldt-Universität zu Berlin.

Type: six “rings” plus an outer “jacket,” no rivets. No roots survive but two relatively intact tooth crowns plus portions of two other tooth crowns were in place in 1909 (Hoffmann-Axthelm 1970: fig. 5).

Origin: Hellenistic necropolis (Tomb 18) at the Fondo Gradavola (on the property of Signore Luigi Nobile) at Teano dei Sidicini (c.32 km northwest of Caserta), Province of Caserta in the Campagna. (Excavated February 1907.)

Jaw part: maxillary C–C (although generally said to be mandibular).

Date: fourth–third century BCE (Guerini 1909: 79, via C. Dall’Osso). Waarsenburg (1991a: n. 8) suggests c.300 BCE.

Sex: female (via tomb contents; confirmed by tooth size).

Dimensions: total dimensions: L. 47; W. 12; H. 4.5 mm (tapering at distal ends).

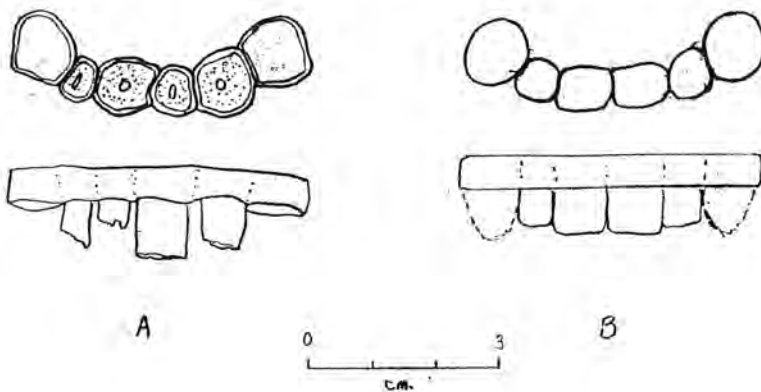


Figure 5.16 The Teano appliance, Becker’s reconstruction (inferred from Hoffmann-Axthelm 1970: fig. 5), four views. Photographs as illustrated by Rath (1958: fig. 2) are probably from the FEHR copy rather than the original find.

Description: constructed of six attached rings, no rivets, originally retaining portions of four teeth. Presumably fashioned to stabilize four loose incisors. The two end rings, empty when collected, presumably were looped over canines as anchors. The extant teeth, almost certainly those of the original wearer, have become much more fragmentary since 1910–1911. The unusual construction is unlike any other example. The six-ring welded construction is surrounded by a single gold-band “jacket.” The double layer measures approx. 0.2 mm thick. Later photographs reveal a premolar inserted into the left lateral incisor position.

References

- Bliquez 1996: 2653, “P,” figs 22–25.
 Casotti 1927: 628 (typographical error in pagination), fig. 5 (blurred front and top views).
 Dieck 1911: no. 114.
 Dieck 1934: VIII.
 Gàbrici 1910: cols. 44–45, 70, fig. 25.
 Guerini 1909: 78–79, figs 25–26 (figures reveal only three dental crowns in place).
 Hoffmann-Axthelm 1970: 83, fig. 5.
 Hoffmann-Axthelm 1985: 78–79, fig. 66 (two views); (Hoffmann-Axthelm 1981: fig. 66; shown as maxillary in all cases).
 Marz 2010.
 Naso 2011: 152, no. 15: cited as “*collocazione attuale sconosciuta.*”
 Rath 1958: 11, fig. 2 (photograph of a copy in Cologne).
 Schweitzer 1936 (catalogue, March).
 Sterpellone and Elsheikh 1990: 137, fig. 65.
 Strömberg 1919: 26, figs 5–6 (from Guerini 1909).
 Sudhoff 1926: 93, figs 53a and b.
 Waarsenburg 1991a: n. 5: no. 15, and n. 8.
 Weinberger 1948: 146–147, fig. 51.

History and context

“Teaunum of the Sidicini” (an Oscan/Italic tribe of central Italy) is situated north-northwest of Naples and 32 km northwest of the town of Caserta in the ancient region of Campania. Teaunum, situated on the fertile slopes of an extinct volcano, was a waypoint on the Via Latina. As the influence of Etruscan culture, strong in Campania through the Archaic period (c.650–480 BCE), waned, the Oscan-speaking Sidicini came under Roman domination c.334 BCE.

Portions of a “Greek-Roman” necropolis near Teano were excavated in February of 1907 from the land of Signor Luigi Nobile. Guerini’s (1909: 78) earliest published description states that this prosthesis was found in the jaw of a woman’s skeleton in one of six *loculi* (niches) cut into the walls of a tomb constructed and roofed with stone slabs. The tomb held many vases and a gold necklace that went missing soon after discovery. Only the skull with this appliance and one “shin-bone” were recovered. At the time of excavation, a stylistic date of c.400–200 BCE was assigned based on the grave goods.

Signor Nobile retained this appliance until 1910 when it was acquired by Professor W. Dieck for his private collection of objects relating to the history of dentistry (cf. Becker 1999a). In 1911 the Teano appliance, identified as mandibular, made an impression on visitors to the International Hygiene Exposition in Dresden (exhibit no. 114). By then someone had replaced (?) a lateral incisor (the left, if identified as a maxillary appliance; see Marz 2010: 45), or perhaps inserted a tooth into that space that previously had been empty. The origins of the two apparently intact dental crowns (if maxillary, the left incisors) now seem questionable, and perhaps represent recent insertions.

In 1934 Dieck arranged to transfer by sale, perhaps through his will, his entire collection of dental antiquities to the Zahnärztlichen Universitätsinstitut, Berlin. Dr. I. Marz (2010) provides the historical sequence of events (also Dieck 1934: VIII):

Bemerkenswert ist, dass sich unter den Objekten auch ein aus einem campanischen Grabe bei Teano herrührendes antikes aus Gold mit natürlichen Zähnen gefertigtes Zahnersatzstück befindet, welches vor c.20 Jahren für 1700 Mk. angekauft worden ist.

It is worthy of note that among the objects there was included, originating from an ancient Campanian tomb, a denture made of gold with natural teeth, which was bought for 1700 Marks around 20 years ago.

The purchase price is that paid by Dieck; transfer of his collection was completed about 1935, a year after his death. A catalogue of the “*Dieck Sammlung*,” dated March of the following year (Schweitzer 1936), suggests that at least one copy of this appliance had been made by 1936 (see Appendix II). The text of that catalogue describes a copy of the Teano appliance as follows (via Dr. I. Marz):

Antikes Ersatzstück, 2400 Jahre alt (Original im Geldschrank) Nachbildung
Ancient prosthesis, 2,400 years old (Original in the Coin cabinet) Reproduction.

Dieck, who dates the original prosthesis to the third century BCE, notes that:

Da neben diesem Ersatzstück sich auch Schmucksachen befanden, unter ihnen ein Diadem aus gepunzten Goldblättern, so ist anzunehmen, dass die Trägerin sehr vornehmen oder zumindest reichen Standes war.

Since near this denture pieces of jewelry were also found, among which was a diadem of beaten gold leaves, it is likely that the woman who wore them was of noble birth or at least of an affluent class.

Gàbrici (1910) dates the necropolis at Teano dei Sidicini to the Hellenistic period. He describes Tomb 18 (27) [sic] (1910: cols. 69–70), noting that the two associated teeth included a rootless lateral incisor that now appears lost (see Bernhard Weinberger’s illustration 1948: 146–147, fig. 51). The dental appliance, he says, was “*fatta mediante due fascette d’oro, che cingono di dentro e di fuori quattro*

denti del cadavere” (“made using two strips of gold, which bind on inside and out four teeth of the cadaver”).

Gàbrici’s 1910 description is the first to note that construction involved “*due fascette d’oro*.” His description and illustration (fig. 25) indicate that this series of rings enclosed six spaces, with four central incisors still in place, stabilized and anchored using the canines. At least some portions of the four incisor (?) crowns, and bits of root are shown. Guerini’s 1909 photograph reveals only three crowns still in place. One of the crowns (left lateral I) shown in prints made from glass plate negatives appears to be a replacement tooth without a root, probably not original to this appliance. Both the color and type of deterioration of the root suggest a different origin, although differential decay within the tomb is entirely possible. Data kept with the Wellcome copy in London (see Appendix II.B) is quite erroneous, but suggests that the fourth tooth had been lost by the time this copy was made, probably by Guerini. Possibly Guerini’s copies were fitted with complete teeth while largely rootless examples were found with the original. Marz notes (personal communication) that a copy of this appliance is in the same institute in Berlin where the original is held, and that a second copy was then, in 1988–1996, in Köln (see Appendix II.C). Possibly both copies were made by, or at the time of Guerini’s work in reproducing all the known examples.

Guerini indicates that the appliance was meant to support the left central to right lateral incisors, reflecting the loss of one of these teeth by the time that he saw and copied it. Gàbrici also notes (1910: col. 45) that

[s]imili dentiere si rinvennero in tombe di Conca, dell Agro Falisco, di Corneto, e se ne conservano nella raccolta Barberini del Museo di Villa Giulia e nei depositi del Museo Archeologico di Firenze.

[s]imilar dentures were found in tombs at Conca [ancient Satricum], in the *Ager Faliscus*, and at Corneto [Tarquinia] and they now are preserved in the Barberini collection of the Villa Giulia Museum and in the storage collections of the Florence Museo Archeologico.

His listing provides an indication of which appliances were well known in 1910 (cf. Guerini 1909).

The six dental spaces involved cannot be identified with certainty from the illustrations, but the large rings at the ends suggest maxillary canines and a maxillary placement. Detailed measurements and examination clarify some of these problems.

Michelsoni (1976: 284) says that the copy of Teano that he saw in Berlin was damaged. Rath (1958: 11, fig. 2) also refers to a copy in Berlin (cf. Waarsenburg 1991a: n. 5, no.15), then in the collection of Professor Carl-Ulrich Fehr. Earlier Rath also notes that he saw various copies at the Forschungsinstitut für Geschichte der Zahnheilkunde, Köln. Rath’s illustration depicts an appliance in which parts or all of four dental crowns are present, two of which appear severely decayed. No roots are evident.

Techniques of construction

The Teano appliance is made of six separate rings, welded in a series to form a slight arc, similar to the Valsiarosa example. Weinberger (following Guerini 1909) provides an entirely fanciful description of the way in which the Teano piece was made, suggesting that he never examined the Teano example, and was unfamiliar with gold-working or construction techniques used for the other gold-band appliances. The most recent laboratory study of this appliance (Marz 2010) appears based as much on the recent literature as on first-hand observation. Marz's observation of what appear to be the two layers of gold, first described by Gàbrici (1910) led Becker to scrutinize the available illustrations assumed to be of the original appliance (Marz 2010: figs 2.6 and 2.7). Becker suggests that six individual rings were formed separately each to fit the specific teeth intended to be surrounded. Then they were "tacked" together in a series using a punch. Then the entire set of rings was outlined by another gold band, which was fitted to the exterior of what became the internal ring-set. This technique seems odd, despite adding strength to a simple set of rings. The over-engineering might be due to a naïve goldsmith following verbal orders from a new client in an attempt to create a particularly stabile appliance. Since there are no rivets and no indication, even in this latest study, of false teeth, a device to stabilize loose teeth could be strengthened by such a design.

The Marz (2010) illustrations, like Hoffmann-Axthelm's photographs, confirm that no rivets were used in this appliance. The "roots" of the four original teeth (2I–I2) clearly have largely rotted off, although the I2 appears to have some traces. This suggests that the four "central" teeth were merely loose, and that this appliance was intended to hold them in place. Quite possibly the "anchor" teeth were also present when this appliance was found, and the discoverer pulled the appliance, and any teeth therein, from the jaw, where the more firmly rooted "anchor" teeth remained.

Unfortunately, during Becker's 1988 and 1989 visits to (East) Berlin, the appliance was not available for study. The story of that visit, and the search for this appliance, reads like a Cold War parody that is, sadly, too true. While an informative lesson in life behind the Iron Curtain led Becker to within 10 meters of the safe in which the appliance was held, permission to see the piece was denied. Some years later, with the encouragement of Dr. Ernst Künzl, Becker contacted Professor J.-F. Roulet (Freie Universität Berlin), who kindly indicated (Roulet to Becker, personal communication February 7, 1994) that this appliance is in the Charité clinic. Professor Roulet requested that Professor Zuhrt at the Charité clinic oversee the request, and later in 1994 Professor Zuhrt asked Dr. Ilona Marz to provide a set of prints of the Teano appliance, taken from glass negatives dating from c.1912. Although these prints are extremely good, a description derived from photographic prints is far inferior to one taken through careful direct observation. Attempts to clean this appliance appear not have involved preservation of all of the fragmentary elements of the surviving dentition.

The size and shape of the central four “rings” are not as would be expected: e.g. a pair of spaces for the central incisors flanked by slightly smaller spaces for the lateral incisors. Instead, we find the next largest opening after the canines appears to be that meant for the 2I, with the space for the 1I appearing nearly as large. The space for the 1I appears to be the third largest, and that for the 12 appears to be the smallest in the series. However, the shapes of all of these elements may have been distorted; only direct examination will provide crucial information regarding its construction.

“The dentition”

The position of the left lateral incisor (2I) is shown as occupied by a tooth crown lacking the color, wear pattern, in-tomb decomposition pattern, and residue apparent on the other teeth. In addition, this crown occupies a space that probably once held the 2I, but the crown (in the photograph) may be a first premolar. It is likely that this relatively unworn crown was added to the appliance from another source. This tooth retains its entire crown, and the dentine within, completely intact, with just the root missing. From the photograph it seems likely that this root was attached whenever this tooth was found (certainly not in its present location), but that it was dislodged by a blow, and not by natural decomposition within a tomb.

The left central incisor (1I) is represented by just the enamel shell of a crown, together with what appear to be remnants of the dentine within. None of the root appears to have survived, but some portion of the neck may be obscured within the appliance. The labial surface of this 1I appears intact, but what may be a pattern of wear is evident on the lingual aspect, suggesting an overbite that led to wearing down into the dentine. This suggests that a relatively old individual (well over 50 years at death) is represented. The extensive wear on the 1I may explain why so little of the right incisors now remain. Both these teeth retain only “shredded” aspects of the base of the enamel crowns, all only evident above the level of the appliance as illustrated. Hidden within the appliance, and extending to a slight degree “above” it are the nearly equal surviving portions of both roots, probably not original to these spaces in the appliance.

Dr. Marz (personal communication, July 25, 1994) had hoped that the organic material surviving in association with the Teano appliance would enable a precise date to be determined. Marz notes that experts involved in the study or evaluation of this piece over the years included C.-U. Fehr and a colleague identified only as Dr. Blum. The two copies now known may have been made by or for either of these scholars. Professor C. U. Fehr, of Berlin, is listed as the source of the excellent illustrations used by Sterpellone and Elsheikh (1990: 37, fig. 65).

17 Praeneste (modern Palestrina)

Present location: Villa Giulia Museum, Rome (inv. no. 13213).

Type: band surrounding four tooth spaces, central pair riveted.

Origin: Praeneste (ancient Palestrina).

Jaw part: maxilla; 2I–12 (only three teeth are present).



Figure 5.17 Praeneste appliance.

Source: drawing by Tim Downs.

Date: not known.

Sex: female.

Dimensions: measured by Becker, June 1988 and June 1999. See Table 5.7 for rivets. L. 31; W. varies from 2.7 to 2.9 mm.

Description: simple band surrounding all four maxillary incisors, with two rivets fastening replaced or replacement central incisors. Central incisors probably were teeth removed from the owner and filed down. The canine displayed with this appliance may derive from the same person, but this tooth did not originally serve as an anchor tooth for the band. See Table 5.6.

References

- Baggieri 1998: 67 (this prelude to Baggieri 1999 depicts the appliance in an inverted position; erroneously suggests that five spaces are bridged).
Baggieri 1999: 37, fig. 6.
Baggieri 2001: 323–324, pl. XXXVIII, g (repeats Baggieri 1999).
Bliquez 1996: 2655, “T,” fig. 31.
Bonfante 1986: 250, n. 63 (noting Helbig 1969: 827, no. 2957).
Della Seta 1918: 444.
Emptoz 1987: 553 (no. XV).
Helbig 1969: 827
Henzen 1855: 75.
Menconi and Fornaciari 1985: 94, 97 fig. 9 (excellent photograph).
Micheloni 1976 (illustrates Praeneste, but text describes Valsiarosa).
Naso 2011: 151 no. 13.
Pot 1985: 37–38, fig. 2.
Sterpellone and Elsheikh 1990: 125, fig. 61; 142, fig. 70 (illustrations are quite good).
Tabanelli 1962: 93.
Torino *et al.* 1996: 121–122.
Torino *et al.* 1997: 538–539.
Van Marter 1886: 57.
Waarsenburg 1991a: n. 15.
Weege 1913: 354–355 lists “Garg E 5758”); 371 notes “n. 1768d.”

History and chronology

The excavations of Giorgi and Mattia in the Colombelle necropolis west of the city of Palestrina (Henzen 1855: 75; see also Braun 1855), were just outside the city wall. The necropolis of Latin Praeneste yielded rich finds including “Princely Tombs” (*Tombe principesche*) of the mid-seventh century BCE and others the Archaic through Hellenistic periods. Pot (1985: 36) cautions that the fourth-century BCE date sometimes associated with this appliance is not supported by any known evidence.

The Praeneste appliance was seen by Van Marter in 1885 in the library of the Barberini Palace in Rome. In 1908 it was acquired, with the Barberini collection, by the Villa Giulia Museum (Waarsenburg 1991a: n. 15). Waarsenburg transcribed from the Villa Giulia inventory book: “*acquistata dal Prof. Elia Volpi, Agosto 1908. Buono no. 49.*” Waarsenburg includes Villa Giulia inventory numbers for items 12755 to 13658, including pieces that may relate to the tomb group within which this appliance was found. The Praeneste appliance, inv. no. 13213, is described as “*Quattro denti con fascia d’oro, Lire 120*” (“four teeth with gold band, 120 *Lire*”). This suggests that four teeth were then with this appliance, although they may have been added after being recovered from its tomb. For data on the history of the Villa Giulia Museum see Pelagatti (1985), Rizzo (1985), and Sforzini (1985).

Della Seta (1918: 444) includes this example in his description of the items in the Collezioni Barberini at the Villa Giulia Museum:

Dentiera. Una fascetta d’oro unisce l’incisivo centrale superiore di destra col canino di sinistra. In mezzo vi sono i due denti rimpiazzati che sono tenuti fermi da due bullettine.

Denture. A small band of gold joins the upper right central incisor with the left canine.

In the middle there are two substituted teeth that are held tight by two little tacks [i.e. rivets].

Quoting Della Seta (1918: 444), Tobias Dohrn (“Kunstliches Gebiss,” in Helbig 1969: 827) lists the Praeneste piece as no. 2957 (cf. Bonfante 1986: 250, n. 63), with its “Helbig number” in the third edition “1768d.”

Description

In 1999 the Praeneste appliance, a maxillary pontic in the Museum of the Villa Giulia was on display in Room 34, as item no. 10 in a case featuring sporadic finds from the Praeneste necropolis, the gold band mounted vertically, with the now empty space for the actual I2 at the superior position.

The configuration of this appliance and the evaluation of the three teeth now held in, but not necessarily original to it suggest that it was worn by a female at least 35 to 45 years old at the time of death (see Becker 2002a). The evaluation of age, however, is based solely on dental wear and the presumption that these teeth belonged

to the original owner. Since foods such as white bread, lacking in the coarse husk materials, cause lower rates of dental wear than bread made from less finely sifted flours, we might assume that these high-status individuals show somewhat lower rates of tooth wear—but in the absence of a skeleton, caution is indicated.

This is a simple band that surrounded four maxillary teeth, with the central pair, possibly the owner's, riveted into place. The presence of a canine when the appliance was on display led some to believe that this band enclosed the maxillary left canine, the left lateral incisor and both central incisors, using the canine and right central incisor as anchor teeth. In fact, almost certainly this appliance spanned the four maxillary incisors (compare the Liverpool appliances).

The "apparent" I1 (certainly the actual I2 position) now is empty, the tooth missing and presumed lost. However, Becker found that part of the tooth on display (perhaps all of its crown) had been in the actual I1 position, but then was shifted to the actual I2 socket. The two riveted central incisors are human and match the surviving anchor tooth in general size, color, and even degree of wear. The canine might have come from this person, although it was not originally displayed with the appliance. At least one tooth displayed in the slot for the I1 was actually the original I2 anchor tooth that, when split, could be placed in the I1 position. The other tooth used in making this display version of the appliance may be a natural canine, but was not originally within the band, but put into the bridge when it was recovered. This conclusion was independently reached by Menconi and Fornaciari (1985).

A canine now within the appliance, represented by the crown only, has a mesial-distal length of 7.5 mm and a buccal-lingual diameter estimated at 8.8 mm. Most of the dentine within the enamel of this crown has decayed away, probably during burial. The lateral incisors that served as anchor teeth are not evident among the teeth displayed with this appliance.

The teeth used in the spaces of the 2I and I1 are represented only by crowns, with much of their interior dentine lost, as in the case of the canine. This precludes determining if the roots of the central incisors had been filed or cut down before being riveted into the band. The I1 is quite narrow for a central incisor, with a mesial-distal breadth of 6.3 mm. The buccal-lingual diameter is suggested by the length of the rivet, about 5.2 mm or less. Incisor shoveling is quite evident on the I1, about level 2 to 3 (0–4 scale; cf. Becker and Salvadei 1992). The tooth fragments in the position of the I1 are problematical since they seem to be a single tooth split in two. Although the I1 appears wider than I1, Becker suspects that the two halves of this

Table 5.6 Configuration of teeth in appliance no. 17 (Praeneste)

<i>Apparent</i>	<i>Actual</i>	<i>Tooth in place</i>
Left C	2I	Canine, in 1999
2I	I1	Fragments
I1	I1	Split tooth
I1	I2	Empty

Table 5.7 Rivet dimensions of the Praeneste appliance

Tooth (replacement)	Rivet length	Rivet head diameter (as peened) labial	Rivet head diameter (as peened) buccal
2I	5.8	2.5/2.0	2.3/1.8*
1I	6.0	2.4/2.2	2.5/2.3

* Peened into the depressed area of the shoveled surface.

“crown” may not belong to the same tooth, or may be all or part of the “missing” 1I noted above. Slight shoveling appears in the margin of the distal portion, which is now firmly set in the bridge. The mesial margin of this broken tooth, however, is loose within the band. Either this half has been poorly glued to its other half, or it is not the mesial half at all. Becker suspects that what is set in the mesial position within the location of the 1I is actually a portion of the 1I (right central incisor), which has been restored in the band incorrectly.⁵

The gold ribbon forming the band is quite thin (0.3 mm or less) and very narrow, ranging unevenly from 2.7 mm wide along the anterior surface of the incisors to 2.9 mm at the left lateral incisor. Many of the irregular bends may be a result of modern forces adjusting the gold to these tooth surfaces. The original band need not have been so curved to adjust from the narrow buccal-lingual (B-L) diameter of the 2I to the very wide B-L diameter of the left lateral incisor. As Pot (1985: 37) points out, this band passes across the teeth close to the centers of the crowns, or much higher than is the case with the Satricum appliance, which had a different construction and function.

The replacement teeth were held in place on the band with gold rivets (rivet diameters not visible). The lengths of the rivets, their heads carefully and delicately peened (see Table 5.7), are just sufficient to hold the band tightly to the teeth. The process of affixing the rivet in the band was one of the more delicate aspects of the production of these appliances.

18 Satricum



Figure 5.18a Satricum appliance, without natural tooth.

Source: drawing by Tim Downs.



Figure 5.18b Satricum appliance, as published.

Source: drawing by Tim Downs.

Present location: Villa Giulia Museum, Rome (inv. no. 12206).

Type: band spanning five spaces with an artificial hollow gold tooth attached at the center.

Origin: Conca (Borgo Le Ferriere, ancient Satricum), northwest Necropolis, Tumulus C, 'Tomb' XVIII, Grave C 11, the "*tomba dell'oro*."

Jaw part: maxillary, 2I–C. (false "tooth" at I1).

Date: seventh century BCE. Waarsenburg (1991a: 320; 1995: 366) suggests 700–650 BCE.

Sex: female (Mengarelli 1898; Becker 1994a).

Dimensions: measured in Villa Giulia Museum by Becker June 1988 and June 1999. L. 29, W. 1.8 (mesial)–2.2 mm (distal). Gold tooth height 10.2 mm (2.9 x 0.2 x 0.02 cm; Waarsenburg 1995: 372 no. 18.50). Baggieri (2001: 325) gives the total weight of the appliance as 1.2 grams.

Description: This gold dental appliance is the only example in which the replacement tooth also was fashioned from gold and then fastened to a very thin band surrounding five dental spaces, almost certainly all four incisors and one canine. This also is the earliest example known, probably made between 700 and 650 BCE.

References

- Ampolo 1980: 177, 185–187 (regarding the date).
- Baggieri 1998: 67 (prelude to Baggieri 1999 includes an inverted (?) illustration).
- Baggieri 1999: 36–37, fig. 5.
- Baggieri 2001: 325–326, pl. XXXVIIb, graph 2.
- Baggieri and De Lucia 1993: 16–17, fig. 1 (from Della Seta 1918).
- Becker 1994a.
- Bliquez 1996: 2655–2556, "U," figs 32–35.
- Boissier 1927: figs 6 and 7 (well drawn).
- Brown 1934: figs 6a and 6b.
- Casotti 1947: fig. 7 (fig. from Dall'Osso 1907).
- Casotti 1957: 675–676 (data derives from Piperno (?))

- Colonna 1974: 315; 1988: 308.
Dall'Osso 1907.
Della Seta 1918: 250–251 (see n. 20).
Della Seta 1928: 304f.
Emptoz 1987: 556 (no. XX).
Ginge 1996: 21, 39–40.
Gnade 1986: 125, fig. 214.
Guerini 1904: 280.
Guerini 1909: 101, figs 31, 32.
Menconi and Fornaciari 1985: 94–95, 96 figs 7 and 8 (excellent illustrations).
Mengarelli 1898: 169–170, figs.
Naso 2011: 151 no. 14.
Piergili 1906–1907 (incorrectly attributed to lower jaw).
Piperno 1913: 4, fig. (via Della Seta).
Pot 1985: 36, fig. 1.
Proskaur 1979 (2): 13 (mentioned only).
Sterpellone and Els Sheikh 1990: 137–138, fig. 66.
Strömberg 1919: figs 7–8 (from Guerini 1909).
Sudhoff 1926: 93, fig. 54 (clearly a copy, shown in a mandible).
Torino *et al.* 1996: 121–122.
Torino *et al.* 1997: 539–540.
Waarsenburg 1994: 319–321.
Waarsenburg 1995: 360–372; 365–367; catalogue entry p. 372 no. 18.50, pl. 72.
Weege 1913, Vol. II: 354–355.
Weinberger 1948: 147, fig. 54.

History of the Satricum appliance

The present crossroads hamlet, Borgo Le Ferriere, within the region south of Rome once called *Latium Vetus* (“Old Latium”) includes a necropolis of the city of Satricum (for archaeological history of the area see Waarsenburg (1994: 314–329; 1995: 10–30). Waarsenburg also provides a complete description of the archaeological context from which this prosthesis was recovered. R. Mengarelli identified the person in this tomb as a woman according to diary entries he made in 1940, more than 40 years after he excavated a burial (Tomba XVIII) from the center of Tumulus C in 1898 (Mengarelli 1898: 169–171; Waarsenburg 1994: 319–329; 1995: 374–377; Gnade 1986: 125; Ginge 1996: 21). His evaluation of gender based on associated artifacts now is independently supported through examination of the teeth associated with the appliance (Becker 1994a).⁶ The monumental tomb type and grave goods indicate a family of aristocratic status.

Carefully combing excavation records (daybooks, *Giornale di Scavo*), Waarsenburg was able to identify the original burial of this appliance as “Tomb” XVIII. This was not a single tomb, but a group of several tombs with varying numbers of burials, all found beneath the large Tumulus C at Satricum. Della Seta (1918: 249) had identified this context, overlooked by later authors (Waarsenburg 1995: 360, n. 947). The burial complex held grave goods ranging from the second

half of the eighth to the later seventh century BCE (Waarsenburg 1995: 361). The appliance was found on a skeleton buried in Grave C 11, the only burial containing gold (although looting may account for this). Grave C 11 held two inhumations and one cremation burial, with one body apparently dressed in a costume decorated with gold appliquéés and amber ornaments, and still wearing this gold dental appliance. The various offerings from these three funerals were not differentiated spatially when the tomb was excavated. They included three Etrusco-Corinthian kylikes (painted wine cups) and another painted vase, as well as impasto cups of earlier style (inv. C 11.8f.) Metal goods included a sheet-bronze vessel, a bronze fibula, and various ornaments, including some of amber. The impasto vases are fine quality seventh-century ceramics; the latest objects are the painted vases, all now lost. Etrusco-Corinthian wares began in the late seventh century and flourished in the sixth.

Mengarelli's approximate date for this tomb, at c.600 BCE, has been accepted by many scholars (cf. Della Seta 1918: 250–251; Della Seta 1928: 304; Colonna 1977; 1988: 308). Ampolo (1980: 177) dates the tomb specifically to 620–610 BCE, but Waarsenburg (1994) suggests an even earlier date, between 700 and 650 BCE, based upon painstaking review of the inventories and extant finds. The deposition of the woman with gold-ornamented robe (*roba di parata*) is characteristic of Latial period IVA, or the first half of the seventh century BCE (Waarsenburg 1995: 349, n. 914; 366). The Satricum appliance clearly is the earliest as well as the most securely dated of any of the known Etruscan pontics, and of course, it would have been made and worn *before* the time of death and burial.

Waarsenburg (1991a) determined that many authors such as Casotti (1957: 676) identify all gold dental appliances considered to be of “better quality” as Roman in origin. On that very subjective basis alone they suggest a later date. We point out that this example appears to be prototypic, and a precursor to Etruscan riveted and other examples, and therefore support Waarsenburg's evaluation of an early date.

Detailed description of the Satricum appliance

The possibility that this appliance was worn in the mandible is suggested by the small size, but it could have been intended for the maxilla of a small woman.

The Satricum pontic is a unique example of a variant form of gold dental appliance. It was constructed in two parts. First, a thin gold band (ribbon) was fashioned, 29 mm long, 0.2 to 0.25 mm thick, varying in width from 1.8 mm at the mesial end to 2.2 near the distal end.

This looped around five dental spaces. Attached at the center of the band is a hollow false tooth fashioned to replicate a dental ‘crown’ (the portion of a tooth that projects above the gum) 10.2 mm high. The mesial-distal breadth of this false tooth is 6.1 mm, the buccal-lingual breadth is 5.0 mm. Both maxima appear slightly above the relatively constricted lower margin of this “crown.” The base measures 5.8 × 4.8 mm, replicating the constricted neck of a natural tooth. The false tooth is made from two thin swaged (molded by hammering into a form)

gold plates (labial and buccal) welded together to form a hollow false tooth that is neither a dental “cap,” nor a true crown as in modern dental terminology.

While the shape of this false tooth resembles a mandibular incisor, it may have replaced a canine. The form may be a result of the manufacturing process rather than a skillful representation of the missing tooth. The bends now in the band suggest that it originally surrounded the two teeth on either side of the gold “capsule.” The band also fits low on the teeth, or at the neck, much lower, for instance, than the band in the appliance that secures the Praeneste teeth (cf. Pot 1985: 37).

Mengarelli correctly notes that this hollow false tooth was made of two thin parts, but suggests that it was designed to imitate the left lateral maxillary incisor, an evaluation with which Becker originally concurred, but since has come to question. Guerini (1909: 101) believes that the piece is made from two plates of gold stamped out and then soldered together. More likely, the pure gold band and false tooth were assembled by welding. This shell of the false tooth may have covered a relatively hard material, now lost or dissolved in the tomb, as suggested by Pot (1985). No evidence indicates that anything more than a single false tooth had been secured to this band.

The Satricum false tooth is welded, at its base, between the labial and buccal (lingual) surfaces of the gold strip. The weld between the tooth and the band on the buccal surface has separated and the band itself has come apart where it had been joined at the midpoint of the labial surface of the false tooth. This separation suggests that the bonding point of the band and the weld that attached the false tooth to the band were at the same location, and may have been achieved in a single step. Guerini (1904: 280) notes that he had made “an exact reproduction” of this appliance (see “Copies” in Appendix II). What has become of Guerini’s copy remains unknown.

Discoloration evident at various junctures along the band appliance probably relates to the manufacturing process, which at this early period may have involved true soldering rather than welding. The dark color patches may be metals or fluxes used in such a process rather than impurities in the gold. The suggestion that a silver-rich solder was used to join various aspects of this appliance (Baggieri 2001: 326) has not been tested. No modern attempts to repair this appliance are known, but such attempts could be another possible source of these discolorations.

The “real” tooth associated with the appliance

No skeletal material is known to have been preserved from the excavations that yielded this appliance (Becker 1994a). The crown of a single tooth, a mandibular first premolar, appears in some early photographs of the Satricum appliance, at the left distal end of the band (Weinberger 1948: fig. 54; Baggieri and De Lucia 1993: fig. 1) appears in some of the early photographs of this appliance (Mengarelli 1898: 169). Guerini’s (1909: figs 31, 32) photographs show the appliance without a human tooth, suggesting that the human tooth now there was only later glued into the band. Whether it is the same tooth has not been demonstrated. Guerini also notes that he saw this piece only a few months after its excavation, at the

“Museum of Pope Julius in Rome” (then the new Museum of the Villa Giulia), but the date of his photograph is unknown.

On the basis of close examination of this premolar crown, and comparison with numerous other dental series in central Italy, Becker suggests that this is a mandibular first premolar of a mature adult female (see also Pot 1985: 36). Only a small portion of the root of this premolar remains, but most of the root appears to have dissolved while still in a tomb. The area around this tooth remnant, at the base of the crown, appears to be covered with modern glue or some agent that may have been used to conserve the bit of root still present, or to hold the tooth within the band for display or conservation. Even if this fragment of tooth is original to the Satricum appliance, belonging to the person who wore this device, it may not be in the correct place within the band. More probably it never was within any of the four spaces of the band, but was recovered from elsewhere. Even less probable is that this tooth has been relocated from the place in the band in which it was found to another space.

The gold replacement tooth

The band has slight undulations that appear to bracket spaces for four healthy teeth, two flanking each side of the false tooth. If the human premolar is a 1PM and is in its correct position within the band, then the appliance would have enclosed the spaces from 1PM to I1. Baggieri and De Lucia (1993) suggest that the false tooth would have replaced the mandibular 2I, since it has the morphology of a small incisor. Although the two mesial spaces within the band are small, or about the size of mandibular incisors, the distal spaces are of the same size. Furthermore, the false tooth is as large as any of the empty spaces, and there is no reason to believe that this unique false tooth was accurately modeled on the tooth it replaces.

Clearly this false tooth was *not* meant as a tight fitting cover or “crown” to a worked or filed tooth since the neck of the “tooth” is constricted and thus could not be seated properly over a decaying tooth, as Waarsenburg (1994) and some others suggest (cf. Pot 1985: 36). Mengarelli (1898) wrongly believed that this hollow false tooth was meant to be a “crown” to cover a carious incisor, an idea that Guerini (1909: 101) seems to have accepted. This idea may derive from Talmudic references, which simply err in their translation, or otherwise distort any possible relationship between false and rotted teeth and are not founded on actual dental possibilities.

The gold used to form the hollow false tooth must be relatively pure, and therefore relatively soft. If such a thin shell had been used as a “crown” over a decayed tooth, it would have collapsed as the tooth below it decayed. Thus this hollow gold “tooth” must have been intended to fill a vacant space. Clearly this gold “tooth” was meant only as a decorative spacer. The use of two pairs of “anchor” or “post” teeth, one pair on *either* side of the gold tooth, may reflect early technological caution and inexperience. Another possibility is that the adjacent teeth had been loosened due to periodontal disease. Most likely the appliance was made as an ornamental replacement for a tooth that had been deliberately removed.

Summary, interpretation

Excavated from a tomb within Tumulus C of the Satricum necropolis, this unique gold appliance is the earliest known, probably made between 700 and 650 BCE and buried no later than c.630 BCE. It is the only example in which the entire replacement tooth was fashioned from gold and welded to a very thin band that surrounded five dental spaces. All four maxillary incisors and one canine were enclosed by this band, but which canine cannot be determined. It was worn by an adult female of 50 years of age or older (Becker 1994a), of elevated social status in her community.

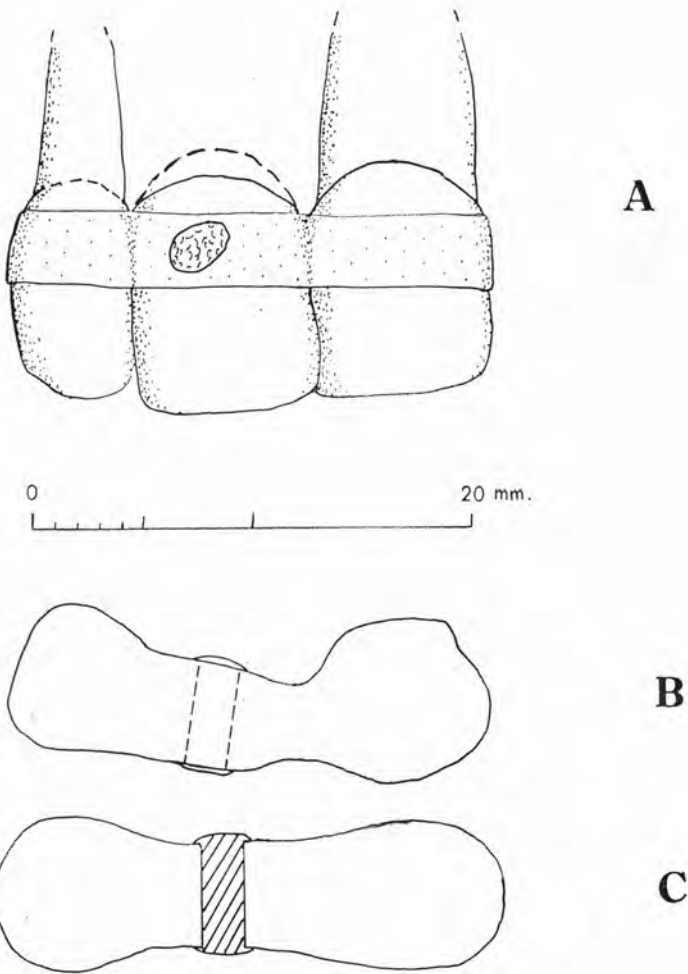
This appliance, unique in technique and early date, presents two scenarios. One is an evolutionary pattern, with this appliance using a tooth made from gold, while later versions re-used the owner's cut-down teeth or teeth from other humans, animals or other substances. The second views this early piece as one of a kind, commissioned especially by the wearer. Great variety in approaches to design and manufacturing technique may be expected, due to the individuality of goldsmiths and wearers. Our sample, with a maximum of 21 examples, is too small to permit generalizations about techniques of manufacture or evolution of the tradition.

Several authors have inferred that the Satricum appliance, perceived as more "complex" than the simple bands, is therefore of later date and Roman manufacture (Sudhoff 1926: 93; Weinburger 1948: 148). Casotti (1947) even fabricated a false origin for this piece to confirm his "conclusion" that it is Roman. Tabanelli's (1963) omission of this example from his "Etruscan" series suggests that he, too, may have believed it to be Roman (see Waarsenburg 1991a: n. 12). The archaeological context tells us that exactly the reverse is true. As the oldest dated dental appliance, the Satricum piece provides possible insights into the development of dental appliances. Later Etruscan appliances tend to anchor a false tooth only to the immediately adjacent teeth. This "prototype" may have been "overdesigned" simply because the makers were as yet uncertain as to how much support would be needed to steady a replacement tooth. Using a long band also provides more margin for errors in calculating the length needed.

Della Seta (1918: 250–251) errs in presenting both the archaeological and descriptive data, citing Weege (1913, II: 355) and Piperno (1913: 4, fig.). Baggieri (1998: 67; 1999) echoes Della Seta by illustrating the Satricum appliance upside down, identifying the false tooth as a left (mandibular?) premolar, and using imprecise terms such as *capsula*, commonly translated as "dental crown," for the false tooth. His use of *saldato* for the join has been incorrectly translated as "soldered" or "welded" in the English version (Baggieri 1999).

The date of the Satricum appliance, as the earliest known dental bridge, certainly reflects the evolving social structure of the urbanizing peoples of central Italy during the era of the *Tombe principesche* (see Ampolo 1980: 185–187, and his n. 96 referring to CLP: 34).

19 Bracciano



Figures 5.19a–c Bracciano appliance, drawings by Becker: (a) anterior view (adapted from Teschler-Nicola *et al.* 1994a); (b) superior aspect of the Bracciano appliance (without the teeth), adjusted from Teschler-Nicola *et al.* (1994a) using their dimensions; (c) the Bracciano appliance as it may have been originally constructed (without the teeth in place). (Note: Naso 2011: 147 fig. 65 provides a color frontal photo.)

Present location: Department of Anthropology, Museum of Natural History, Vienna (Wiener Naturhistorisches Museum), inventory no. 24.286.

Type: simple band, with one rivet holding a false tooth.

Origin : “Etruscan tomb” near Lake Bracciano.

Jaw Part: maxillary I1–I2 (three spaces).

Date: seventh century BCE (?).

Sex: female (?) aged 35–40 years (Becker); 25 to 40 years (according to wear patterns, per Teschler-Nicola *et al.* 1998: 61).

Dimensions: L. 24 (est.), W. 3.5–3.8 (depicted as c.3.0) mm. Becker: gold band at least 3.5 to 3.8 mm in width, and 0.24 to 0.30 mm thick; Teschler-Nicola *et al.* (1998: 61) suggest a band width of 2.4 to 3.0 mm (3.9 mm at overlap); diameter of rivet approx. 2.55 mm.

Description: simple gold-band appliance, of somewhat mysterious provenance, spans three dental spaces and holds a surviving cut-down right central incisor. Most unusually, both anchor teeth survive. The gold of the band is 0.24 to 0.30 mm thick, much thicker than in most bands. The diameter of the gold pin is indicated as 2.0 to 2.5 mm; either measurement appears large, prompting questions of authenticity.

References

- Naso 2011: 147 fig. 65; 151 no. 5.
 Teschler-Nicola *et al.* 1994a.
 Teschler-Nicola *et al.* 1994b.
 Teschler-Nicola *et al.* 1998.

History, provenance

If the seventh-century BCE date of this Etruscan dental bridge, suggested by Teschler-Nicola and colleagues (1998), is accurate, it would be among the earliest known, comparable in date to the Satricum example. The date, like the sex of the wearer, is said to be based on “some accompanying archaeological artifacts” of 700–600 BCE found in “an Etruscan tomb near Lake Bracciano north of Rome” preserved in the Department of Prehistory of the Vienna Natural History Museum (Teschler-Nicola *et al.* 1998: 57). In 1998 it was said to have been “found a few years ago,” or prior to 1994, but no details of excavation or finds have been published. Teschler-Nicola *et al.* stated in 1998 “We have incomplete information as to the arrangement of the grave goods in the tomb” (1998: 61). If the finds were obtained and transported to Vienna prior to 1970 they may have been studied in accordance with international laws. We include some of the published data here in the interests of a complete catalogue of Etruscan appliances, but we have not seen any of this tomb material. Study of the documents of the previous owner of these materials may enable scholars to determine a more precise location for the tomb from which they came and the excavations that brought it to light.

The Bracciano appliance may provide new information that confirms earlier inferences regarding the technology employed in the fabrication of Etruscan prostheses. In addition to macroscopic examination of this appliance, Teschler-Nicola and colleagues (1994b) used “radiological and light-microscopical [analytical]

techniques.” The tooth structure was examined with a scanning electron microscope (DSM 962 ZEISS, Germany). Scanning electron microscopy (SEM) was used in conjunction with a Link eXL microanalysis system (Oxford Instruments Ltd., UK) to determine precise metallic composition (at 20kV, with X-ray signal collected for 500 seconds; Teschler-Nicola 1994b). The X-ray provided by Teschler-Nicola *et al.* (1998: 62, fig. 6) indicates that the rivet used is approximately 2.5 mm in diameter, or much larger than other examples.

Description of the teeth

The Vienna team estimated age and sex by size and wear on the teeth, determining the female wearer of the appliance was between 25 and 40 years at time of death. This bridge had the unusual feature, when found, of retaining both anchor teeth as well as the “false” tooth. The “false” tooth riveted into this appliance is the right central incisor, believed by Teschler-Nicola *et al.* (1994b: 28) to be the person’s own tooth with a filed root. The anchor teeth are the right lateral and left central incisors, the entire crowns of which survive intact, but with only small portions of the neck areas. The roots are entirely deteriorated, a common condition in the dentition of individuals buried in Etruscan chamber tombs (Becker 2004). Damage to the roots has not resulted from postmortem removal of this device, as Teschler-Nicola and her colleagues suggest (1994b). More likely little if any of the skull remained intact in this tomb.

Both anchor teeth show “moderate shoveling” (cf. Becker and Salvadei 1992), and the left central incisor also has a lingual tuberosity. None of the observations make reference to this clear example of a typically Etruscan lateral incisor trait (see Pinto-Cisternas *et al.* 1995).

Teschler-Nicola *et al.* (1994b: 28) noted that the right central incisor (the “false” tooth) was similar to the anchor teeth in “form, size, color and degree of dental abrasion.” Color, however, can be derived from staining in the tomb context, and is not as useful a diagnostic trait as would be metric or morphological data. If the false tooth is actually a tooth from this person, shoveling might be expected, however, the lingual surface of this false tooth “is partly ground off . . . to achieve a flat surface for the fixation of the gold band” (Teschler-Nicola *et al.* 1994a: 132). Their study of the false tooth by “light and scanning-electron microscopic investigations” indicates that “all parts seem to be artificially treated” (Teschler-Nicola *et al.* 1992: 57, 55) including the lingual surface. Unclear is whether the articular surface of the crown also had been ground down, as the authors later suggest that both central incisors share the same degree of “dental abrasion.” This information contradicts the statement regarding “all parts” being artificially treated. In any case, such extensive treatment of “all parts” of this tooth would seriously reduce its identifiability, and reduce our ability to compare the two central incisors.

Most of the “false” tooth root had been removed, with the remnant rounded and smoothed. The degree of dental abrasion on the natural teeth suggests to Teschler-Nicola and colleagues an age at death greater than 25 years. Based on

Becker's observations of high status Etruscans from this area we suggest an age at death of 35 to 45 years. Teschler-Nicola *et al.* (1994a) infer that the dental metrics (Teschler-Nicola *et al.* 1998: 61, table 1) indicate that a female is represented, using for their comparative sample an Early Bronze Age population (Teschler-Nicola 1992). Becker (1994b) concurs with her conclusion (Teschler-Nicola *et al.* 1994a) regarding the wearer's sex (Becker 1992a).

Technical aspects, construction

The appliance is formed from a simple gold band varying from 3.5 to 3.8 mm in width, and 0.24 to 0.30 mm thick. These dimensions suggest a coarser example (wider and thicker) than the Poggio Gaiella appliance (no. 5). The band was made into a long oval by overlapping the ends of a long strip for some 8 mm and welding along this overlap (cf. Teschler-Nicola *et al.* 1994a). These authors indicate that the welded section of the appliance is "between the right medial and lateral incisor," but whether placed on the labial or lingual (buccal) surface is not stated, nor is it evident from the drawings. Where this overlap has been welded, it ranges up to 0.39 mm in thickness. Important in the Teschler-Nicola observations is that "[O]ne of the broader scratches shows signs of repair, obviously done by the goldsmith," a point to which we shall return (Teschler-Nicola *et al.* 1994a: 132). Note that these authors depict the superior margin of this maxillary appliance as shorter than the inferior margin. Unless post-excavation damage has occurred this could only be achieved if there was post-insertion molding of the appliance around the anchor teeth.

The rivet holding the false tooth is depicted (Teschler-Nicola *et al.* 1994a) as just under 2 mm in diameter or approximately the same width as the diameter of its peened ends. This is highly unlikely, and the diameter is far greater than the approximately 1.0 mm of the pins used in other appliances. Drawings of the Bracciano appliance showing the placement of the rivet are not in agreement, leaving the exact pattern of riveting uncertain. Also, despite the high technology applied in the study of the Bracciano appliance, no attempt appears to have been made to verify the rivet diameter. Larger holes are not only more difficult to drill, but in this case they would leave both the artificial tooth and the band to which they are affixed more fragile.

The most significant contribution of the study by Teschler-Nicola and her colleagues (1994a, reprised 1998) is in the analysis of the gold used to make the Bracciano appliance (cf. Becker 2003). The authors made "10 examinations from different regions of the gold band and the rivet" but do not indicate the loci of these tests (1994a: 132). The gold component was found to be from 92.2 to 98.6 percent (by weight; mode c.97 percent), with silver varying from 0.74 to 7.0 percent (mode c.2.2 percent) and copper varying from 0.00 to 0.8 percent (mode c.0.7 percent). These figures reveal that the gold employed in making this appliance was exceptionally pure (Becker 2003); compare the results of analyses of Liverpool I and II (nos. 13–14).

Discussion and conclusions

Teschler-Nicola (*et al.* 1994b: 28) conclude that “the functionality” of this appliance

is indicated by lesions, such as tiny chips, situated on the occlusal margin along the gold band and the presence of very small scratches (–0.5 m [sic]) on the labial as well as on the lingual surface of the band.

They infer “that this prosthesis was in use for at least several years.” While this inference is reasonable, it is not at all likely that this “seems to be primarily a functional device.” The soft quality of the gold led to the scratches and distortions, including the large one that supposedly required repair. An appliance fashioned from this quality of pure gold would be easily bent out of shape by very slight pressure. Thus the possible use of this person’s own incisor as a false tooth might *require* that the occlusal edge be filed down to *prevent* it from being in alignment with the other teeth, where it would be torn apart by normal use! The kinds of damage evident on this appliance suggest that it was kept in place during mastication, but the purity of the gold could *not* tolerate even the simplest masticatory functions—in other words, the wearer had to chew very carefully around the cosmetic appliance!

20 Tanagra (Greece)

Present location: National Archaeological Museum, Athens. Lambros Collection, no. 358.

Type: compound band made of two joined loops (no rivets).

Origin: Tanagra (Boeotia) Greece.

Jaw part: maxillary, C–C (6 spaces)

Date: fourth–third century BCE (?)

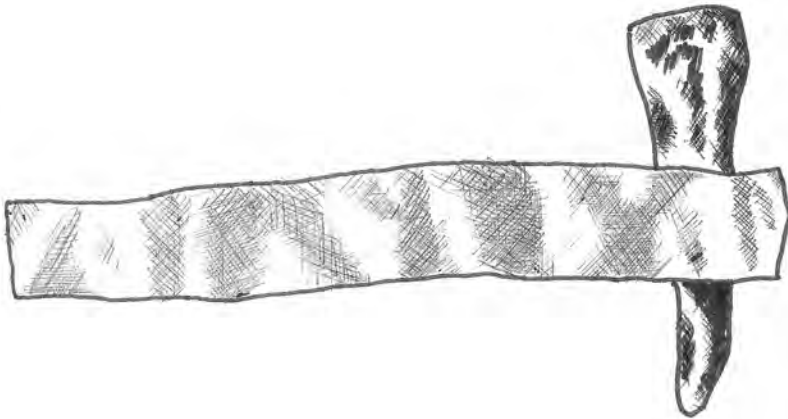


Figure 5.20 Tanagra appliance, drawing after Weinberger 1948: fig. 39.

Sex: female?

Dimensions: L. 43.3, W. 4.3–5.6 mm.

Description: gold band appliance designed to surround six maxillary (?) teeth; almost certainly the four incisors and both canines. The two long gold loops that were welded at the “center” were formed from two strips each nearly 25 mm long and 0.2 mm thick. The ends for each strip may have overlapped for about 6 mm before being welded to form an elongated oval. Then the bands were welded to form a single appliance (cf. Poggio Gaiella). Given the apparently unusual flexibility of the alloy used for this example, the fitting must have been extremely delicate. The possibility that this had been a mandibular appliance cannot be ruled out.

References

- Becker 1997: 87–88, figs (written prior to direct observation).
 Bliquez 1996: 2648, “G,” fig.12.
 Deneffe 1899: 26–27, 59 (no illustrations).
 Emptoz 1987, no. IX.
 Guerini 1909: 60, fig. 9 upper.
 Koch 1909: 6.
 Lufkin 1948: 56.
 Micheloni 1976: 264.
 Sudhoff 1926: 63, figs 39b and c.
 Weinberger 1948: 119, fig. 39 upper.

History

The Tanagra dental appliance is one of only two of this type known from outside Italy. It may have been worn by an Etruscan immigrant or made to Etruscan specifications. Very little is known of the tomb goods and other conditions of this find. One hundred years ago Platschick (1904–1905: 241), although apparently unaware of the Tanagra example, predicted that Etruscan dental appliances would be found in Greece.

Deneffe (1899: 26–27) published the first reference to this appliance, believing that it had been recovered from excavations at ancient Tanagra, near Greek Thebes (*Thēbai*), a site 55 km northwest of Athens. Deneffe’s description had been provided to him, we presume, verbally, by Dr. Lambros, who told Deneffe that this piece was bought from a private collector in Boeotian Thēbes, where it had been deteriorating. Deneffe notes that but one incisor was left, and later illustrations show one intact tooth (with root) within the appliance. The intact state of the tooth suggests a later substitution. Illustrations depict a narrow band, which has been crushed flat (pinched together) except for one end, where it encloses a tooth (see Sudhoff 1926; Lufkin 1948; Weinberger 1948). From the illustrations the tooth appears to be a maxillary 2I, but Becker does not believe this tooth to have been in the appliance when it was found. It had probably been inserted after excavation to indicate the inferred function of this band.

Deneffe (1899) describes an oval band 5 mm wide, but illustrations suggest slightly *less*. Deneffe, noting an absence of rivets, believes that it was affixed

around four incisors to hold the loose centrals in position. Micheloni (1976), stating that the band measures 70 mm long (referring to the total length of the gold ribbon before forming the oval?), gives the width as varying from 5 to 6 mm.

Description

Direct observations of the Tanagra appliance were made by Becker at the National Archaeological Museum in Athens in June, 2002 through a permit issued by the Ministry of Culture. The kind co-operation of Dr. Rosa Proskynitopoulou greatly facilitated this research. Unfortunately, arrangements to secure an XRF analysis of its metallurgical content were blocked by Museum officials. The appliance was apparently recovered in its present flattened condition, but what taphonomic processes or conditions led to this crushing (distortion) remain speculative. At present a single human tooth is encircled within the gold band at one extreme end of the appliance.

Published illustrations give the impression that this appliance was a simple, one-loop band that bounded the upper four central incisors. It actually has a compound or two-loop construction, formed by fashioning two units or “long gold rings” that together encircled six (maxillary?) teeth, the canines as well as the incisors. Each of the two units of the appliance extended from the distal surface of the canine to the medial point in the dental arch, between the central incisors. This two-loop configuration would provide a much tighter fit around these six teeth than could be achieved by fashioning a single loop band.

The tooth now found in the band is tightly pinched into place, but rotated some 90° from any likely original orientation. In fact, the buccal (labial) surface of the tooth now faces the “right” end of the band rather than either the “front” or the “back” of the appliance. For the purposes of this description the end of the appliance with tooth will be considered as the right end, and the “front” of the appliance is thus arbitrarily determined. With the appliance in this position, the catalogue number (“A. 358”), carefully written in black ink, appears inverted at the lower left corner of the band.

Two lines of evidence strongly imply that the association of the incisor now in the Tanagra appliance occurred after the excavation and recovery of this artifact. First, this is the only tooth now in association with the appliance and this tooth shows no signs of the deterioration that commonly occurs in the environment of ancient chamber tombs. Since all six teeth encircled by this appliance, both canines and four incisors, must have been present in the owner’s mouth at the time of death, the recovery of only one perfectly intact tooth would be unlikely. Additional evidence relates to the present configuration of the band. Becker found the band to be highly resilient and not easily distorted, as might be expected in a pure gold strip. The tooth, somewhat obscured by the gold band, is missing a portion (approximately one-third) of the crown, but appears to be a left central maxillary incisor with slight shoveling (1 on a scale of 0–4). At present the tooth measures 21.5 mm long; the apical (root) tip is slightly damaged. An original length of c.23.0 mm is suggested here. The crown of the tooth has a vertical fracture and a portion of the “right” side (when looking at the labial surface of the

tooth) has been lost (cf. Copenhagen, no. 4). The breadth of the surviving piece of crown is 3.9 mm, and the maximum probable breadth of the dental enamel of the crown may have been c.5.8 mm, that point falling just below the level of the encircling band. A lateral measurement made from the left side of the crown to the exterior surface of the band on the right side (rear surface of the “encircling” part of the band) is approx. 5.8 mm. The breadth of the dental crown where it now emerges from the band appears to be about 5.4 mm, from which point a slight expansion of the crown takes us to its widest point.

Age and sex

The degree of wear seen on the articular surface of the incisor is slight to medium, implying an age of perhaps 45 ± 10 years at death. Without a comparative population of appropriate social class, though, this estimation of age must be considered as tentative. The estimated breadth of the incisor, *if Etruscan*, indicates a female. The estimated length of the tooth is less diagnostic of sex. As noted, it is not likely that this tooth is original to the appliance. In the selection of a “random” tooth to be part of a “display” or even to indicate function of this artifact, there is an approximately 50 percent chance that it came from someone of the same sex as the original owner.

The appliance: a band formed from a pair of long loops

The color of the surface of the appliance is largely a pure yellow commonly associated with gold, but several irregular blotches of reddish- or “copper”-colored “highlights” are easily observed. Whether these reflect metallurgical variations in composition remains to be investigated. Quite possibly the surface hammering of the gold to form the basic band led to differential surface enrichment of the gold (Becker 2003).

The composite appliance, formed by two relatively long loops, is now largely flattened. The “right” loop retains some slight degree of physical separation over its entire length. The other, or “left” loop, has a portion of its length so tightly “crushed” together that Becker could not visually detect the two thicknesses of the band. At present the greatest length of this “crushed” appliance, at point “N” on the “lower” margin of the band, is 43.3 mm. The length at “M” on the “upper” part of the band, at the present tooth root direction, is only 0.1 mm less, or a length of 43.2 mm.

The width of the gold strip varies from 4.3 to 5.6 mm, narrowest on the left end. Seven points (A through G) have been located along the “upper” margin of the “anterior” part of the band, with widths listed in mm at these locations.

- A 4.3
- B 4.8
- C 5.4
- D “6.0” (represents the total width of the two overlapping portions of one band)
- E 5.3 (only the anterior width; total width at this point including overlap is 5.8 mm)

F 5.6

G 5.4

In the anterior view of this appliance there appear two faintly visible “lines” evident between the points D and E. While both lines appear to be scratch-like in origin, the “line” at point E, which crosses the entire band, almost certainly represents the point of overlap in the band. This overlap remains evident despite the hammering or welding that was used to form the original loop. The line that appears closer to point D does not cross the entire band and is angled in such a way as to further suggest that this line may be only a surface scratch.

The thickness of the gold strip or strips used to form this appliance could not be measured, as there is no location at which a precision instrument of the types available could bracket both surfaces. At point W, where two parts of the appliance have been crushed or beaten together, the metal in the two parts of the band has a total thickness of c.0.4 mm, suggesting that at this point the individual dimensions are approximately 0.2 mm thick. These figures may be compared with the details of band thickness from other appliances.

Bilateral variations in mesial-distal tooth lengths are normal in any population (cf. Becker 2002a). Thus the lengths of each of the two loops in a well-fitted appliance such as this would be expected to reflect these slight differences. A simple band appliance surrounding six teeth would best be fitted with a two-loop system. While more difficult to fit among the teeth in a healthy maxilla, a two-loop system could provide a tighter and more stable appliance while an “open” or single-loop band around six teeth would be relatively unstable.

21 Sardis (Ancient Lydia)

Present location: Private collection, Ankara, Turkey. Present owner Dr. Ilter Uzel.
Type: band encircling four teeth; the two central incisors are replacements each held by one rivet; second incisors serve as anchor teeth.

Origin: said to have been discovered by a Turkish farmer in a tomb at Salihli, the area of ancient Sardis (Lydia, Anatolia), shortly before 1986 when it was first published. Details of the tomb context are cast into doubt because of the apparently illegal excavation/collection of the burial.

Jaw part: securely attached to original mandible—anchored to mandibular lateral incisors.

Date: earring and wreath said to match grave goods of Teano appliance, dated by Terzioğlu and Uzel to third century BCE (unsubstantiated).

Sex: female? Currently in what is said to be the original female skull; small tooth size (apparently).

Dimensions: gold band L. 48; W. 4 mm.

Description: an unusual mandibular prosthesis formed from a single strip of gold that Terzioğlu and Uzel measured at 48 mm long; hammered into a loop. The central incisor tooth sockets appear to have been fully resorbed (healed), perhaps

after the appliance was made. If correct, this would provide important evidence for evulsion and replacement of these teeth with an Etruscan-style pontic. Only the buccal ends of the rivets are illustrated, revealing irregular peening.

References

Terzioğlu and Uzel 1986.

Terzioğlu and Uzel 1987.

(Incorrectly cited as from an Etruscan tomb, by Teschler-Nicola *et al.* 1998: 64.)

Circumstances and description

The prosthesis, along with a skull and mandible in which it is said to have been found, is presented only in two brief articles by dentistry professors from the University of Istanbul and the Gülhane Military Medical Academy in Ankara. The appliance has not been made available for scholarly study. Such restriction of access has been known to occur in cases in which scrutiny of the evidence would reveal that this is a modern attempt to create an “archaeologically significant” object.

The complete skull (missing many teeth) was said to have been found in a tomb, with one gold earring and one small gold wreath, which led the authors to identify the deceased as female. They imply that it is similar to the appliance from Teano (third century BCE), stating that radiographic and other analyses show the skull to have an asymmetrical nasal bone (as do many/most persons today).

Five natural teeth are preserved with the skull; the owner later “found” the upper front teeth (Terzioğlu, and Uzel, 1987: 112, fig. 10). According to Terzioğlu and Uzel, tartar deposits on the replacement teeth show that the appliance was worn constantly. They maintain that the design and technical features of the appliance are of Etruscan manufacture.

Gold band 4 mm wide, 48 mm in length. As illustrated, two replacement teeth, carved from human teeth (or possibly bovine teeth, according to the authors) are each set with a rivet within a gold band; they are anchored by the two second incisors, wrapped with the gold band. The replacement teeth project slightly below the gold band, and their irregular, convex edges must have abutted the gum uncomfortably—or else the gum had already receded. The publications imply that the band held the false teeth above the gum so that it did not press on the tender gums, and thus was helpful for appearance and speech but not for biting or chewing. Given the evident loss (postmortem) of most of the other teeth, it seems odd that this appliance should have remained in place.

The authors cite Herodotus’ claim of Lydian origins for the Etruscans to explain the presence of a person wearing this prosthesis in the Lydian capital of Sardis. While the tale of Lydian migration to Italy has been debunked (Briquel 2013), there certainly were intensive commercial interactions and travel between Etruria and Lydia during the Archaic period (seventh–sixth centuries BCE) (Demetriou 2012: 82–83). By the third century BCE, though, conditions were very different.

PART II: CATALOGUE OF ANCIENT WIRE DENTAL APPLIANCES

The history of dentistry and oral medicine in the Near East is a subject only recently addressed by diligent scholars. Musitelli (1996: 213–217, n. 12–22) provides an excellent summary of the general topic with suggestions regarding future directions of research. Relevant to Etruscan ornamental dental appliances is the evidence from the Eastern Mediterranean and Levant of a completely different use of gold and silver wire as ligatures to retain loose teeth in place. This important aspect of dental medicine appears to have developed in the Eastern Mediterranean and Levant, passing into Western dental medicine. Unlike much medical knowledge in the Arabic world that derived from Greek and Roman traditions, the use of wire dental appliances may have had beginnings in the area of the Levantine coast and Egypt, to judge from archaeological evidence. Relationships with early Hebrew medical knowledge remain to be explored in greater depth, but wire ligatures were found to be so useful that the technology survived well into the twentieth century.

This tradition of fashioning dental appliances from gold or silver wire is documented from only seven examples, identified as W-1 through W-7 below. Six examples were retrieved from archaeological contexts along a small part of the eastern Mediterranean, two from Phoenicia, three from Egypt and one from Greece (Becker 1997). Carroll's (1972) comprehensive study of the process by which wire was manufactured in antiquity is of considerable value in understanding the technology involved in fabricating the wire used in these appliances (also Formigli 1979). (See description in Chapter 2).

Origins

In an attempt to demonstrate what they believed to be an Egyptian origin for all of these pieces Iskander, Harris and Farid (1978: 107) noted that the Phoenician Sidon I example was found with an impressive array of "Egyptian" artifacts. The Sidon II appliance also had Egyptian faience figurines with it. In fact, by the later first millennium BCE, much faience of Egyptian appearance (scarabs, amulets, beads) was actually produced in Phoenician factories in the Nile delta, Levant, and/or Phoenician colonies. Such amulets and faience vessels were avidly sought across the entire Mediterranean, so their presence cannot be construed as diagnostic of ethnicity. There is no evidence for an Egyptian origin for these appliances. Wiring to support loose teeth was mentioned in Greek medical texts (a fifth–fourth-century BCE Hippocratic work, reiterated in the first century CE by Celsus). Greek and Roman doctors knew about the treatment, yet most of the meager evidence has been found in the Levant and Egypt, with one example in Greece, and one in Rome.

While a case can be made that wire appliances were "Phoenician" in origin, the extant specimens are dated from the end of the fifth century through the

Hellenistic period or later—none approach in date the Etruscan example(s) of the second half of the seventh century BCE. Although some authors believe that all seven known examples may have been influenced from the Western tradition of dentistry, namely, the Etruscan gold-band appliances, no prototypes exist. The distribution and manufacturing techniques of the wire appliances suggest a Levantine origin for this distinctly medical procedure, apparently also informed by Greek medical traditions. More is involved in their design and installation than would be necessary for Etruscan band appliances.

No dental prosthesis of any type was known in Egypt until the Hellenistic (Ptolemaic) period. The literature on Egyptian examples is summarized by Bardinnet (1990), who provides references to his own thesis (1977) and the paper by Rombauts and Monier concerning Egyptian dentistry (Bardinnet 1990: n. 10).

W-1 Sidon I (sometimes referred to as “Gaillardot”)

Present location: Musée du Louvre, Paris, inv. no. AO 5777.

Type: gold wire, holding two (?) replacement teeth together with four others.

Origin: Sidon (Saïda), Lebanon. Tomb XI, Room 1, fosse b (1861) (Phoenician).

Jaw part: mandibular (???), C–C, holding replaced incisors I1 and I2.

Date: fourth century BCE or later (intruded into an earlier tomb? See Clawson 1933: 155).

Sex: female.

Dimensions: L. 34 mm (est.), W. variable, but generally three widths of wire.

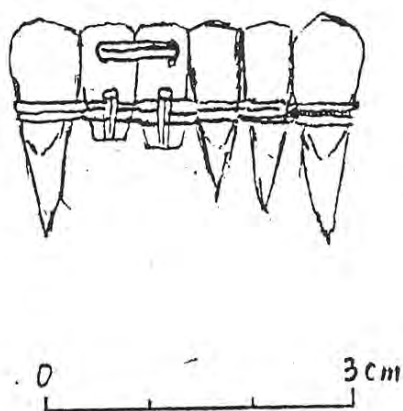


Figure 5.21a Wire appliance W-1 (Sidon I), drawing after Hoffmann-Axthelm 1970: fig. 2. Drawing by Becker.

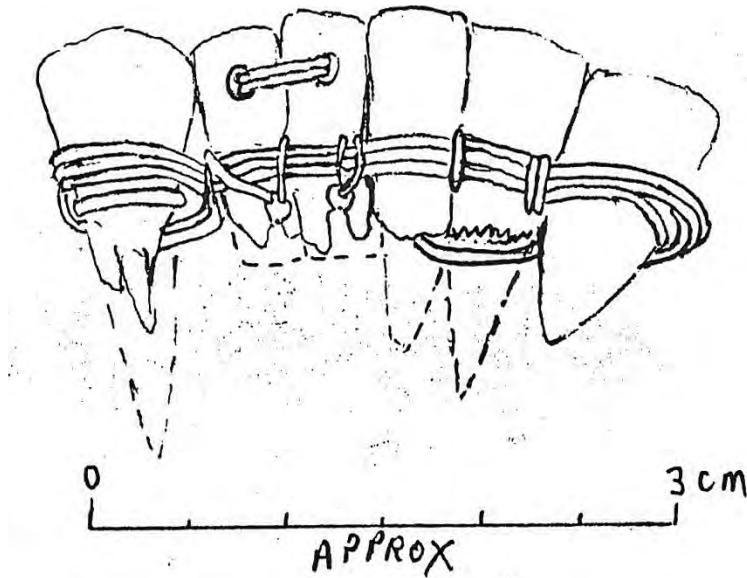


Figure 5.21b Schematic view of the buccal surface (?) of appliance Sidon I, after Clawson (1933: 156, fig. 8c). Drawing by Becker.

Description: gold wire pontic, used to replace both mandibular right incisors that had been lost or deliberately removed. These incisors were re-placed in position using wires that pass through two holes drilled front to back (buccal-lingual) in each tooth. Two horizontal ligatures pass through the crowns of these drilled teeth, and two vertical sets of wires pass through the neck hole and around the cut-down base of each replacement tooth. These lower vertical bindings pass over the three wires that pass horizontally around six teeth, including both canines. The wires that surround all six teeth were tightened by running wires between the teeth, and then pulling the external horizontal wires in towards the center. Where and how the ends of the gold wire were tied off is not indicated by any illustration.

References

- Asbell 1948: figs 1 and 2.
- Becker 1997: 79–80, figs.
- Bertolin 1955: figs 1–2.
- Bliquez 1996: 2645, “A,” figs 1 and 37 (no. 8), using data via A. Caubet (Louvre).
- Bobbio 1958: figs 2–4.
- Brown 1934: figs 3a and 3b.

- Casotti 1927: 625, fig. 1 (drawing and photograph).
 Clawson 1933: 155–157, figs 8c, 8d; 1934: 25–26, fig. 3 (the usually careful Clawson could provide only a poor schematic sketch of this appliance).
 Deneffe 1899: 52, 59, 83–88, fig. 8.
 Emptoz 1987: 546, 548–549, figs 4 and 5 (no. VII). (fig. 3 is incorrectly labeled).
 Farrar 1888: fig. 9.
 Filderman 1931 (with 5 figs).
 Guerini 1894a: 396 and fig. 9.
 Guerini 1894b: fig. 9.
 Guerini 1909: 29–31, figs 3–4.
 Fontan 2004: 293 no. 92, catalogue entry by Elisabeth Fontan, with recent color photos. Gives Louvre inv. no. AO 5977.
 Hoffmann-Axthelm 1970: 81, fig. 2 (superior).
 Hoffmann-Axthelm 1985: 38–39, fig. 22.
 Iskander *et al.* 1978: 106–107, pl. V, B (from Weinberger 1946).
 Koch 1909: 8–9, fig. 6.
 Lufkin 1948: 49, fig. 15.
 Quintarelli 1946: 36, fig. 5.
 Rath 1958: fig. 1 (photograph of a copy in Köln).
 Renan 1864: 462–463, 472–473 with poor sketch of appliance, 484 (updated inventory of finds).
 Ring 1985: 28, fig. 19, and on p. 6 (excellent photographs).
 Sterpellone and Elsheikh 1990: 124, fig. 60. Depicts a plaster reproduction said to be in a Roman museum (see Appendix II). On p. 134, fig. 62 center reproduces two different drawings of this same appliance.
 Sudhoff 1926: 34, fig. 26 (drawing and photograph).
 Van Marter 1886: 60, fig. 4.
 Weinberger 1946: fig. 4; 1947: fig. 5; 1948: 100–103, fig. 32.
 Weiss 1989: fig. 30 (from Renan 1864).
 Woodforde 1967: fig. on p. 10.

History and context

The Sidon I appliance is one of two wire prostheses recovered from the Phoenician necropoleis at Sidon (modern Saïda), in southern Lebanon. Sidon was, since the Bronze Age, a major Canaanite and Phoenician city, prospering again through the Hellenistic and Roman Imperial periods. Some Phoenician-period tombs (sixth–fourth centuries BCE) appear to have been re-used in the Roman era.

(Joseph) Ernest Renan (1823–1892) was with the invasion team sent by Napoleon III into the Lebanon, ca 1860. His monumental *Mission de Phénicie* (1864) published exciting discoveries at several Phoenician cities, including the first account of the discovery of the Sidon I appliance, based on Gaillardot's journal (daybook) of the excavations in 1861. Finds were consigned to the Louvre and Gaillardot updated the inventory once they were received in Paris (Renan 1864: 484). Elisabeth Fontan (2004: 293, no. 92) gives the name of the necropolis location as Magharat Tablun.

Working in the southeast necropolis on the outskirts of Sidon, on April 14, 1861 Gaillardot began excavating the pit (*grande caveau* XV) in which on May

14–24 workmen cleared several linked, rectangular chambers. The tomb at issue, 8 meters below the surface, was labeled Tomb XI, Room 1, fosse b. (In 1901, Charles C. Torrey of the American School of Oriental Research (Jerusalem) would excavate a nearby tomb and discover the Sidon II appliance (no. W-2). Gaillardot suggested that the ceiling had collapsed before the tomb could be looted, preserving somewhat sparse finds and fragmentary skeletons. The room held two trenches (*fosses*), one on each long side, which may have held disintegrated wooden coffins.

“*Fosse b*” held several grave goods mixed in the sand filling along with the portion of what was thought to be a woman’s upper jaw with the wire appliance in place. Also recovered were: 1 “teardrop” (vase) in white glass; 1 blue glass bead; 2 copper coins (bronze?) illegible; 1 small copper disc with suspension hole (not cited in updated inventory); 12 small figurines/amulets in faience depicting Egyptian gods, all pierced to be strung as a necklace; 4 small faience amulets in the form of fantastic animals; 1 scarab, pierced for suspension; 1 small pierced ornament, described as “(phallus?)”—perhaps a *djed*-pillar amulet; 1 heavy iron finger ring “*forme chevalière*”; and 1 ceramic vase (variously described as small or large).

Gaillardot’s final inventory also cites four mirrors, but this might be a confusion with mirrors found in adjacent burials. No images of the objects were published, and the descriptions are not very useful: the glass beads and faience amulets are types favored from the eighth century BCE through the Ptolemaic period. A very likely date, given the proximity of similar tombs and finds would be the fourth or third century BCE. Since the tomb was near that in which the other wire appliance was found, within a well-dated marble sarcophagus (of the fourth quarter of the fifth century BCE), it is tempting to assume a date around 400–350 BCE, but this could be improved if the associated offerings can be more closely dated. Elizabeth Fontan (2004: 293, no. 92) dates the appliance c.400 BCE. She states that an X-ray study showed that an abscess had infected the bone of the lower jaw.)

Description

Renan’s published description (1864) of this wire dental appliance was provided by Dr. Gaillardot, who actually discovered the piece and acted as Renan’s medical assistant. The fragmentary “upper jaw” (actually mandible) that Renan found (there is no mention of a sarcophagus or complete skeleton) held a gold-wire appliance holding two canines and four incisors. The two right incisors were held tightly in place with wire wound through holes drilled in them. Clearly the roots of the owner’s live teeth encircled by this appliance had deteriorated while in the tomb, having almost completely rotted away. The surviving teeth as well as the cut-down and re-placed incisors in this pontic appeared quite similar when excavated.

The canines and incisors bound by the gold wire are said to derive from “*une portion de mâchoire supérieure de femme*,” but the teeth as illustrated appear

to be mandibular. Since none of the roots are intact, the bound set of teeth may have been in a position that was ambiguous to an archaeologist—no mention was made of the rest of the skull or skeleton, and other burials had only fragmentary bones found in the sandy fill. The specific configuration of the crowns of these teeth, however, leaves no doubt as to their original location in the jaws of this person. Two human incisors are held in place as false teeth. Clawson suggests that they derive from another individual, but they may be the owner's own teeth, lost through a blow, if not deliberately removed.

Filderman (1931: 336) provides the best and most complete description of the Sidon I appliance itself, including detailed measurements of the teeth and (correctly?) suggests that this is mandibular (Clawson 1933, 1934). Gaillardot (in Renan 1864: 423–473) had (incorrectly?) identified this piece as maxillary (Deneffe 1899: fig. 8; Hoffmann-Axthelm 1970: fig. 2), but correctly identified the wearer as a female. Although Clawson (1934: 26) is correct in noting that no gender-related morphological variations can be detected, absolute size may be used to identify gender in some populations (Becker and Salvadei 1992).

One or more complexly wound wires are used to suspend two “replacement” teeth in position. The replaced, or artificial teeth each have two holes drilled through them (buccal-labial) to allow the gold wires to attach these to the sound teeth. The roots of the replacement teeth have been cut off.

The I1 and I2 (probably the natural teeth) are suspended above the area from which they were lost, possibly due to periodontal disease. In some illustrations the I1 also appears to be replaced. Weinberger (1948) reviews related data with copies of the early drawings and two new photographs, but possibly errs in believing these to be maxillary. Weinberger suggests that the wire had been taken off these teeth and later replaced, with the teeth not returned to their proper order. Weinberger (1948: 102) also believes that at least one of the replacement teeth comes from another individual. Hoffmann-Axthelm's (1970: fig. 2) illustration appears to show a left canine in the appliance, which may derive from another person. Van Marter (1886, repeated by Lufkin 1948: 50) suggests that both “replacement” teeth come from another person. Ring (1985) erroneously believes the replacement teeth are ivory. Fontan (2004) presumably follows this in stating the material is ivory. She states that X-ray studies revealed an abscess affecting the bone, but without references.

The wire binding these teeth passes around the anchor teeth at the base of the crowns. A hole for the wire is drilled B-L through the center of the crown of each “replaced” tooth. This wire also appears to pass through a second hole drilled in each tooth, near the base of the crown. Two turns of wire are visible on the buccal surface, passing between the holes near the centers of the crowns, and this wire (or wires) must be evident on the lingual surfaces of these two replacement teeth. How they were “tied off,” whether they were connected to wire holding all six teeth, and if there are two or three wires involved in this prosthesis would require further detailed study of the actual appliance.

W-2 Sidon II (sometimes called “Ford” or “Torrey”)

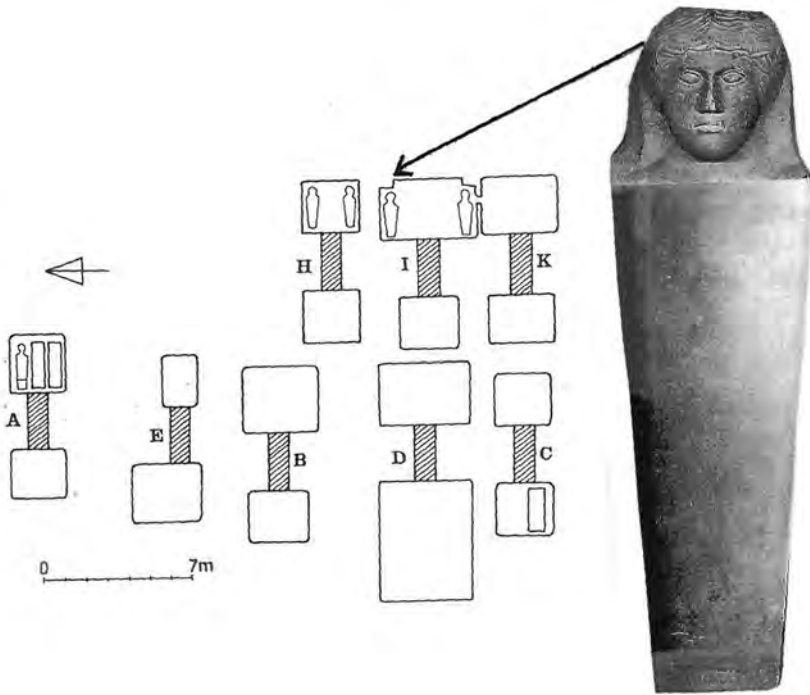


Figure 5.22a Plan of the tomb in which the appliance was found, within the Sidon necropolis (near the tomb of Sidon I), after Clawson 1933: 142; inset: approximation of the sarcophagus in which the appliance was found.

Present location: Archaeological Museum of the American University in Beirut, Lebanon. Catalogue no. 5998. (Sarcophagus that contained the skeleton is in Beirut, National Museum no. 4366).

Type: gold-wire brace.

Origin: Sidon (modern Saïda), Lebanon, ‘Ain Hîlweh necropolis, Tomb I, East room.

Jaw part: mandible, C–C.

Date: fifth century BCE.

Sex: male.

Dimensions: L. 35 (est.), W. 3–4 (?) mm.

Description: This gold-wire retention device was woven around and between six teeth, probably the mandibular canines and incisors. The detailed pattern of this wiring, made from a single gold strand over 100 mm long, and technique of manufacture are well illustrated by Clawson (1933).

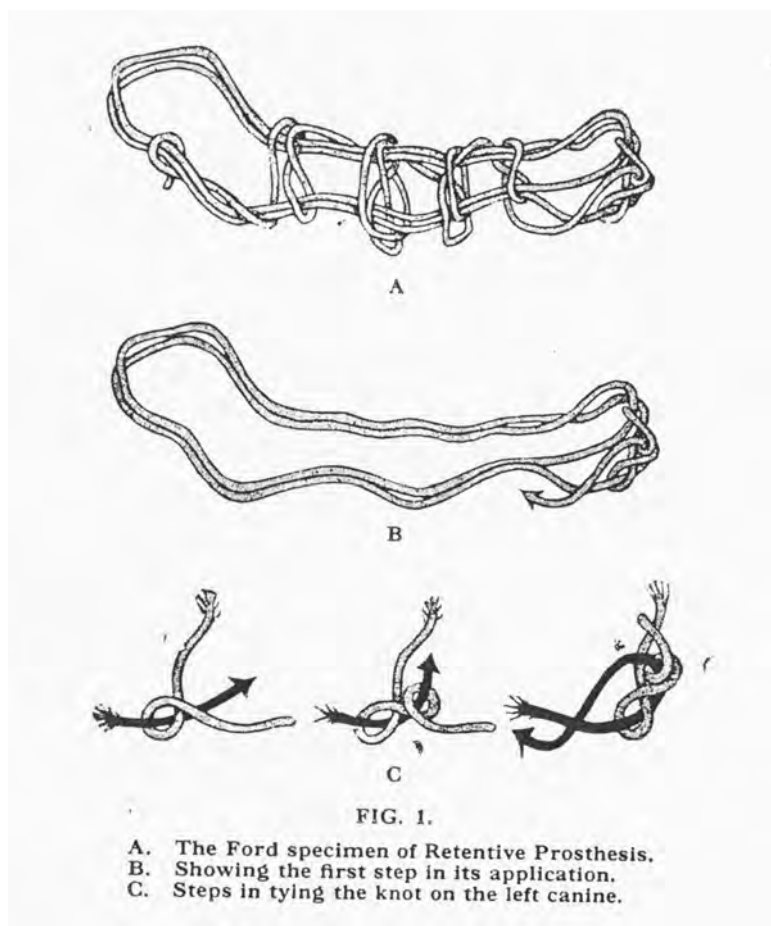


Figure 5.22b Sidon II appliance, after Clawson (1933: 147, fig. 1), showing it as it was removed from the teeth, plus drawings showing details of the ligation.

References

- Asbell 1948: 816, figs 5 and 6.
 Badre 1986.
 Badre 2004: with recent color photograph of appliance in mandible.
 Becker 1997: 80–81, figs.
 Bertolino 1955: figs 4, 5.
 Bliquez 1996 (noted under his “A”).
 Bobbio 1958: figs 5–7 (from Bertolino 1955).
 Clawson 1933: 142–143, 147, 150–153, 159; figs 1 and 8e (very detailed account with archaeological drawings of excavation area and appliance, as well as photographs; figs 1–3).
 Clawson 1934: 27–31, figs 5–6c.

- Deneffe 1899: 52, 59, 83–88, fig. 8.
Emptoz 1987: no. VIII.
Ginestet 1927: figs 1, 2 (poor illustrations).
Guerini 1909: 29 (mentions it, only).
Fontan 2004: 292–293 no. 91
Hoffmann-Axthelm 1970: fig. 3 (the lower illustration, the appliance removed from mandible, is either the reverse or the underside of Clawson's illustration (1933: 147, fig. 1a).
Hoffmann-Axthelm 1985: 38–39, fig. 23 (from Torrey 1920: figs 14, 15).
Iskander *et al.* 1978: 107–108, pl. VI A.
Ring 1985: fig. 18, and page 28.
Tabourian 1932 (cited by Clawson 1933: 160, n. 10, but not located for this project).
Torrey 1920, vol. I: 14–15, figs 14–15.
Weinberger 1946: fig. 5.
Weinberger 1947: fig. 6; 1948: 103–106, figs 33–34.

History and context

The Sidon II appliance was discovered in January, 1901 during excavations by the American School of Oriental Research in Jerusalem in the 'Ain Hilweh necropolis, which lay about one mile southeast of the boundary of the ancient Phoenician city of Sidon. Tomb I containing this gold-wire appliance (plan and section in Clawson 1933: 142; reproduced by Lembke 2001: 14 fig. 6) lay immediately east of Tomb XI, where another wire dental appliance (Sidon I) was found by the French expedition of 1860. Tomb I (reused and joined to adjacent Tomb H in antiquity) was a rock-cut chamber tomb accessed by a deep plastered shaft; the larger of two rooms, on the east, had a niche on each side to accommodate a marble anthropoid sarcophagus. The tomb later held bodies in wooden sarcophagi, when the necropolis was reused during the Roman Empire, but the pair of Phoenician-era marble sarcophagi remained undisturbed. One sarcophagus portrayed a youthful male, the other was interpreted as his wife, but Lembke (2001: 15, 131–133, no. 41) has noted the rather short hair, suggesting that this sarcophagus also depicted a male (see Clawson 1934: pl. VII; and Torrey 1920: 3, fig. 2; 14–15, figs 14–15; 20, fig. 18).

"Anthropoid no. 8," the male sarcophagus, held a well-preserved skeleton that had been wrapped in linen (presumably many of the occupants of these sarcophagi had been mummified). (See Lembke 2001: 129, no. 29 for full analysis and additional references on the sarcophagus and its style.) With successive cycles of flooding of the tomb, the linen had disintegrated except for some tatters adhering to the sides of the sarcophagus (Torrey 1920: 13, fig. 12, upper right corner illustrates a fragment of the textile). The two sarcophagi seem never to have been disturbed and no. 8 was said to have contained a gold seal ring "of excellent Greek workmanship" (Torrey 1920: 21, not illustrated). The sarcophagus (2.15 m. long) is smooth with only the head carefully carved, with classical Greek features and short, wavy hair, and bosses for handling on shoulders and foot. On the foot was incised the Greek letter Δ, a mason's mark indicating East Greek, or Ionian,

workmanship (Lembke 2001: 107–109), and traces of red paint remained in the man's hair. Stylistically, the sarcophagus shows the influence of Greek art on the Egyptianizing, mummiform format favored by the Phoenicians.

Marble sarcophagi are not common, but have been found in other Phoenician cities and in the colonies of Cadiz and Soloeis/Solunto (Sicily), where they may denote high-born personages. Similar sarcophagi in terracotta or wood imitated these forms for less expense. Certainly, the extremely costly (in terms of manpower and time) tomb and rare sarcophagi attest to their occupants' importance in Persian-occupied Phoenicia. Only the very affluent could have availed themselves of such gold prostheses. The necropolis area that held this tomb and that of the Sidon I appliance probably was used by members of related families, and may represent the earliest context for wire appliances: Katja Lembke, in a survey of all Phoenician anthropoid sarcophagi, has dated the one which held the dental appliance to the fourth quarter of the fifth century BCE, and placed the companion sarcophagus just slightly later, in the first quarter of the fourth century (Lembke 2001: 129 no. 29; 131–132, no. 41; 98–99, figs 20–21, pls. 16a–d and 22d—she does not mention the dental appliance).

This dental prosthesis was found on the jaw of the skeleton identified as a middle-aged adult male (Clawson 1934: 27; Asbell 1948: fig. 7). For this information Clawson (1933: 160, n. 4) cites T. Reinach ("Necropole" pp. 145–178, presumably Hamdî and Reinach 1892, describing finds in this necropolis). Leila Badre (catalogue entry for Fontan 2004: 293) notes that the skeleton was that of a man of uncommon body size, whose lower front teeth were affected by periodontal disease, occasioning the need for bracing with wire. Clawson (1933: 160; 1934: 27) identifies this appliance as the "Ford" specimen, naming it for an American physician who had lived in Sidon for over 20 years, but who had died by 1933. Ford's will left this appliance to the Museum of the American University in Beirut. We do not know the details of the apportionment of the finds from this excavation. Clawson (1933) provides the best description. Dr. C. F. Torrey, first Director of the American School of Oriental Research (ASOR) in Jerusalem (established 1901) often has his name associated with this appliance, because ASOR was the agency directing the excavation of these tombs (Torrey 1920). Along with W-1, this prosthesis (in a mandible) was shown in 2004 in the exhibition "*Hannibal ad portas. Macht und Reichtum Karthagos*" held in Karlsruhe, Germany, at the Badisches Landesmuseum.

Description

Clawson (1933: 143, photographs and fig. 1) shows two views of this appliance as it had been removed from the jaw. These views indicate that the Sidon II appliance had enclosed six teeth, with the wire completely encircling each of the four teeth on the left (?), but with the two teeth on the right (?) bound as a single unit with no wire passing between them. The appliance is fashioned from a single strand of pure gold wire, 24 gauge, and weighs just over two grams (Clawson 1933: 146; 1934). Clearly this was meant to hold loose teeth in place;

teeth that Clawson believes had been loosened by extensive pyorrhea. Gold wire was looped around six teeth (C–C). The mandibular 1PM had been lost before death, and Clawson (1933: 149; 160, n. 10) suggests that Tabourian believes that the lost tooth was the result of an accident. Interestingly, the only decay noted in the other teeth appears in the right (first?) premolar (Clawson 1933: 149). Serious periodontal resorption, most strongly noted around the central incisors, shows a bilateral symmetry in its degree around the lateral incisors and declines past the area surrounding the canines, to a trace around the molars. All of the bones are well preserved and said to be tinted with a greenish color, possibly from proximity to copper salts (Clawson 1933: 151). Clawson also provides a summary of the associated grave goods (from Torrey 1920).

Clawson (1933: fig. 1) illustrates the pattern used to weave the gold wire around the teeth, describing the weaving in some detail (Clawson 1933: 146–147), noting, “salivary [calcareous] deposits had overlain the appliance at this point and had to be scaled away before the splint [appliance] could be taken from the teeth.” Clawson (1934: 24) notes that Ginestet (1927: 17) errs in describing this piece as composed of two separate wires. Asbell (1948: 816) provides fine illustrations of the teeth and sarcophagus.

W-3. Alexandria

Present location: Greco-Roman Museum, Alexandria, Egypt (appliance now lost?).

Type: gold wire.

Origin: Ibrahimia (Ibrahimieh), near Alexandria, Egypt.

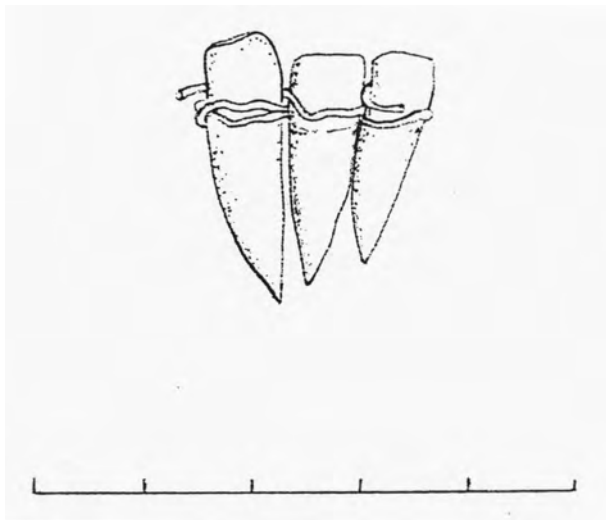


Figure 5.23a Wire appliance, W-3 (Alexandria), showing buccal surface of the right canine and incisors (after Weinberger 1948: fig. 40). Drawing by Becker.

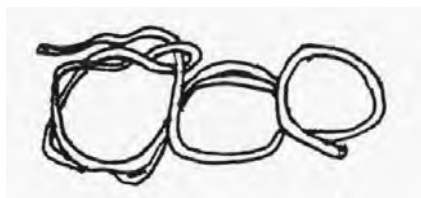


Figure 5.23b Alexandria appliance, a superior view, adjusted from Weinberger (1948). Drawing by Becker.

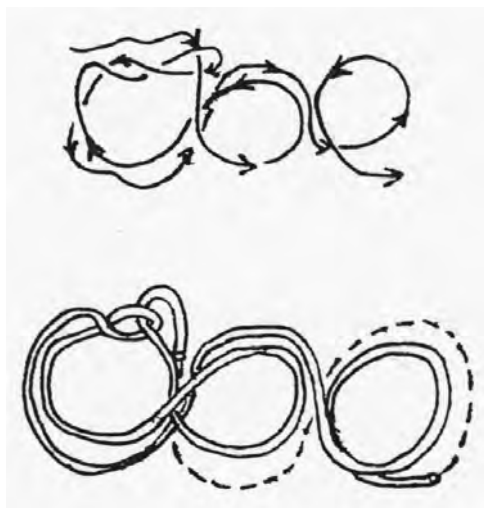


Figure 5.23c Alexandria appliance, diagram and sketch, showing Becker's suggested method of ligation. Drawing by Becker.

Jaw part: mandible, I1–C.

Date: Roman period (first century CE).

Sex: ?

Dimensions: L. 25 mm, W. variable (loosened wire).

Description: thin gold wire, 1 mm or less in diameter, has been used to stabilize a mandibular left lateral incisor by binding it to the two adjacent teeth. No holes are drilled in any tooth. The wire passes twice around the canine but now not completely around the incisors. Originally it probably did complete a second loop around the central and lateral incisor, but a small piece of wire, perhaps 21 mm long, has broken off.

References

Bardinet 1990: 2554, fig. 3 (after Weinberger).

Becker 1997: 82–83, figs.

Brown 1933: 584; 1934: 829.

Iskander *et al.* 1978: 105–106, pl. V, A (drawing from Weinberger 1946).

Lufkin 1948: 41 (citing Ruffer 1921: 314).

Ruffer 1921: 314 (but, see below).

Weinberger 1946: fig. 7.

Weinberger 1948: 119, fig. 40.

History and provenance

The origins of the Alexandria appliance remain vague. Ruffer (1921: 314, cited in Lufkin 1948: 41) appears to refer to this appliance as a series of teeth bound with a gold wire, found in a Roman tomb in Egypt by a Professor Breccia (obviously about or before 1921).⁷ Earlier, Ruffer (1920) appears to have referred to the same appliance, as a set of artificial teeth, suggesting a denture. Since the Greco-Roman Museum was founded in 1892, this appliance probably was acquired by the Museum between that date and 1920.

Brown (1933: 584) was given some information about the Alexandria piece by a Mr. Aloriani of the Greco-Roman Museum in Alexandria (below), so it may have been there in 1933. Weinberger (1946: fig. 7) stated that it was found in a Roman grave of the third century CE at Ibrahimia. Some years later Iskander and colleagues (1978: 106) could not find it. More recently Bardinet (1990) tried to locate it, but also was unsuccessful. An attempt by Becker to clarify its location with the Director of the Greco-Roman Museum, Dr. Doreya Said (Becker to Said, personal communication, July 1, 1994) was unsuccessful.

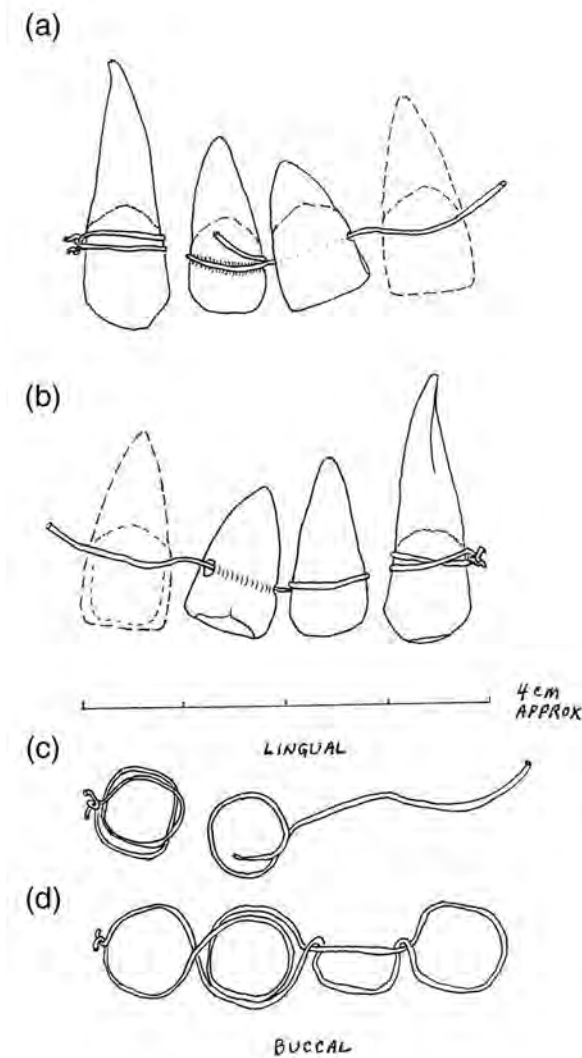
In reply to Brown's (1933: 584) enquiry, Mr. Aloriani of the Greco-Roman Museum indicated that this mandibular appliance included three natural teeth (designated as 3, 2/1) connected with gold wire. Weinberger's drawing (1948: fig. 40) seems to indicate that the teeth are a mandibular right central incisor to right canine. He notes that a single piece of gold wire around their gingival margins binds these three teeth, the middle one of the three, or the I2, most probably being supported by the appliance. Weinberger says that "signs of abrasion" are present on the teeth, suggesting that this is "due undoubtedly to the mechanical action of the gold wire." He states that "The teeth are considerably worn and over calcified, with tartar extending halfway down the roots." Dr. L. Avrouskini of Alexandria had sent photographs to Weinberger, who stated stated that the photographs "show the accompanying drawing to be correct" (Weinberger 1948: fig. 40).

Description

In Becker's redrawing of Weinberger's illustration of the Alexandria appliance it became evident that no means of providing tension or a secure bond is evident between the incisors. However, if the end of the gold wire at the lingual-distal aspect of the canine, as shown by Weinberger, is turned in the *opposite* direction and passed in front (buccal) of the lateral incisor and then behind (lingual) to the central incisor, it would meet the other end of the wire. At that location the ends

could be twisted together to lock the appliance in place and provide the needed tension to retain a loose lateral incisor in place. The present wire may be sufficiently long to achieve this closure if all the slack now seen were removed by pulling. It seems likely that a small section of the wire may have been lost.

W-4 El-Qatta



Figures 5.24a–d Wire appliance W-4 (El-Qatta): (a) labial surface, with teeth; (b) lingual (buccal) surface, with teeth; (c) configuration of the wire, as inferred by Becker from Harris *et al.* (1975); (d) a possible arrangement of the wire appliance, as suggested by Becker. Drawings by Becker.

Present location: believed to be at the “Cairo Museum,” inv. no. 8/75:2/1 (or 82/751).

Type: gold wire support for a re-placed maxillary central incisor.

Origin: a mastaba of the Old Kingdom, reused in Roman period, in the cemetery of El-Qatta (near Imbaba, according to Harris *et al.* 1975: 401) about 40 km northwest of Cairo.

Jaw part: maxilla, II–C?? (but only I1, I2, and C survive).

Date: Ptolemaic? (32–31 BCE) or Roman era

Sex: uncertain.

Dimensions: L. 43 mm (broken), W. variable (loosened and broken wire).

Description: The two pieces of gold wire assumed to have been part of a single original length of wire now are associated with three maxillary teeth. A right canine is surrounded by two loops of gold wire, twisted closed at the distal side. A right lateral incisor is surrounded by a single loop of wire, which then passes through a hole drilled laterally through a right central incisor that has been cut down for replacement in this jaw. On the mesial exit from the drilled hole the wire now extends 17 or 18 mm. No other tooth was recovered but Becker reconstructs this appliance as having encircled the now-missing left central incisor for use as an anchor tooth. The fact that the root of the drilled tooth has been filed down suggests that this appliance was intended to re-place a central incisor in the Etruscan fashion.

References

Bardinet 1990: 2554, fig. 2 (a poor sketch).

Becker 1997: 83–84, fig.

Harris *et al.* 1975 (includes four figures and an extensive but imprecise description).

Harris and Ponitz 1980: 50, fig. 3.5 (an error-filled summary).

Hoffmann-Axthelm 1985: 29, figs 15–16.

Iskander *et al.* 1978: 108–112, pl. VI, B and also pl. VII through X.

Ring 1985: 36, fig. 28 (an enlarged, but indistinct photograph.)

History and provenance

The archaeological location known as El-Qatta is located on the edge of Egypt’s Western Desert. The site is primarily known for a series of “mastaba” tombs (“bench”-shaped burial structures) of the Old Kingdom, probably Fourth Dynasty. Iskander and colleagues, including the excavator (1978: 109) indicate that the superstructure of Tomb no. 90, the mastaba from which the appliance derives, had been re-used for burials during the Roman period.

The appliance was found in 1952 by Dr. Shafik Farid (Exc. Reg. 58), but was overlooked until 1974. Details of the archaeological context have never been published (but see Iskander *et al.* 1978: 109). The wired series of three teeth are said to be from the crushed skull of a burial in Shaft 5 (of 6). Harris (*et al.* 1975; also Iskander *et al.* 1978) seem to imply that this appliance has an Old Kingdom date because it was associated with a mastaba (see Iskander *et al.* 1978: pl. I)

of that period, but re-use of such funerary structures was quite common. Ring (1985) suggests an Old Kingdom mortuary prosthesis, made to be fitted to a mummy. No Egyptian “mortuary prostheses” of any date are known, however (see Chapter 2).

Bardinet (1990: 2554) carefully points out that the burial chamber from which the El-Qatta appliance was recovered had been reused in the Roman period. This appears to be a true dental prosthesis, but it most certainly dates to the Roman period. Leek (1967, also 1981) provides the best evaluations of the dates of these artifacts. The errors made by Harris and his colleagues are similar to those made in their evaluation of the Junker amulet.⁸ The quality of the archaeological record for El-Qatta leaves a great deal to be desired. The lack of any scale in the published illustrations, and the absence of measurements of any of the associated teeth renders impossible an evaluation of gender.

Both Quenouille (1975) and Emptoz (1987) refer to a gold-wire appliance that they believe to be located in Cairo, but their commentaries are not sufficiently specific to enable us to identify it. Both of these authors probably refer to the El-Qatta piece, which Emptoz dates to the fourth–third centuries BCE; the basis for the date is not explained. Becker’s attempts to locate it in Cairo in 1993, through the efforts of Professor Krishna Kumar (West Chester University), were not successful. Attempts to trace the history of this appliance, through a program of letter writing, have been equally unrewarding.

Description

The evaluation of the El-Qatta appliance as a dental prosthesis is based on the teeth involved as well as on the apparent technique of fastening. Direct descriptions are available from Harris *et al.* (1975) and are repeated by Iskander *et al.* (1978). The latter account provides a somewhat more clear evaluation and much better illustrations, but the overall presentation is still wanting.

Iskander *et al.* (1978: 111) believe that the presence of “calculus” (tartar) “on the cuspid and the substituted lateral incisor demonstrate that the bridge was worn by the patient for [a] relatively long period of time.” Whether this “calculus” is true dental detritus or actually a coating that formed on all the objects in the tomb need not be considered here.

The teeth recovered appear to be a right central incisor, a right lateral incisor, plus a “cuspid” (canine). This worn canine, “around which has been placed a double strand of gold wire twisted into a knot on the distal surface” (Iskander *et al.* 1978: 110), is generally believed to be an abutment (anchor) for a bridge. Only the photograph (Iskander *et al.* 1978: pl. IX) provides evidence that this is a *single* wire, in the form of a crossing-over of the wire that is evident on the buccal surface of the canine. Iskander and colleagues (1978: 110) believe that the canine “has a gold loop in the wire that was twisted around it on its distal surface. To this loop was attached by a hook the gold wire on which were placed the two teeth just described.” The other anchor tooth, the left central incisor, is said to have “been lost with the broken parts of the skull” (Iskander *et al.* 1978: 404).

In fact, no hook and loop arrangement is evident around the canine in the illustrations provided by Iskander *et al.* (1979). A hook and loop arrangement would not be a possible attachment method for a dental appliance using soft gold. Furthermore, why the “loop” would be on the “distal” surface of the tooth when the replaced teeth abut this canine’s mesial surface cannot be explained. Clearly what we have here, called a “loop,” is a typical twist of the wire ends used to “tie off” the wire appliance (see Harris *et al.* 1975: 402, fig. 4).

What is evident from both of these texts is that the two components were not described *in situ*, and that the arrangement or order of the teeth is a matter of some speculation. Iskander *et al.* (1978: 111) clearly states that “this prosthesis was originally an anterior bridge. The arrangement of the three teeth as presented to us must have been, therefore, changed since its discovery or *post mortem*.” They then conclude that this appliance included two false teeth bound to two living teeth, a pattern that may be possible. However, if this appliance held two replacement incisors, then we must ask why only one of the two “replacement” incisors has been drilled (see below)? Considering the nature of this find and the early “descriptions,” another scenario may be suggested. The drilled central incisor may be the only replacement tooth, but it was anchored to the lateral incisor as well as to the canine and left central incisor.

One illustration of the El-Qatta appliance (Iskander *et al.* 1978: pl. VI, B) clearly indicates that someone has linked these teeth by bending, as well as straightening, various parts of the gold wire, a procedure that has destroyed the evidence for the way in which the wire originally related to these teeth. It depicts a length of gold wire that might have been sufficiently long to encircle two replacement teeth. Our understanding of the specific construction arrangement of the El-Qatta appliance may be distorted because the original author failed to provide a careful description of the appliance before someone tampered with it. We have no idea how the “appliance” was “constructed” before it was “assembled” in the pattern generally seen in various illustrations. The label for fig. 2 of Harris *et al.* (1975: 403) states that: “The lingual view of the bridge *assembled according to the author’s* [sic] *contention* that the cuspid was a right maxillary abutment tooth” (emphasis added).

The other “two teeth placed on a long gold wire completed the specimen” (Harris *et al.* 1975: 402). One of these

two teeth appears to be a central Incisor [sic] where a hole has been drilled in a mesial distal direction through the crown. The root of the tooth is said to have been scraped to give the root a smooth and artificial appearance. Also, a buccal groove is said to have been prepared parallel to the hole to accommodate the gold wire from outside.

(Harris *et al.* 1975: 402)

Harris (*et al.* 1975: 402, figs 2, 4) indicate that the central incisor has had extensive wear, and is extremely “shoveled” (about which more below). In fact, the groove is on the labial surface, as they correctly note in fig. 3 (caption) and again

on p. 404. However, the wire is *not* shown encircling this incisor, but only passing through it. The “groove” on this central incisor (I1) is described by Iskander and colleagues (1978: 110) as located on the buccal surface, but on the following page (p. 111) it is said to be “a labial groove.” The photographs provided do not help solve this problem as no “groove” is evident on any of these teeth. *Drawings* of the right central incisor, however, do seem to indicate a groove on the labial surface of the crown.

The possible lateral incisor is not likely to be a second “false” tooth secured by this appliance. Iskander *et al.* (1978: 110) state that “The other tooth has the gold wire wrapped around it with a labial groove prepared to accommodate this wire” (see also Harris *et al.* 1975: 402, 404). This tooth would appear to be a lateral incisor, and these authors state that the root has been scraped giving it an artificial morphology. This supposed reconfiguration of the I2 is less certain. Iskander *et al.* (1978: 110) also note that “[I]n addition, considerable tartar is found on the gingival two thirds of the root.”

Bardinet (1990) points out that the technique of mesial-distal drilling, assumed to hold one tooth in place (the I1) for the El-Qatta appliance, is like that found with the Tura El-Asmant silver wire appliance (no. 15), another example where only one tooth has been replaced. Bardinet suggests that El-Qatta has a similar date, probably during the Ptolemaic period, or during the Roman period in Egypt. Bardinet’s fig. 2 shows the three teeth of this El-Qatta appliance (I1–C.), so well illustrated by Harris and his colleagues. No mention is made by Harris *et al.* (1975) of how this appliance functioned.

Regarding the shoveling observed on this tooth, this non-metric dental feature is relatively common in central Italy and probably elsewhere, but the trait is rarely mentioned in dental studies in the Mediterranean region. While noted as common in central Italic populations, the exact incidence is not known for other regions. Professor Joel Irish (2006) provides a review of up to 36 morphological dental variants in 15 Egyptian samples, Neolithic through Roman, revealing shoveling incidence greater than 40 percent in some groups, while others have none. The high incidence groups suggest that the person at El-Qatta with this appliance was most likely a local resident.

W-5 Tura El-Asmant

Present location: Iskander *et al.* (1978: 105) state that this appliance is at the Centre of Research and Conservation of Antiquities, General Organization of Egyptian Antiquities, Cairo, “but it will be shortly exhibited in the Cairo Museum.”

Type: silver wire.

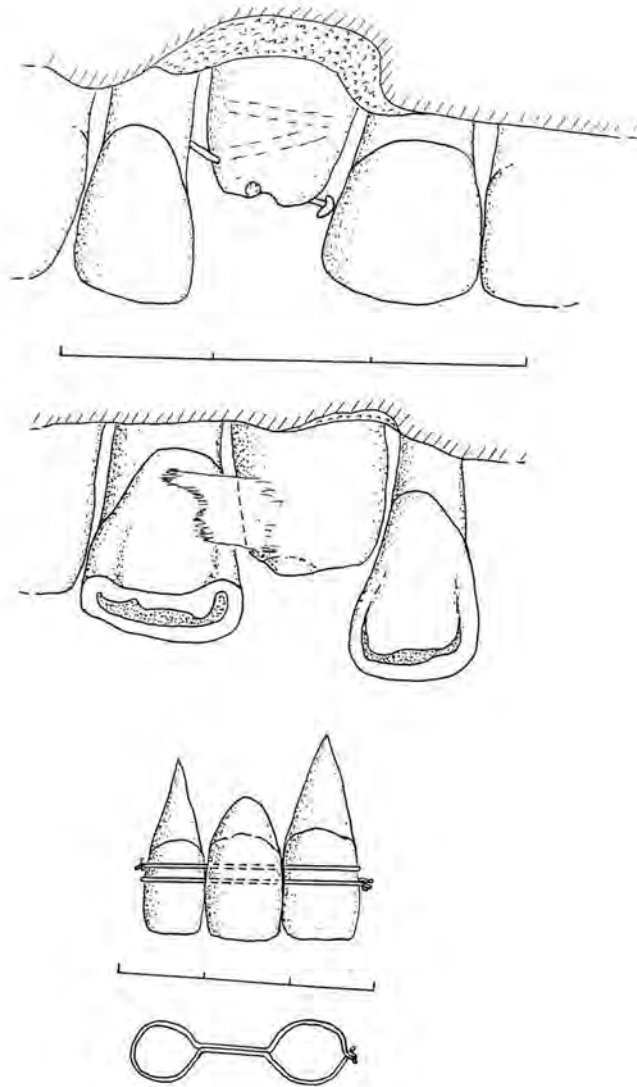
Origin: Tura El-Asmant, Egypt.

Jaw part: maxilla (I1–I2)

Date: Ptolemaic (323–31 BCE)

Sex: male??

Dimensions: L. 27 mm, W. two lengths of silver wire, each less than 1 mm diameter.



Figures 5.25a–d Wire appliance W-5 (Tura El-Asmant): (a) labial view, after photographs by Quenouille (in Iskander *et al.* 1978: pl. III, IV); (b) lingual view (cf. Figure 5.25a); (c) and (d) a possible technique for affixing the Tura El-Asmant wire appliance. Drawings by Becker.

Description: This bridge supports between its natural adjacent teeth a re-placed maxillary right central incisor, with root filed down. The re-placed tooth has had two holes, one above the other, drilled through the mesial-distal aspects. A single silver wire, about 75 mm long, passed around one anchor tooth, with both ends

then passing through a single hole in the re-placed tooth and dividing again to surround the opposite anchor tooth where it was then twisted closed. The second silver wire, of about the same length, passed in the opposite direction at the level of the second drilled hole and parallel to the first wire, similarly passing around the opposite anchor tooth and tied off with a twist. These two wires at no point were in contact.

References

- Bardinet 1990.
 Becker 1997: 84–86, figs 5a and 5b.
 Iskander and Harris 1977: 85–90, pl. I–III.
 Iskander *et al.* 1978: 104–105, pl. III and IV.

History and provenance

During archaeological rescue excavations of a burial area at Tura El-Asmant during the 1952/3 season, Mr. Mohammed abd El-Tawab El-Hetta recovered over 100 mummies from 182 tombs of the Ptolemaic period. Tura el-Asmant, in the region/state of Al Qahirah, is today a small town near the eastern bank of the Nile, south of Cairo and Giza, and east of Saqqara. The exact location of the archaeological site is not specified; salvage excavation was initiated in response to the expansion of “the Portland Cement Company.” At the time Mr. El-Hetta was a co-excavator with Zaky Youssef Saad at nearby Helwan, which lies only a few kilometers south of Cairo.

Simple graves (T-1 through T-182) held mummies described (Iskander and Harris 1977: 86) as exemplifying an embalming technique noted by Herodotus (2.88) as the third, and least expensive, method of mummification. This implies that the tombs at Tura El-Asmant were not those of families of the highest status. This is an important observation, since it is the only case where there is a possible indication of a dental appliance being available to other than the elite members of society.

Tomb T-121 (Iskander and Harris 1977: 86; but Iskander *et al.* 1978: 104 identify the tomb as T-127) held a mummy whose skull was associated with traces of what has been interpreted to be silver wire. The skull was sent to the Department of Antiquities in Cairo for conservation and study. The subsequent location of the skeleton and/or the “dental appliance” is not provided by Iskander and Harris (1977), and has not been determined by Professor Krishna Kumar’s attempt on behalf of Becker to locate the skull or the appliance in Cairo in 1993.

Description

Testing revealed that silver, with slight copper impurities, had been used in some form of “wire,” but that this metal, or perhaps alloy, had corroded into something like silver chloride. Iskander and Harris identify the metal as “horn silver,”

which is a heavy halide mineral composed of silver chloride (AgCl). This is an extremely important observation. If this material reflects the use of a silver wire dental appliance, this would suggest that inexpensive silver appliances (see Iskander and Harris 1977: 88–89 for ancient values of silver) may have been used by affluent but non-aristocratic Egyptians of the Ptolemaic period. This would also confirm those texts of Ptolemaic and later date, which discuss silver dental appliances in contexts that would suggest that they were commonly available. Quite probably the silver in such prostheses has deteriorated and therefore gone unrecognized in most cases.

The illustrations provided by Iskander and Harris (1977) suggest that this is the skeleton of an adult (young?) with all teeth present, except for a “false” or replacement tooth wired in place. Iskander *et al.* (1978: pl. III, IV) include two excellent photographs taken by Dr. Jean-Jacques Quenouille (cf. Quenouille 1975). These plates offer detailed views of only the three teeth involved plus the adjacent alveolus. The photographs clearly indicate that the I2 and I1 are shoveled, which on a scale of 0–4 could be noted as 2 and 1 respectively. This non-metric dental trait is discussed above (no. W-4). The considerable wear on the lingual surface of these teeth also reflects a severe malocclusion (overbite).

What appears in the photographs to be a rotted stump of the right central incisor (I1) is situated in a severely abscessed area of the alveolus, or has a necrotic depression surrounding the root area. The illustrations provided by Iskander and his colleagues (1978) are not sufficiently clear to permit a secure evaluation, but the photograph appears to indicate a male. A detailed examination of the teeth and of this skeleton is needed.

A maxillary right central incisor (I1), which may have been lost due to trauma or to deliberate ablation, appears to have been placed back in the mouth (or replaced with another human incisor). The replacement may have taken place after some interval of time, while the injury (?) healed from the loss. Two holes (“or a bifurcate hole”: Iskander *et al.* 1978: 105) are said to have been drilled through “the lower half of the crown” of the replaced or replacement incisor, through which silver wire was passed and then tied. The relationship of these two drilled holes and the pattern of wiring and tying the replacement tooth to the flanking incisors is unclear from even Quenouille’s excellent photographs (in Iskander *et al.* 1978: pl. III, IV).

The dental health of this individual, as represented by surviving dentition, is good. If the traces of metal discovered represent a wire dental appliance, one possible method by which this tooth may have been suspended in place would have been to use two wires and to place the two drilled holes one above the other. Each wire would pass from one flanking or support tooth, through a drilled hole and around the opposite “post” or support tooth, then back through the same drilled hole. When the wire emerges it would pass around the opposite side of the first flanking tooth, and the ends would be “tied off” or twisted together near the distal margin of the post tooth.

Iskander and Harris (1977: 87) note that the root of the replacement or replaced right central incisor “is considerably shorter than that of the left” central incisor.

We concur in suggesting that this tooth could be the owner's original tooth with a root modified after loss for replacement in the space. The pattern of construction of the Tura El-Asmant dental appliance appears similar to that used in the El-Qatta appliance.

W-6 Eretria

Present location: National Archaeological Museum, Athens, inventory no. 733.

Type: gold wire.

Origin: Eretria, on the island of Euboea, Greece.

Jaw part: mandibular I1–PM1 (?).

Date: fourth century BCE (Clawson 1934: 25).

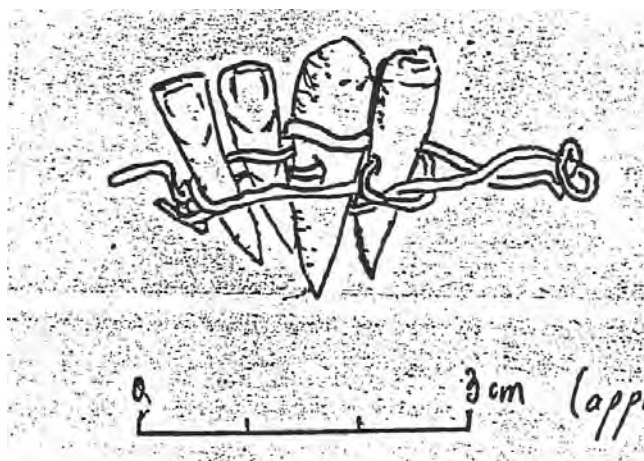


Figure 5.26a Wire appliance W-6 (Eretria). View of the lingual surface suggesting that teeth had been drilled (after Clawson 1933: 156, fig. 8b; also Weinberger 1948: fig. 39, lower). Drawing by Becker.

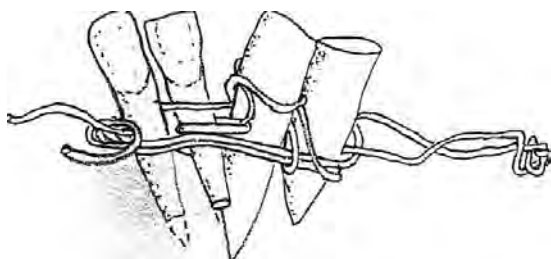


Figure 5.26b Becker's reconstruction of Eretria as a wire appliance with no drilled teeth. Drawing by Becker.

Sex: female ??? Becker (1997: 86) originally suggested that this was a male, but now questions that evaluation.

Dimensions: L. 49 mm (est.); W. variable (2–4 widths of wire).

Description: The poor original drawing and subsequent unavailability of this gold-wire appliance render difficult any efforts at description. This appears to have been a classic use of wire to stabilize loose teeth, but whether the four teeth apparently depicted were original to this artifact is another of the many unknowns. Clawson's (1934) effort to depict this appliance and the teeth associated with it reveals a confusing tangle that appears to represent a single strand. Our efforts to decode the path of that strand around the teeth do not result in a coherent or reasonable appliance such as would have been the effort of any practitioner. We can only assume that the appliance is incomplete as is.

References

- Becker 1997: 86–87, fig. 6.
Bliquez 1996: 2645, “B,” fig. 2
Clawson 1933: 158, fig. 8b; 1934: 25, fig. 2. This illustration, possibly taken from Guerini (1909) seems to depict two drilled teeth, emphasized by the lack of surviving roots on them.
Emptoz 1987: 551, fig. 9 (illustration depicts an extremely poor *copy* of the Eretria appliance, possibly a copy in Paris; see Appendix II).
Guerini 1909: fig. 9, lower.
Lufkin 1948: fig. 16.
Papabasileiou 1903: 65.
Sudhoff 1926: 63, fig. 39a (this is *probably* Eretria).
Weinberger 1946: fig. 6 (this figure appears to be the Eretria appliance, but may depict the Alexandria example; see Weinberger 1946: fig. 7).
Weinberger 1947: fig. 7 (same as 1946: fig. 6).
Weinberger 1948: 118, fig. 39 lower (this illustration derives either from Clawson (1933) or Guerini (1909)).

History and context

The Eretria prosthesis, apparently discovered in 1902 (Papabasileiou 1903: 64–65), is said to have come from a necropolis excavated at Eretria, where at least one burial (not that with the appliance) was found with a gold coin (stater) of Alexander the Great placed in the mouth. No details were published on the tomb in which the appliance was found, however. Clawson (1934: fig. 2) derives his depiction of this example from Guerini (1909: 60). Emptoz (1987) repeats the belief that it came from the skull of a woman, probably deriving his evaluation from the Papabasileiou text. The present disposition of related skeletal material, if any was recovered, is unknown. Micheloni (1976: 264) suggests a similarity between this example and the Sidon pieces (nos. W-1, W-2), and now we can link this form of appliance with others known from the Eastern Mediterranean and Rome.

Description

Four teeth appear entwined (bound?) by this wire, which extends off at either end of the appliance as if other teeth may have been involved. The four teeth involved might be the right central incisor through the right first premolar, but only if these are the original teeth that are depicted and if they are shown in proper order. Becker has redrawn them with an estimate of their relative size, as derived from published drawings. The presence of four teeth appears to be the only detail on which there is general agreement among those early authors describing this appliance, but all this information may have been copied from a single original source. As in many other cases, earlier descriptions were so uniformly poor that readers cannot be certain of anything. We cannot determine from the literature if the teeth are all natural, or were bound in place, or if one or two are “artificial,” or at least lost and replaced in correct position by drilling through them and suspending them with wires, as is the case with the Sidon I appliance. Although Clawson’s rough drawing (1934: fig. 2) may appear to represent one or more holes drilled through the canine, this is an unlikely tooth to be replaced after loss. Our reconstruction (Figure 5.27b) suggests that no drilling had been used to put in place this gold-wire appliance. The two visits made by Becker to the National Archaeological Museum in Athens (see no. 20) failed to locate this example.

Without detailed direct examination, no conclusions can be drawn at this time, but it would be very unlikely that holes would have been drilled in any teeth still in position, and these teeth appear to have intact roots. In some illustrations at least the canine appears to be pierced, but the total number of teeth originally involved is not evident. The original appliance probably spanned six teeth, but the anchor teeth are no longer associated with the fourth teeth illustrated. These four teeth usually appear to be gracile when depicted, thus might derive from a woman. However, dentition among Greek populations appears uniformly gracile, and gender variation has not been specifically demonstrated for Greek populations, so it remains possible that the individual wearing this appliance was male.

The Eretria prosthesis was formed through a complex pattern of wiring, but whether the wires pass through holes drilled mesial-distally through the teeth at the base of the crown, or only pass around the teeth remains unclear. Whether the two teeth in the middle (I2 and C.) may be the only ones drilled also is not clear. Quite possibly all four teeth are not even original to the appliance, but are replacements used for display, or in the process of making a copy of this appliance. If these are original teeth, the anchor teeth to which they might have been tied would have been the I1 and the PM2, which are not present.

Interpretation

Perhaps the presence of this appliance reflects some technological/dental connection between Greece and the eastern/Levantine practice of using wire dental appliances. The Hippocratic texts of the Classical era (fifth–fourth centuries BCE)

mention wiring to stabilize loose teeth. It is also late enough in date (early Hellenistic, if we may date it by the gold stater placed as Charon's coin in the mouth of someone in the same necropolis) that many traders, diplomats, slaves, mercenaries and their families, were circulating across the Mediterranean. It is true that Etruria was intimately associated with Euboea since the Iron Age (eighth century BCE) through the settlers of the colony of Pithekoussai in the Bay of Naples. Ancient literary sources (Strabo 5.247C) note that gold was being purveyed there (processed, probably into diadems for the burials of native Italian wives of the original Greek and Levantine colonists (although there are no gold sources on or near the island itself). Archaeology has opened new vistas on these early, shared efforts at trade and colonization, with the excavation of burials of families of mixed ethnic background: Greeks from Eretria, Chalkis, and Corinth, men and women from Phoenicia and North Syria, Italic persons (Daunian, Latin, Campanian, etc.) and Etruscans who all married and raised (and often buried) numerous children. (See Ridgway 1992: 111–118; Buchner 1979; Buchner and Ridgway 1993; Becker and Donadio 1992; Becker 1995b.) The grand cooperative enterprise of Etruscans and Euboean Greeks occurred in the Iron Age-Archaic periods, the era of Etruscan band-type appliances.

Professor Larry Bliquez (personal communication 2002) wisely urges caution in identifying any dental appliance found outside of Etruria or Phoenicia as deriving from one source or another. He suggests that Greeks may have derived such technology from their Levantine trading partners, and employed the technology at home. This might be consistent with Lucian's Roman-era testimony, which certainly notes the existence of these devices, probably in the Greek world. Becker emphasises, however, that there would have been cultural inhibitors on the use of such appliances in Greek society. Certainly the Greeks would not wish their wives to bedeck themselves with the kinds of ornamentation to be seen on those Etruscan women present during dining sessions, who made the visiting Greeks so uncomfortable by their presence.

W-7 Collatina (Rome)

Present location: Soprintendenza Archeologica di Roma, Palazzo Altemps, Rome.

Type: wire.

Origin: Rome, Collatina Necropolis, Viale della Serenissima (Municipio V di Roma).

Jaw part: mandible, spanning five teeth.

Date: Imperial Era (first–second century CE?).

Sex: female.

Dimensions: L 39; Extant W. 10 mm.

Description: This simple gold-wire dental prosthesis from the Collatina Necropolis holds one re-placed tooth, a right central incisor in an otherwise normal mandible. It is fashioned from a single length of gold wire, less than

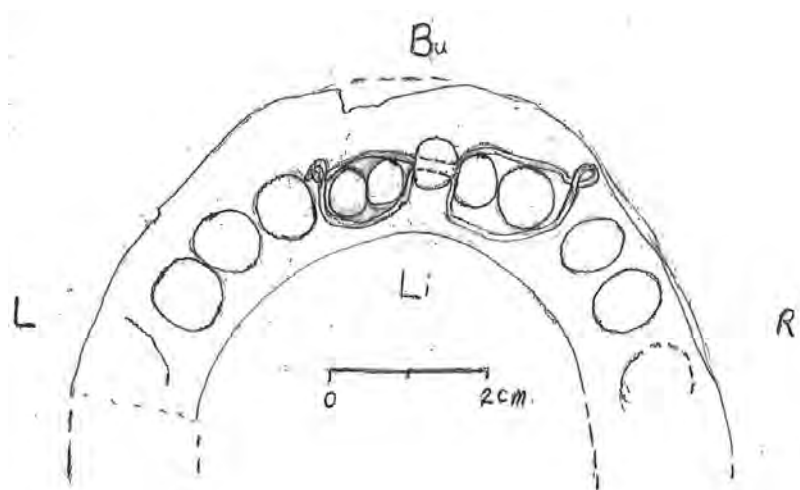


Figure 5.27 Roman wire appliance from Collatina necropolis, Rome. Drawing of placement in jaw as found, by M. J. Becker.

1 mm in diameter, with an estimated length of perhaps 100 mm. The appliance as wound in place measures 39 mm long and now approximately 10 mm across the lingual-buccal plane, probably due to post-mortem distortion. Originally this appliance probably measured only 8 mm across, the distance needed to surround the right canine. The wire passes around two teeth on each side of the single “replacement” or re-placed tooth, held in place by wire passing through a mesial-distal hole.

References

- Catalano *et al.* 2007.
Minozzi *et al.* 2007.

Context

The complete reports on the excavations from the Collatina Necropolis remain in production, but Simona Minozzi kindly provides us with an important glimpse of an object of great interest. Excavations in a necropolis near the Viale della Serenissima (Municipio V di Roma, the southeast sector of Rome) during the year 2000 brought to light the only archaeologically documented gold-wire appliance found in Rome, or anywhere on the Italian peninsula. The necropolis of the Viale delle Serenissima is now identified as the Collatina Necropolis, relating it to the ancient Via Collatina, which

passes nearby. An impressive stretch of the original surface of this ancient road has been found nearly intact in the area of Rome's Serenissima train station (Buccellato *et al.* (2008).

The full extent of the Collatina necropolis, as with most of Rome's sprawling ancient burial areas, remains unknown; it is now the largest known necropolis of the Imperial Age (31/27 BCE–476 CE) within the area of modern Rome. Excavations were directed by Dr. Stefano Musso, then an Inspector for the Archaeological Superintendence of Rome, and its director in 2014.

The specific archaeological context of this cremation burial is provided only in Catalano *et al.* (2007), who describe a "Tomba Monumentale" identified as "M1 T.1." The cremation with this appliance is said to have been at the center of the tomb, where the cremation itself took place. This suggests that the tomb was built around the pyre. In addition to mausoleums, most of the "tombs" from the Collatina Necropolis are said to be inhumations. Over 2,200 burials have been recovered (Minozzi *et al.* 2012: 269), but among them the cremations encountered are said to be few (Buccellato *et al.* 2008: 63). Specific numbers have not been tabulated, but the three cinerary urns that are mentioned appear to be only a small percentage. The burned bones from these various cremations are to be studied by Elena Santandrea (Buccellato *et al.* 2008: 86, n. 42).

Recognition of the importance of the find of a gold-wire dental appliance in peninsular Italy led to the rapid publication of this object by a team of scholars (Minozzi *et al.* 2007). The archaeological context has been roughly dated first–second centuries CE, or early in the Imperial Age. Specific details regarding this individual burial have not yet been published more precisely. The bones are described as representing "the partially cremated remains of an adult woman" (Minozzi *et al.* 2007: e1). The focus of the poster presentation and article is on the gold appliance and the mandibular corpus that holds the teeth found bound with gold wire. This is the only cremated bone from this burial yet described. A photograph of the complete surviving *corpus mandibularis* appears in Catalano *et al.* (2007: fig. 1). The publications and their illustrations provided the primary sources for the evaluation that follows.

The use of photographs to evaluate the age and sex of an individual is a notoriously risky business. Given that only the body of the mandible is illustrated, any statement made here regarding age and sex must be considered with caution. The apparent presence of robust third molars might be an indication that this person might be a high status female, but the evaluation of sex of a single individual without others from that population is extremely difficult (Huntsman and Becker 2013). The absence of apparent dental decay should be compared with others in this population, but the extent of periodontal deterioration suggests an adult of perhaps 50 years of age ($\pm$ 10 years) at death.

Becker interprets the illustrated evidence associated with this appliance as indicating the product of a complete and perfectly normal cremation, probably

at a temperature of 900–950°C. The presence of an un-crushed mandibular corpus within this cremation burial presents one of two possibilities. First, that the cremated remains had been buried in a large container (sarcophagus, large chest, wooden coffin, etc.) or; second, that only selected portions of the *ossilegium* (the bones that normally survive cremation) had been chosen for deposit in this grave. The second alternative is less likely than that the burned bones lay where they were burned as indicated by the authors.

Minozzi *et al.* (2007) carefully note that the typical cremation damage to the crowns of teeth that were in place at death applies to this individual. All the missing crowns were lost due to fire damage in a process that Becker has described as “exploding.” They add that, “the jaw was largely preserved from the third left molar to the second right premolar. The single tooth held in place by this gold “bridge” does not manifest the type of fire damage common to teeth that were living at the time of death. This seems as if the “replaced” tooth was a dry (non-living) inclusion, supported by gold wire, and not subject to the kinds of spalling or explosive damage to the crown as the living teeth. Whether this “replacement” was the owner’s original tooth re-placed in this position is uncertain.

The melting point of pure (24 ct) gold is 1063°C. Silver melts at 961°C. The pyre temperatures of most cremations in antiquity did not exceed 950° C. It is not surprising that the gold dental appliance survived the process; we might expect more of them to be found, if they had been popular.

The detailed illustrations in these publications focus on the appliance and omit some important evidence concerning the surviving mandibular corpus and the condition of the individual teeth therein. The descriptions below derive from the published photographic evidence (Minozzi *et al.* 2007: page e2, figs a–c).

The dentition

Some evidence of periodontal deterioration is evident, suggesting an age of perhaps 30 to 35 years. Roots of the left third and second molars are in place (Catalano *et al.* 2007: fig. 1). The root and neck of the first molar appear firmly anchored. Aside from the missing (“exploded”) crowns, the left premolars appear to have been healthy at death. The authors’ observation of a “peculiar abrasion of the buccal surface of the left first premolar” warrants further study. The socket for the left canine reveals a relatively healthy tooth, aside from the periodontal inflammation. The root itself is not present, having been lost post-mortem, and almost certainly post-cremation.

The left lateral incisor served as an “anchor” for the dental appliance. Despite its precarious position, next to an extremely loose central incisor, it maintained this small pontic during life and even after cremation and burial. A root canal, as suggested by the authors, is not visible. The “left central incisor is not preserved.” Becker strongly disagrees with the authors’ belief

that it “is certainly ‘artificial,’ as suggested by the space apparent in the gold wire” (Minozzi *et al.* 2007). If it were “artificial” or the original tooth held in place by the wire, the tooth would have been drilled and the wire would have been pinched to pass through the hole, not simply passed around the tooth. The space for the left central incisor is evident in the wire of the prosthesis. However, the area of the socket of the left central incisor reveals extensive periodontal deterioration. The same deterioration certainly caused the right central incisor to fall out, probably providing the “replacement” tooth, as the authors also believe. Becker believes that the left central incisor was extremely loose in a deteriorating and shrinking socket, but that it had not yet fallen out. In that loosened condition it would not have served as an anchor tooth, requiring that the wire appliance incorporate the left lateral incisor to support the pontic.

The right central incisor is the dental space for which this pontic was constructed using the person’s own lost tooth to create this appliance. The re-placed tooth has been drilled at the base of the enamel with a mesial-distal hole, approx. 2 mm in diameter, to pass both the lingual and buccal aspects of the appliance wire through it. Minozzi *et al.* (2007) suggest that the root of this tooth has been filed down to allow it to fit above the gingival tissue without causing further irritation. Note that this tooth is the best preserved of the series in that it retains its crown. This suggests that this dry (dead) tooth was not subject to the same effects of the fire as the others.

The right lateral incisor and canine appear to be angled toward the medial area, probably in response to the periodontal conditions that were causing dental loss of the central incisors. Not mentioned in the publication is a flake of burned bone shown lodged within the gold prosthesis against the lingual surface of the right lateral incisor. The right premolars were present at death, but the roots probably became dislodged from their sockets after cremation. The second premolar socket has what appears to be an abscess at its base. The socket for the right first molar is so extensively damaged as to preclude any evaluation. The extent of the post-cremation bone loss suggests that a tooth had been in place at death. Had that tooth been lost, bone formation at the site probably would have survived combustion. The Minozzi *et al.* (2007) text suggests that the region of the corpus below the second and third molars does not survive, but the illustration in Catalano *et al.* (2007) reveals that the area below both teeth is present, although extensively damaged.

The loss of anterior teeth, or general looseness of the anterior teeth in the elderly is seen here to be associated with extensive periodontal disease, as noted by the authors. Becker’s experience with Italic pre-Imperial populations is that dental loss most often begins with caries among the molars. Anterior teeth tend to remain in place to very old age.

The need to use two teeth on either side to anchor this tooth reflects the extent of periodontal loss that caused this right first incisor to fall out. The immediately adjacent teeth were so compromised as to be incapable of

(continued on p.299)

Table 5.8 Etruscan and related dental appliances

#	Name	Provenance	Date	Sex	Jaw region	Type	*	Comments, location
1	Barrett I	Etruria	? 5th c.	F?	Maxilla I1 plus	Simple band, 3 spaces Stabilizing	NA	Formerly private collection of Dr. Wm. C. Barrett, Buffalo, NY—location unknown, believed to be in Buffalo area
2	Barrett II	Bisenzio, tomb on Capodimonte	500– 480	F	Maxilla (2PM–I1?)	Simple band, 5 spaces Stabilizing	NA	Formerly private collection of Dr. Wm. C. Barrett, Buffalo, NY—location unknown, believed to be in Buffalo area
3	Van Marter	“Lake of Valseno” (Bolsena?)	600??	F??	Maxilla?	Ring series 3 spaces, with central false tooth	NA	Formerly private collection of Dr. Wm. C. Barrett, Buffalo, NY—location unknown, believed to be in Buffalo area Obtained by Dr. James Gilbert Van Marter in Italy
4	Copenhagen	Orvieto?	500–490	F	Maxilla I1–I2	3 welded loops, no rivet	*	National Museum, Copenhagen inv. no. 8319
5	Poggio Gaiella	Chiusine necropolis Poggio Gaiella	?4th– 3rd c.	F	Maxilla 1PM to PM1	Single, complex band Now in 2 pieces	*	Florence, Museo Archeologico, inv. no. 11782 Now in wrong skull and mandible
6	Populonia	Populonia, San Cerbone necropolis (tomb destroyed)	4th c.	?	Maxilla (2I–I2)?	4-ring series, 4 spaces, 1 rivet	NA	Florence, Museo Archeologico, inv. no. 84467—lost in Florence Flood 1966, Minto, Populonia 1943: pl. 48, 1 (mandible)

7	Ghent	“Near Orvieto”	6th c. or later?	?	Maxilla (2I–I2?)	Complex band 3 loops, 4 spaces	NA	(Ghent University Museum, 1899—looted by German invasion forces WWII) attached to skull fragment by canine tooth
8	Bruschi I	Probably territory of Tarquinia	5th c.?	F	Maxilla 4 spaces (incisors)	Narrow band, 4 spaces Stabilizing	*	Museo Nazionale Archeologico, Tarquinia, Ex collection, Count Bruschi-Falgari, Corneto
9	Bruschi II	Territory of Tarquinia?	?	F??	Maxilla (2I–I2?)	4 rings, 1 rivet (1 false tooth, 3 spaces)	NA	Displayed with two teeth Collection, Count Bruschi-Falgari, Corneto Missing since 1916–1925 transfer to Museo Nazionale Archeologico, Tarquinia
10	Bruschi III	Territory of Tarquinia?	2nd c.??	F	Maxilla?	4 braces, 5 spaces, No rivets Stabilizing	*	Museo Nazionale Archeologico, Tarquinia Ex collection, Count Bruschi-Falgari, Corneto Often confused with Corneto I Teeth now associated are recent replacement
11	Corneto I	Tarquinia	530–510?	??	Maxilla? Left C–I2	Braced? band, 5 spaces, 2 rivets, 1 false tooth	NA	Museo Nazionale Archeologico, Tarquinia. Probably excavated 1876–1877 Missing since 1916–1925 (or earlier)
12	Corneto II	Tarquinia	500?	F??	Maxilla? (2PM–right C?)	Compound: 7 rings, 8 spaces, 3 rivets, ivory	NA	Museo Nazionale Archeologico, Tarquinia. Possibly excavated c.1875 Missing since 1916–1925 (or earlier)

(continued)

Table 5.8 (continued)

#	Name	Provenance	Date	Sex	Jaw region	Type	*	Comments, location
13	Liverpool I	Unknown	?	F?	Maxilla 2I–I2	Band with 2 false teeth, 2 rivets, 4 spaces	*	Liverpool World History Museum inv. no. 10334 Ex collection of Joseph Mayer Collected before 1857
14	Liverpool II	Unknown	?	F? 45+ yrs	Maxilla 2I–I2	Band for 2 false teeth, 2 rivets, 4 spaces	*	Liverpool World History Museum inv. no. 10335 Ex collection of Joseph Mayer Collected before 1857 Anchor teeth may be replacements
15	Valsiarosa	Valsiarosa necropolis of Falerii Veteres, Tomb 20	4th c.?	F?	Maxilla 2I–I2	4 welded loops, 4 spaces, 1 rivet	*	Museo Archeologico del Agro Falisco, Civita Castellana, previously Villa Giulia inv. no. 1515 Displayed since 1880s on ancient skull that is not original owner
16	Teano	Teano, Tomb 18, Fondo Gradavola necropolis, Campania	4th– 3rd c.	F	Maxilla C–C	6 rings, no rivets Stabilizing?	NA	Institut für Geschichte der Medizin, Universitätsklinikum Charité, Humboldt-Universität, Berlin; Excavated 1907
17	Praeneste	Praeneste, Sporadic find in necropolis	?	F 35–45 yrs**	Maxilla 2I–I2 3 teeth	Band, 4 spaces Central pair riveted	*	Ex collection Professor W. Dieck Villa Giulia inv. no. 13213, on display In 1885 was in Palazzo Barberini, Rome

18	Satricum	Satricum, Borgo Le Ferriere, Tumulus C	7th c. c.630	F 50 yrs	Maxilla 2I-C	Band, 5 spaces Gold tooth (I1)	*	Villa Giulia inv. no. 12206
19	Bracciano	Near Lake Bracciano	7th c.?	F?	Maxilla 1I-12	Band, 3 spaces, 1 rivet holds false tooth	NA	Vienna, Museum of Natural History inv. no. 24.296 (acquired or catalogued c.1990), apparently part of a tomb group in museum Athens National Museum: Lambros Collection inv. no. 358
20	Tanagra	Tanagra, Boeotia	4th-3rd c.?	F? 45±10 yrs**	Maxilla C-C	Compound band of 2 joined loops, no rivets Stabilizing band, 4 loops, 2 rivets (in 2 false teeth)	NA	Private collection, Ankara, Turkey. Present owner Dr. Ilter Uzel
21	Sardis	Salihli "ancient Sardis" found by farmer in a tomb	3rd c.??	F 25-40 yrs	skull + mandible 2 central incisors false; anchors 2 second incisors			

Note: where noted, sex has been assigned based on (small) tooth size, determined by the spaces in appliances.

* Autopsied by M. Becker.

NA Not available for examination or lost.

** Age estimate based on dental wear or extant teeth.

Table 5.9 Typological details of Etruscan gold-band appliances

#	Name	Provenance	Date	Sex/age	Gold design	# Rivets	False: material	Living teeth present	Type/ design
1	Barrett I	Etruria	5th c.	F?	Simple band				3 spaces Stabilizing
2	Barrett II	Bisenzio	500	F	Simple band				5 spaces Stabilizing
3	Van Marter	Bolsena?	600	F??	Ring series		?		3 rings Replacement
4	Copenhagen	Orvieto?	500	F	Welded loops				3 rings Stabilizing
5	Poggio Gaiella	Chiusi	4th–3rd c.	F, 25–40	Complex band			Wrong skull	4 rings, 8 teeth Stabilizing
6	Populonia	Populonia	4th c.	?	Ring series	1	1		4 rings Replacement
7	Ghent	Orvieto	6th c.??	?	Complex band				3 rings, one double-size Lost
8	Bruschi I	Tarquinia	5th c.?	F	Narrow band				4 spaces Stabilizing
9	Bruschi II	Tarquinia	?	F??	Ring series	1	1		4 rings Replacement
10	Bruschi III	Tarquinia	2nd c.?	F	Braces?	No			5 spaces, braced Stabilizing band
11	Corneto I	Tarquinia	530?	??	Braced band?	2	2		5 spaces, band Replacement

12	Corneto II	Tarquinia	500?	F??	Compound rings	3	Ivory	7 rings Replacement
13	Liverpool I	??	?	F?	Band	2	Human	Band, 4 spaces Replacement
14	Liverpool II	??	?	F? 45+	Band	2	2	Band, 4 spaces Replacement
15	Valsiarosa	Falerii	4th c.	F?	Welded loops	1	1 at II	4 rings? Replacement
16	Teano	Teano	4th–3rd c.	F	Ring series			6 rings Stabilizing?
17	Praeneste	Praeneste	?	F, 35–45	Band	2	2	Band, 4 spaces Replacement
18	Satricum	Satricum	630	F, 50	Band		Gold	Band, 5 spaces Replacement
19	Bracciano	Bracciano	7th c.?	F?	Band	1	1	Band, 3 spaces Replacement
20	Tanagra	Boeotia	4th–3rd c.?	F? 45	Compound band			Band, 2 loops Stabilizing
21	Sardis	Sardis, Turkey	3rd c.?	F 25–40	Band	2	2	Mandibular, band Replacement

Note: where material is not noted, false teeth are missing but attested by presence of rivet(s).
Shading denotes appliances missing or not available for study.

Table 5.10 Technical details of Etruscan gold-band appliances

<i>Appliance</i>	<i>Band type</i>	<i># Rings</i>	<i>Total # teeth/ replacements</i>	<i>Replacement Teeth</i>	<i># Rivets</i>	<i>Purpose/ conditions</i>
Barrett I	Simple, narrow	—	3 spaces maxillary	No	None	Stabilize (poor info)
Barrett II	Simple, narrow	—	5 spaces maxillary	No	None	Stabilize
Van Marter	—	3 rings	2 + 1 replace maxillary	R central incisor	None	Replacement human tooth
Copenhagen/ Orvieto?	—	3 rings	2 + 1 replace maxillary	R central incisor lost	None	Replacement; anchors worn
Gaiella (Chiusi)	Complex band	4 “rings”	8 teeth maxillary	No	None	Stabilize; skull substituted
Populonia	—	4 rings	4 maxillary incisors	Central incisor lost	1 rivet	Replacement; fragments of the maxilla remain
Ghent (Orvieto?)	—	3 ring— one double	4 incisors	No	None	Stabilize; a tooth substituted
Bruschi I Caere/Tarquinia	Simple, narrow band	—	4 maxillary teeth	No	None	Stabilize loose teeth
Bruschi II Tarquinia	—	4 rings	3 + 1 replace	Central incisor lost	1 rivet	Replacement
Bruschi III Tarquinia	Braced band	—	5 teeth—L canine to R lateral incisor	No	None	Stabilize; teeth recent inserts
Corneto I Tarquinia	Braced band	—	3 + 2 replace	Central lateral incisors—1 lost	2 rivets	Replacement unknown material

Corneto II Tarquinia	—	7 rings compound rings	8 spaces = 5 + 3 replace maxillary	1PM; I1 and I1 one lost	3 rivets	Replacement 2 “teeth” carved in one piece (animal tooth)
Liverpool I	Band	—	2 + 2 replace maxillary	2 central incisors	2 rivets	Replacement; human teeth
Liverpool II	Band	—	2 + 2 replace	2 R incisors lost	2 rivets	Replacement; only anchors remain
Valsiarosa Falerii	—	4 rings welded loops	3 + 1 replace maxillary	Central incisor lost	1 rivet	Replacement; skull substituted
Teano	—	6 rings	4 + 2 spaces maxillary	No	None	Stabilize; probably original teeth overbite, worn
Praeneste	Band	—	2 + 2 replace maxillary	2 central incisors	2 rivets	Replacement human teeth
Satricum	Band	—	4 + 1 replace maxillary	Central incisor	None	Replacement; hollow gold tooth
Bracciano	Band, compound	—	2 + 1 replace maxillary	R central incisor	1 rivet	Replacement; owner’s tooth, filed down?
Tanagra	Compound band	—	6 spaces, canines, incisors	No	None	Stabilize, 2 loops
Sardis	Band	—	2 + 2 replace mandible	2 central incisors	2 rivets	Owner’s teeth filed down; tartar on both replacements

Note: shading denotes appliances now missing or unavailable.

Table 5.11 Wire-appliances (Hellenistic and Roman periods; not Etruscan)*

Item	Provenance, location	Date	Sex/age	Gold design	False: material	Living teeth present	Type/ design
W1. Gaillardot	Sidon, Tomb XI, Room 1, fosse b (1861 excavations) Louvre	1st 1/4 4th c. BCE	F	Wire	2 human incisors—I1, I2 2 holes drilled in each	4 C to C roots not intact	Mandibular?? Wire binding 6 teeth
W2. Ford	Sidon, Tomb I, 'Ain H ilweh necropolis, East sarcophagus American University Museum Beirut	5th–4th c. BCE	M? middle aged	Wire brace	No	6, with decay etc. C to C	Mandible
W3. Alexandria	Ibrahimieh, Tomb Greco-Roman Museum, Alexandria	Roman 1st c. CE	?	Wire	No	3: I1 to C? Poor condition	Mandible
W4. El-Qatta	El-Qatta Tomb no. 90 reused mastaba, shaft 5 Cairo Museum?	Roman	?	Wire	1 tooth drilled	3 teeth wired series I1 to C? anchor on R canine and C incisor	Maxilla

W5. Tura el-Asmant	Tomb T-121 in Ptolemaic necropolis Cairo/Museum etc.?	Ptolemaic	M?? young adult?	Silver wire	R central incisor returned to mouth (drilled) root modified Human teeth, drilled	I1 to I2 necrotic abscess R central incisor	maxilla severe malocclusion 2 teeth drilled— or caries?
W6. Eretria	Tomb at Eretria Athens Nat.Arch. Museum	4th c. BCE	F???	Wire		4 teeth—I1 to P?M1? originally spanned 6 teeth I2 and C are drilled	Mandibular complex pattern wiring— originally binding 6 teeth, 2 of them replacements (drilled, wired)
W7. Rome	Tomb in Collatina necropolis, partially cremated skeleton	Imperial Roman	F 50?	Wire	Original tooth: R central incisor, root missing (drilled, wired)	R canine thru L lateral incisor:	Mandibular: adjacent teeth damaged by periodontal disease, so anchored to next teeth

* All gold wire except where otherwise noted.

Table 5.12 Measurements of ancient appliances

Item	Length	Width	Thickness	Length of strip	Buccal-lingual	Rivets	Type	# Teeth spanned	Details
Barrett I	20	6	0.2 to 0.3?	53	6	—	S	3	Lost?
Barrett II	25	2.0 to 2.5	0.2 to 0.3?	60	6 to 7	—	S	4	Lost?
Van Marter	25	4	—	—	5	—	R + 1	3	Lost?
Copenhagen	21.6	5.5 to 6.2	—	NA	5.1 to 6.7	—	R + 1	3	* 2 strips
Poggio Gaiella	42	2.82 to 3.3	0.12 to 0.17	*90 and	—	—	S	8	Lost
Populonia	(25 to –30)	—	—	—	—	1	R + 1	4	Lost
Ghent	27	2.8	—	—	—	—	S	4	Lost
Bruschi I	32	1.9 to 2.1	0.20	74	—	—	S	4	NA
Bruschi II	29	2.1	—	—	—	1	R + 1	4	NA
Bruschi III	36.8	2.6 to 2.8	—	83	—	—	S	5	NA
Corneto I	28	5 to 6	—	—	—	2	R + 2	5	NA
Corneto II	62.5	4.0 to 4.3	—	—	—	3	R + 3	8	NA
Liverpool I	30.9	3.0 to 3.1	0.3	?	?	2	R + 2	4	Most complex
Liverpool II	28.2	4.9 to 5.2	0.2	?	?	2	R + 2	4	
Valsiarosa	38	5.0 to 7.0	?	?	?	1	R + 1	4	
Teano	47	4.5	—	—	—	—	S	6	NA
Praeneste	31	2.7 to 2.9	?	?	?	2	R + 2	4	Unique 2-layer construction
Satricum	29	1.8 to 2.2	?	?	?	—	R + 1	5	Gold tooth
Bracciano	24	3.5 to 3.8	0.24 to 0.30	?	?	1	R + 1	3	
Tanagra	43.3	4.3 to 5.6	0.2	*2 × 25	?	—	S	6	* 2 strips
(Sardis)	18	4.0	—	—	—	2	R + 2	4	NA

Note: all measurements in millimeters; many approximated or estimated from illustrations (see catalogue).
NA = not available.
Function: S = Stabilize; R = Replacement (R + # teeth replaced); ? = Some doubt as to exact identification.

Table 5.13 Wire appliances*

Item	Length	Width	Thickness	Length of wire	Type	# Teeth spanned
Sidon I	34	Varies		over 100	R + 2	6
Sidon II	35	3 to 4?			S	(6)
Alexandria	25	Varies	1 or less		S	(3)
El Qatta	(43)	Varies			R + 1	3 or more
Tura el-Asmant	27	Varies			R + 1	3? 2 silver wires
Eretria		Varies			R + 2	6
Rome		Varies			R + 1	5

* Measurements in millimeters, mainly from published illustrations.

Function: S = Stabilize; R = Replacement (R + # teeth replaced); ? = Some doubt as to exact identification.

providing supports for this simple pontic. The single wire passes in a single loop around the right canine and then the right lateral incisor before being pinched together to pass through the hole drilled through the re-placed tooth. Upon exiting from this hole (or holes), the two ends of the wire extend around the left central incisor (now lost) and the left lateral incisor that serves as the anchor tooth. The left central incisor must have been quite loose at the time that this appliance was fashioned. The ends of the wire are then twisted, at the distal-buccal position, in a clockwise direction for two and one-half turns, and then end. Not noted in the publications is the extremely loose fit of the right distal aspect of the wire and what seems to be a small loop at the right distal end, at the distal-buccal surface of the canine. This is not clearly revealed in the illustrations but appears to be a tiny loop, of less than 3 mm diameter that was made in the wire at the end near where it begins to pass around the right canine, and then the right lateral incisor. This loop might have served to take up slack in the appliance by allowing the wire to be tightened by twisting it with a small pair of pliers.

Summary

This use of gold wire to fit a replacement tooth is the only example yet known from Rome (or Italy), and was excavated under controlled conditions from a tomb of the first–second century CE in the Collatina necropolis of Rome. A woman who died around 50 years of age was wearing the appliance to replace a mandibular central incisor probably lost to periodontal disease. If such appliances had been common, more examples should have survived. This Roman example of the Early Imperial Age falls within the same medical tradition known from the Hippocratic corpus, and also reflects specimens buried in the eastern Mediterranean, Levant and Egypt.

Table 5.14 Appliances ordered by type and number of replaced teeth, number of teeth spanned

Item	Length	Width	Thickness	Length of strip	Buccal-lingual	Rivets	Type	# Teeth spanned	Details
Barrett I	20	6	0.2 to 0.3?	53	6	—	S	3	Lost?
Barrett II	25	2.0 to 2.5	0.2 to 0.3?	60	6 to 7	—	S	4	Lost?
Ghent	27	2.8	—	—	—	—	S	4	Lost
Bruschi I	32	1.9 to 2.1	0.20	74	—	—	S	4	
Bruschi III	36.8	2.6 to 2.8	—	83	—	—	S	5	
Teano	47	4.5	—	—	—	—	S	6	NA unique 2-layer construction * 2 strips *2 strips
Tanagra	43.3	4.3 to 5.6	0.2	*2 × 25	?	—	S	6	
Poggio Gaiella	42	2.82 to 3.3	0.12 to 0.17	*90 and 42	—	—	S	8	
Van Marter	25	4	—	—	5	—	R + 1	3	Lost?
Copenhagen	21.6	5.5 to 6.2	—	NA	5.1 to 6.7	—	R + 1	3	
Bracciano	24	3.5 to 3.8	0.24 to 0.30	?	?	1	R + 1	3	
Populonia	(25 to –30)	—	—	—	—	1	R + 1	4	Lost
Bruschi II	29	2.1	—	—	—	1	R + 1	4	NA
Valsiarosa	38	5.0 to 7.0	?	?	?	1	R + 1	4	
Satricum	29	1.8 to 2.2	?	?	?	—	R + 1	5	Gold tooth
Liverpool I	30.9	3.0 to 3.1	0.3	?	?	2	R + 2	4	
Liverpool II	28.2	4.9 to 5.2	0.2	?	?	2	R + 2	4	
Praeneste	31	2.7 to 2.9	?	?	?	2	R + 2	4	
(Sardis)	18	4.0	—	—	—	2	R + 2	4	NA
Corneto I	28	5 to 6	—	—	—	2	R + 2	5	NA
Corneto II	62.5	4.0 to 4.3	—	—	—	3	R + 3	8	Most complex

NA = not available; ? = indicates imperfection of data.

Function: S = Stabilize; R = Replacement (R + # teeth replaced); ? = Some doubt as to exact identification.

Notes

- 1 A note in *The Dental Record* (Waite 1885b) erroneously suggests that these prostheses had been described by Waite in the April (1885) issue of the *Independent Practitioner*. The text reveals that Waite actually refers to Van Marter's (1885a) publication.
- 2 Shoveling here indicates the condition of the incisors being "shovel-shaped," dish-shaped in profile so that the back face is concave, the front face flat or convex. The condition is a harmless genetic variation providing a greater articular surface for chewing or use of teeth as tools. Given a large enough sample this trait might indicate familial, regional or ethnic affiliations.
- 3 Post-mortem tooth loss occurs as the skull and its tissues dry during burial. It can be recognized, with some experience, by the clearly defined edges of the tooth sockets: the gums have disappeared, the jaw is dry and teeth with straight roots can easily drop out (often they are shoved back into the wrong sockets for a better display!).
- 4 For background on the Bruschi archaeological collection, see Fabrizio Santi, 2002. "La famiglia Bruschi-Falgari e la sua collezione di antichità etrusche," *Bollettino della Società Tarquiniese d'Arte e Storia* 31: 159–188. The Attic Black Figure vases in the collection, surely imports found in Etruscan tombs, are catalogued by Danilo Nati (2012). *La collezione Bruschi Falgari. Materiali del Museo Archeologico Nazionale di Tarquinia*, 20 (Archeologia 168), Rome: G. Bretschneider.
- 5 Pot, who fails to recognize that that the replaced teeth in the Praeneste appliance are held with rivets with peened heads, also presents an inaccurate drawing of the appliance (1985: fig. 2), as well as an incorrect interpretation of the method of its manufacture. Van Marter (1886: 57) also failed to note the rivets that held the two false teeth in place. Van Marter notes that he saw "four natural teeth" in 1885 or early 1886, but his general description is too poor to determine if one of the anchor teeth was lost only after 1886.
- 6 This Tomba XVIII, sometimes noted as "Tomb 18," is distinct from the Tomb 18 that was recovered during the 1907–1910 excavations at Satricum (Ginge 1996: 171).
- 7 Presumably Evaristo Breccia (1876–1967) who published extensively on Alexandrian archaeology and the Greco-Roman Museum.
- 8 Those scholars who ignored Euler's report, submitted to Junker in 1928 (Junker 1929: 256–257), fail to understand the function of that amulet as well as its date. Euler's report on the Junker amulet, written long before the discovery of the El-Qatta appliance, reflects many of the same problems of interpreting archaeological context and providing an accurate date.